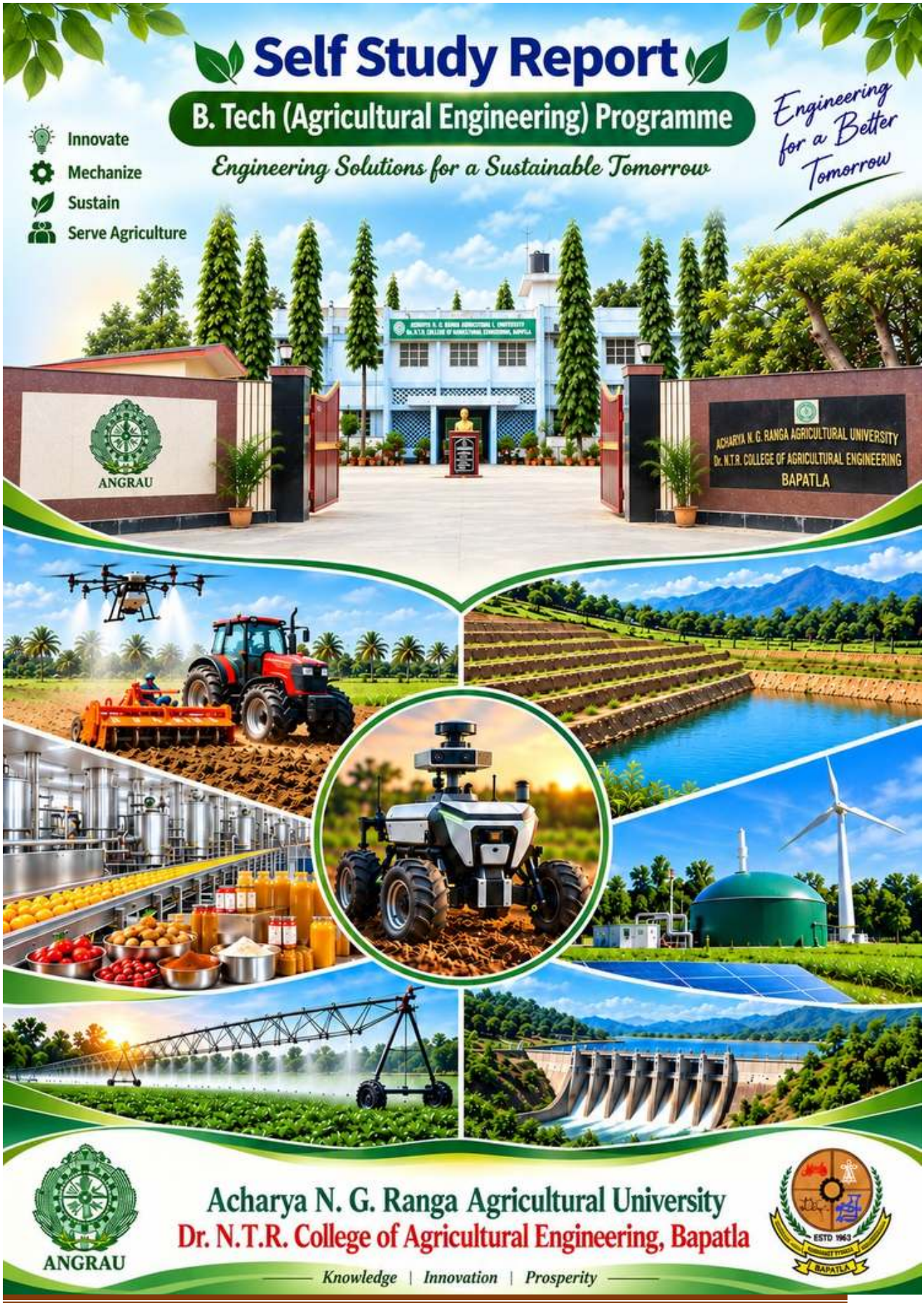

Self Study Report
B. Tech (Agricultural Engineering) Programme
Engineering Solutions for a Sustainable Tomorrow
Engineering for a Better Tomorrow
Innovate
Mechanize
Sustain
Serve Agriculture
ANGRAU
ACHARYA N. G. RANGA AGRICULTURAL UNIVERSITY
Dr. N.T.R. COLLEGE OF AGRICULTURAL ENGINEERING
BAPATLA
ANGRAU
Acharya N. G. Ranga Agricultural University
Dr. N.T.R. College of Agricultural Engineering, Bapatla
Knowledge | Innovation | Prosperity
ESTD 1963
BAPATLA

Self Study Report

Submitted by

Dr NTR College of Agricultural Engineering
Bapatla – 522 101

ANGRAU

Acharya N G Ranga Agricultural University
Lam, Guntur – 522034, Andhra Pradesh

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Self-Study Report for the
B. Tech. (Agricultural Engineering)
Programme

Self-Study Report for the
B. Tech. (Agricultural Engineering) Programme at
Dr. N.T.R. College of Agricultural Engineering, Bapatla

6.4.1 Brief History of the Degree Programme

The Bachelor of Technology (B. Tech.) programme in Agricultural Engineering was inaugurated in August 1983 within the Department of Agricultural Engineering at Agricultural College, Bapatla, with an initial intake of 20 students. In response to the growing demand for education in agricultural engineering, the program's intake capacity was increased to 30 students in the subsequent academic year. In 1990, based on the recommendations of a high-level committee appointed by the Government of Andhra Pradesh, the undergraduate programme was elevated to the status of a full-fledged College of Agricultural Engineering (CAE) in Bapatla. This institution is notable for being the first College of Agricultural Engineering established under Acharya N. G. Ranga Agricultural University (ANGRAU), Hyderabad, India. The College was founded to provide high-quality education, technical knowledge, and practical skills across various disciplines of agricultural engineering. Its establishment aimed to serve the farming community and rural population of the then-unified Andhra Pradesh by promoting engineering interventions in agriculture. Through its educational, research, and extension activities, the College has consistently contributed to enhancing agricultural productivity, reducing labour intensity in farm operations, promoting mechanization, and making agriculture more profitable and sustainable for the farmers.

Over the years, the College has established itself as a leading institution in agricultural engineering education, producing a significant number of skilled professionals who contribute to academia, research organizations, industries, government departments, and entrepreneurial ventures, both nationally and internationally. To date, 1,415 students have graduated from this institution. Numerous alumni have pursued advanced studies and research at prestigious national institutions, including the Indian Institutes of Technology (IITs), Kharagpur, Delhi, Roorkee, Chennai, and Mumbai; the Indian Agricultural Research Institute (IARI), New Delhi; the National Institute of Food Technology Entrepreneurship and Management (NIFTEM); the National Institute of Agricultural Extension Management (MANAGE); the National Institutes of Technology (NITs); and several renowned State Agricultural Universities. The College's relocation to its current campus in May 1994 marked

a significant milestone in its growth and development. Since then, the institution has rapidly expanded, both academically and infrastructurally. Modern laboratories, classrooms, workshops, libraries, research facilities, and advanced equipment have been established to enhance teaching, research, and practical training in agricultural engineering in the college.

Dr. N.T.R. College of Agricultural Engineering has consistently maintained high academic standards, achieving national recognition. The College first received accreditation from the Indian Council of Agricultural Research (ICAR), New Delhi, during 1999-2000 academic year, based on the recommendations of the ICAR Review Team. It has continued to secure accreditation in subsequent phases, with the most recent being in 2019-2020, highlighting its ongoing commitment to excellence in agricultural engineering education and research. Initially part of the Faculty of Agriculture, the College became part of the newly established Faculty of Agricultural Engineering and Technology in 2010, following amendments to the University Act. In 2018, it was renamed as Dr N.T.R. College of Agricultural Engineering (Dr. N.T.R. CAE) in honour of the late Dr. Nandamuri Taraka Rama Rao, former Chief Minister of the combined Andhra Pradesh, who laid the foundation stone for the College building on 24 February 1988. Today, Dr. N.T.R. CAE, Bapatla, is widely recognized as a centre of excellence in agricultural engineering education, need-based research, and technology transfer. The College continues to play a pivotal role in developing competent agricultural engineers and innovative technologies aimed at addressing the challenges of modern agriculture and improving the livelihoods of the farming community.



Fig. 1. Dr N.T. Rama Rao garu, Hon'ble Chief Minister of Andhra Pradesh laying the Foundation Stone of the College Building on 24-02-1988

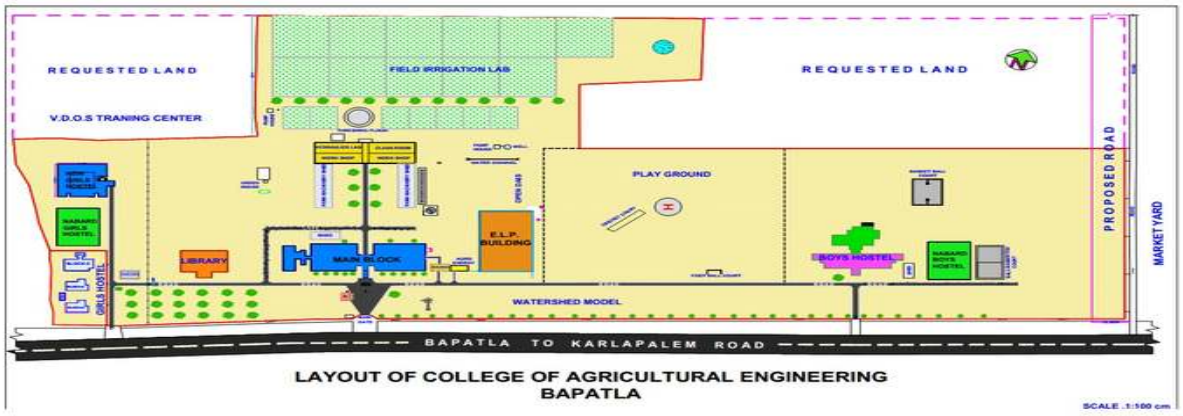


Fig. 2. Layout of College of Agricultural Engineeirng, Bapatla

The courses being offered are based on ICAR's VI Deans' Committee recommended syllabus to gear up towards equipping the students to meet the challenges that are being faced by the agricultural sector with introduction on the application of AI, sensors and robotics. The syllabus includes multi-disciplinary courses, value added courses, ability enhancement courses, skill enhancement courses and student ready courses. The College has state-of-the-art infrastructure and facilities and the academic environment which combines academic rigor with high level of interaction with government departments in serving the farming community of the nation in general and in Andhra Pradesh in particular. The driving force behind the high level of academic excellence and quality education is the dedicated team of over twenty core faculty drawn from different specializations. B.Tech. (Agricultural Engineering) degree programme is supported by seven (7) departments as follows:

A. Farm Machinery and Power Engineering
B. Processing and Food Engineering
C. Irrigation and Drainage Engineering
D. Soil and Water Conservation Engineering
E. Renewable Energy Engineering
F. Basic Engineering Applied Agricultural Sciences
G. Physical Education

The College is located at Bapatla, Guntur District, Andhra Pradesh very nearer to National Highway 214A and Vijayawada-Chennai main railway line and is well connected by rail and roads. The lush green campus is spread over 25 acres with sufficient infrastructure.

VISION

To make Dr. N.T.R. College of Agricultural Engineering, Bapatla an integrated center for academic excellence of agricultural engineering education known for its contribution to agriculture and allied fields in making agriculture more remunerative to farmers.

MISSION

To serve the farming community and enhance their quality of life through excellence in education, research, development and transfer of agricultural technologies.

6.4.2 Faculty Strength

There are enough faculty available to teach all the courses of B. Tech. (Agricultural Engineering) degree. College is having sanctioned strength of 31 faculty positions. Apart from 16 filled in posts of the College, there are 6 Teaching Associates working in the College, a Scientist of AICRP on Farm Implements Machinery Scheme and support from Agricultural College to cater the needs of the academic requirement.

Table 1. Sanctioned and In-Position Faculty at Dr. N.T.R. CAE, Bapatla

SI. No.	Position	Sanctioned Faculty	Faculty in Place	Vacant Positions
1	Associate Dean	1	1	0
2	Professor	5	5+2*	0
3	Associate Professor	8	2+2*	4
4	Assistant Professor	17	8+16*	0
	Total	**31**	**36 (16 + 20*)**	**4**

**Staff from Agricultural College/Research/Extension and Contractual faculty.*

Table 2. Department wise Sanctioned and In-Position Faculty at Dr. N.T.R. CAE, Bapatla

Sl. No.	Department	Assistant Professor		Associate Professor		Professor		Total	
	Faculty / Department	**SS**	**IP**	**SS**	**IP**	**SS**	**IP**	**SS**	**IP**
1	Associate Dean	-	-	-	-	1	1	**1**	**1**
2	Farm Machinery and Power Engineering	5	5+1*	2	-	1	1	**8**	**7**
3	Processing and Food Engineering	3	3*	2	1+1*	1	2	**6**	**7**
4	Irrigation and Drainage Engineering	2	2+1*	1	1	1	1	**4**	**5**
5	Soil and Water Conservation Engineering	2	1+2*	1	-	1	1	**4**	**4**
6	Renewable Energy Engineering	1	1*	-	-	0	1	**1**	**2**
7	Basic Engineering and Applied Sciences	4	7*	2	1*	1	1	**7**	**9**
8	Physical Education	-	1*	-	-	-	-	**-**	**1**
	Total number of posts	**17**	**24**	**8**	**4**	**6**	**8**	**31**	**36**

Staff from Agricultural College/Research/Extension and Contractual faculty
SS - Sanctioned Strength
IP - In Position

6.4.3 Technical and Supporting Staff

The College is sanctioned with 10 permanent positions as technical and supporting staff apart from 26 positions working on Time scale/Outsourcing/Contract basis. The technical and supporting staff of 36 is sufficient to carry out the daily required turnout is much more than the number of positions recommended by the Indian Council of Agricultural Research, New Delhi.

Table 3. Sanctioned and In-Position Office Staff at Dr. N.T.R. CAE, Bapatla

Sl. No.	Sanctioned Post / Category	Sanctioned Strength	In-Position	Vacant Positions	Office Staff Recommended by ICAR
1	Administrative Officer	1	1	-	1
2	Superintendent	1	1	-	20 Clerks, steno, Messenger
3	Senior Assistant	2	2*	-	
4	JACT (Junior Asst cum Typist)	2	1+7*	-	
5	Record Assistant	1	1	-	
6	Office Subordinate	2	2	-	

7	Office supporting staff	-	8*	-	-
8	Messenger/Peon	-	10*		8
9.	Watchman	1			
	Total	**10**	**36**	**-**	**29**

** Staff recruited on Time scale/Outsourcing/Contract*

The College has a sanctioned strength of 16 permanent positions as technical support to laboratories whereas a total of 41 personnel are working on Time scale/ Outsourcing/ Contract basis. The Laboratory staff of 41 is sufficient to carry out the daily required turnout, which is equal to the number of positions recommended by the Indian Council of Agricultural Research, New Delhi.

Table 4. Sanctioned and In-Position Laboratory Staff at Dr. N.T.R. CAE, Bapatla

Sl. No.	Sanctioned Post / Category	Sanctioned Strength	In-Position	Vacant Positions	Laboratory Staff Recommended by ICAR
1	Workshop Supervisor	1	1*	-	27 Workshop Staff, Laboratory Staff, Driver, Technicians
2	Technician (Mech. Gr.I)	3	5*	-	
3	Technician (Mech.Gr.II)	2	2*	-	
4	Draughtsman Grade III	1	1*	-	
5	Carpenter Grade II	1	1*	-	
6	Welder	1	1*	-	
7	Blacksmith	1	1*	-	
8	Turner	1	1*	-	
9	Driver	1	1+3*	-	
10	Workshop Helper	2	2*	-	
11	Luskar	2	2*	-	
12	Watch & Ward Staff	0	8*	-	
12	Laboratory Assistant	-	8*	-	14
13	Computer operator	-	3*	-	
	Total	**16**	**41**	**-**	**41**

**Positions filled with timescale, outsourcing, or contractual employees.*

6.4.4 Classrooms and Laboratories

The College is equipped with sufficient instructional infrastructure to effectively deliver the B. Tech. (Agricultural Engineering) degree programme. Currently, four dedicated classrooms are designated for conducting theory classes for undergraduate students. These

classrooms are spacious, well-ventilated, and benefit from ample natural light, thereby creating a comfortable and conducive learning environment. Each classroom is furnished with essential teaching and learning facilities, including short-throw LCD projectors, audio-visual aids, chalkboards/whiteboards, comfortable student furniture, and other necessary instructional equipment for effective teaching. Furthermore, the classrooms assigned to first-, second-, and third-year students have been upgraded to Smart Classrooms, featuring modern teaching aids such as audio systems and digital presentation facilities. These smart classrooms enhance interactive learning, multimedia-based teaching, and student engagement in lectures and academic activities.

The available classroom infrastructure is adequate and sufficient to facilitate the seamless execution of the four-year B.Tech. (Agricultural Engineering) programme, allowing for the concurrent management of theory classes for all batches without difficulty. In addition to classroom facilities, the College is well equipped with functional laboratories that support practical learning and hands-on training across various disciplines of agricultural engineering. Each department within the College has dedicated laboratories tailored to meet the specific practical requirements of the courses offered. Overall, the College boasts 29 well-established laboratories furnished with the necessary instruments, machinery, models, experimental setups, and advanced equipment essential for conducting undergraduate practical classes, demonstrations, research activities, and skill-oriented training programmes. These laboratories are regularly utilized for conducting practical classes in accordance with the curriculum and recommendations of the VI Deans' Committee. Laboratory facilities provide students with opportunities to gain practical exposure, technical competency, and experiential learning in different domains of agricultural engineering. The existing laboratory infrastructure is adequate and sufficient for effectively conducting all practical courses and laboratory exercises prescribed under the B.Tech. (Agricultural Engineering) degree program.

Department-wise laboratories are furnished hereunder:

A. **Department of Farm Machinery and Power Engineering**

1. Farm Power Laboratory
2. Farm Machinery Laboratory
3. Workshop
4. Soil Dynamics Laboratory

5. Ergonomics Laboratory
6. Field Lab

B. **Department of Processing and Food Engineering**

1. Instrumentation Laboratory
2. Process Engineering Laboratory
3. Food Analysis Laboratory
4. Food Microbiology Laboratory
5. Environmental Control Laboratory
6. Extrusion Processing Laboratory

C. **Department of Irrigation and Drainage Engineering**

1. Field Irrigation Laboratory
2. Field Solar Pump Laboratory
3. Wells and Pumps Laboratory

D. **Department of Soil and Water Conservation Engineering**

1. Watershed
2. Soil and Water Conservation Laboratory
3. Field Lab

E. **Department of Renewable Energy Engineering**

1. Renewable Energy Laboratory

F. **Department of Basic Engineering and Applied Sciences**

1. Workshop with equipment of all manufacturing processes
2. Thermodynamics Laboratory
3. Central Computer Centre
4. Soil Mechanics Laboratory
5. Engineering Drawing Laboratory
6. Theory of Machines Laboratory
7. Strength of Materials Laboratory
8. Engineering Physics Laboratory
9. Applied Electronics Laboratory
10. Electrical Engineering Laboratory
11. Hydraulics and Fluid Mechanics Laboratory
12. Surveying and Levelling Laboratory

Theory classes are conducted in a single batch, but practical classes are conducted in two batches.

Instructional farm details at Dr. N.T.R. College of Agricultural Engineering, Bapatla

Dr. N.T.R. The College of Agricultural Engineering, Bapatla, maintains a 5-acre (2 ha) instructional farm known as the College Field Lab, which is utilized by all departments. The farm comprises sandy loam (2 acres) and sandy soils (3 acres) with excellent irrigation facilities. Approximately 1.5 acres are dedicated to mango plantations, 700 m^2 to hydroponics, and the remaining area is designated for student practical and research experiments. Within this facility, B. Tech. (Agricultural Engineering) undergraduate students conduct field experiments as part of their final year projects, cultivating various crops such as groundnut, leafy vegetables, and maize under different irrigation methods (Drip, Sprinkler) and employing diverse water conservation practices. M. Tech. and Ph.D. students also utilize this field for research on the automation of water application to crops, soil water nutrient studies, and testing of developed farm machinery.

6.4.5 Conduct of Practical and Hands-on-Training

Practical classes are conducted in the respective laboratories according to the class timetable using instruments, equipment, and machinery. The B. Tech. (Agricultural Engineering) degree programme is supported by the following hands-on training courses: To increase students' knowledge and abilities, they are sent to research institutes and large Industries in Andhra Pradesh and neighboring states for the following courses.

1. SRDY 281 Skill Development Training - I (4 weeks)
2. SRDY 381 Skill Development Training - II (4 weeks)
3. SRDY 481 Industrial Attachment/Internship (10 weeks)
4. SRDY 482 Experiential Learning (10 weeks)
5. SRDY 483 Educational Tour (2 weeks)
6. SRDY 484 Project Work and Report Writing (20 weeks)

The Student Rural Entrepreneurship Awareness Development Yojana (Student READY) programme is an essential component of the B. Tech. (Agricultural Engineering) curriculum, aimed at equipping students with practical exposure, experiential learning, entrepreneurial skills, and industry-oriented training, and was evaluated. This programme

involves comprehensive hands-on training, industrial exposure, field visits, experiential learning modules, educational tours, and research projects to bridge the gap between classroom education and real-world applications in agricultural engineering. As part of the program, SRDY 281, SRDY 381, and SRDY 481 trainings are conducted at esteemed ICAR research institutes, Central Government organizations, State Government agencies, industries, and non-governmental organizations nationwide. These trainings provide students with insights into the latest technological advancements, modern engineering practices, research methodologies, and field-level applications in agriculture and allied sectors.

Students receive training at esteemed ICAR institutes, including the Central Institute of Post-Harvest Engineering and Technology (CIPHET), Ludhiana; the Indian Institute of Horticultural Research (IIHR), Bengaluru; the Central Institute of Agricultural Engineering (CIAE), Bhopal; and the Indian Institute of Soil and Water Conservation (IISWC) centers located in Ooty, Bellary, Koraput, Kota, and Vasad. Additional training is provided at the National Academy of Agricultural Research Management (NAARM), Hyderabad, the Central Research Institute for Dryland Agriculture (CRIDA), Hyderabad, and the Central Tobacco Research Institute (CTRI) Regional Station, Kandukur, among others. Furthermore, students undergo specialized training in various Central Government organizations, such as the Central Farm Machinery Training and Testing Institute (CFMTTI), Budni; the Southern Region Farm Machinery Training and Testing Institute (SRFMTTI), Garladinne; the Northern Region Farm Machinery Training and Testing Institute (NRFMTTI), Hissar; the National Institute of Plant Health Management (NIPHM), Hyderabad; the Indian Institute of Packaging (IIP), Hyderabad; the Central Institute of Petrochemicals Engineering and Technology (CIPET), Vijayawada; the Indian Grain Service Institute (IGSI), Hyderabad; the India Meteorological Department (IMD), Pune; and the National Institute of Hydrology (NIH), Roorkee and Kakinada.

In addition, students receive training in various State Government organizations, including WALAMTARI, Hyderabad; Andhra Pradesh Space Applications Centre (APSAC), Vijayawada; AICRP on Farm Implements and Machinery, Hyderabad; Telangana Foods, Hyderabad; and District Water Management Agencies (DWMA), among others. The program also offers industrial exposure through training in leading industries and agribusiness organizations such as Global Green Group, Sadasivapet; Pioneer Seeds, Hyderabad; Srini Food Park Pvt. Ltd., Chittoor; Food Tropicals, Tirupati; Konark Frozen Foods, Vijayawada;

Jaya Lakshmi Rice Mill, Bapatla; Karanam Technologies, Kakinada; Farm Implements (India) Pvt. Ltd., Chennai; ASPEE, Mumbai; TAFE Ltd., Chennai; Escorts Ltd., Faridabad; VST Tillers Tractors Ltd., Bengaluru; Jain Irrigation Systems Ltd., Hyderabad; ITC Limited, Guntur and Mysore; and Kumar Pumps, Tenali. Students are also exposed to developmental and community-oriented activities through NGOs such as WASSAN, Secunderabad; DHAN Foundation, Madurai and Punganur; and Krushi Agri Development Training Centre under AC&BC, Kothagudem. These industrial and institutional training programs equip students with practical knowledge, technical competence, professional confidence, and exposure to advanced technologies adopted in agriculture and allied sectors. Upon completion of the training program, students deliver seminar presentations and undergo viva voce examinations before a panel of experts for evaluation. Such training programs significantly enhance students' employability and assist them in securing placements in reputable organizations and industries.

SRDY 482 is conducted within the college campus under various experiential learning modules aimed at developing entrepreneurial and technical skills among students. The programme includes practical training in the fabrication and manufacturing of farm machinery and equipment such as garden tools, groundnut strippers, monocycle weeder-cum-furrow cleaners, tractor components, tillage implements, weeder rakes, solar-powered sprayers, manually operated cultivators, and multipurpose manual weeders. Students are also trained in mushroom cultivation, leafy vegetable production, cultivation of commercial and floricultural crops, and the development of low-cost agricultural and rural utility devices, such as manual fruit harvesters and low-cost air coolers. Furthermore, students gain hands-on experience in the preparation and value addition of ANGRAU-developed food products, including ragi malt, ragi biscuits, sorghum laddu, oat cookies, mango jam, and other nutritionally enriched food products. The program also includes field surveys and studies on farm mechanization practices in Andhra Pradesh to understand the status and challenges of mechanization at the grassroots level in the state.

SRDY 483 is an educational study tour designed for final-year students to gain exposure to advanced agricultural research institutions, industries, plantations, and processing units in South India. During this tour, students visit esteemed institutions and organizations, including the Indian Institute of Horticultural Research (IIHR), Bengaluru; University of Agricultural Sciences (UAS), Bengaluru; IISWC, Ooty; Tamil Nadu Agricultural University

(TNAU), Coimbatore; Central Marine Fisheries Research Institute (CMFRI), Kochi; Kanan Devan Hills Plantations Company Pvt. Ltd., Munnar; Central Tuber Crops Research Institute (CTCRI), Thiruvananthapuram; House of Kalam, Rameswaram; and National Institute of Food Technology Entrepreneurship and Management (NIFTEM), Thanjavur. This educational tour offers students valuable insights into innovative technologies, industrial operations, research activities, plantation management, food processing technologies and successful agricultural enterprises. Upon returning to the college, students are required to submit a comprehensive tour report documenting their observations and learning experiences. Evaluation is conducted based on the submitted report, seminar presentation, and viva voce examination by an Evaluation Committee.

SRDY 484 encompasses the student research project component, wherein groups of four to five students engage in experimental research on relevant and problem-oriented topics in agricultural engineering. Students design and conduct experiments, develop innovative technologies or prototypes, analyse results, and propose practical solutions that are beneficial to the farming community. At the conclusion of the project, students prepare a detailed project report and present their research findings before the Evaluation Committee, followed by a viva voce examination for assessment. This component of the program aids students in developing research aptitude, problem-solving skills, teamwork, technical writing abilities, and innovation-oriented thinking.

6.4.6 Supervision of Students in UG Programmes

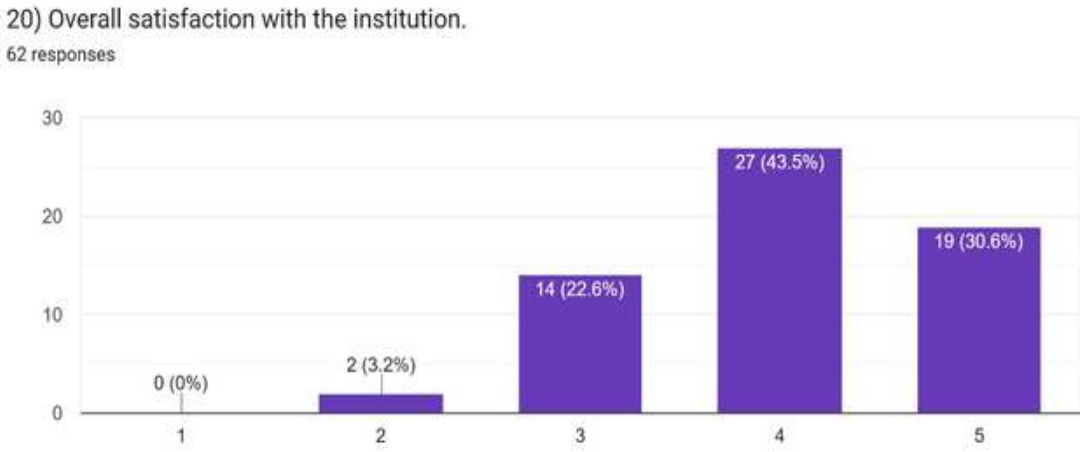
The College has established a comprehensive student advisory and monitoring system to ensure the academic progress, discipline, well-being, and overall development of students. Under this system, each student is assigned a faculty advisor at the time of admission who serves as a mentor and guide throughout the duration of the academic programme. Faculty advisors provide continuous guidance on academic matters, course registration, attendance, extracurricular activities, career planning, higher education opportunities, and preparation for competitive examinations. To facilitate regular interaction, one hour per week is specifically allotted in the academic timetable for advisory meetings. During these sessions, students are encouraged to discuss academic difficulties, career goals, and personal concerns with their advisors, enabling timely counselling and support whenever required. Each student maintains an “Advisory Book” containing details of registered and completed courses, grades obtained,

attendance percentage, and other academic records. Faculty advisors regularly review and update these records to monitor student performance and progress effectively. In addition, advisors maintain a record of their student advisees to ensure continuous mentoring and proper follow-up. To promote punctuality and discipline, the College maintains a register at the main entrance to record the daily entry and exit timings of students. Any deviation from the prescribed timings is reported to the Associate Dean for counselling and necessary corrective measures. The College campus is also equipped with a comprehensive CCTV surveillance system to ensure safety, transparency, and discipline. CCTV cameras are installed in classrooms, corridors, laboratories, examination halls, and other important locations within the premises. The recorded footage is continuously monitored and stored in the system for future reference. Through this integrated advisory, counselling, and monitoring mechanism, the College provides a disciplined, safe, and student-friendly academic environment that supports the holistic development of students.

6.4.7 Feedback of Stakeholders (Students, Parents, Industries, Employers, Farmers, etc.)

Students and their parents are satisfied with the provisions of curricular, co-curricular and extra-curricular activities and programme being handled; and employers and farmers also acknowledge the quality of the students and services rendered at Dr N.T.R. College of Agricultural Engineering, Bapatla. The following is the feedback:

Students: A feedback form is supplied to all students on the last day of the semester final theory examination to capture the data related to the course conduction regarding proportionate completion of syllabus, power point presentations made, videos shown, etc. The data is collected and analysed for improving the teaching capabilities of the students. A feedback form was designed to collect the feedback and opinions regarding academic activities, teaching-learning process, infrastructure, student support, and overall institutional performance for ICAR accreditation purposes. A questionnaire was circulated among the students of B Tech (Agricultural Engineering) and 62 students responded on a scale of 1 to 5 representing poor to excellent respectively. The list of students along with their mobile numbers and emails is available with the college office.


20) Overall satisfaction with the institution.
62 responses
30
20
10
0
0 (0%)
2 (3.2%)
14 (22.6%)
27 (43.5%)
19 (30.6%)
1
2
3
4
5

Parents:

I am very happy to see the academic Growth and practical exposure provided to UG students at Dr. NTR college of Agricultural Engineering the college Gives Strong Technica Training in soil and water conservation Engineering, Farm machinery and power Engineering and Food processing Engineering Through well equipped laboratories, field visits, workshops and hands on learning activities. Students are encouraged to participate in innovative projects and practical problem solving related to modern Agriculture the faculty members are supportive and Guide students in both Academics and Skill development. The college environment is disciplined, safe and Student-friendly, helping UG students build confidence, technical knowledge, and Good carrier opportunities in Agricultural Engineering Sectors.

As a parent, I sincerely appreciate the efforts of you and the faculty in Shaping the future of students. Thank you for your Guidance and support to my child.

Your's sincerely,
CH. Rajeswari
mother of:
CH. Smilypriya
BE/23-018.

Signature
CH. Rajeswari

My daughter, M. Shanmukha Vyshnavi had completed her graduation in Agricultural Engineering in the most prestigious named "Dr. NTR College of Agricultural Engineering, Bapatla (ANGRAU)". My daughter has shown the improvement during her academics only because of the faculty that they have explained each and every unit very clearly. We are very happy with the support of the proffessors who had helped her in theacademics as well as in the projects and in the Trainings too.

We have known that she had changed a lot according to her practical knowledge and the knowledge she gained was helped her to solve the problems at her workplace. We are very thankful to the faculty who have supported her in each and every small things too. She had learn many Technical skills like "auto cad" and "QGIS" and she had shown more interest too.

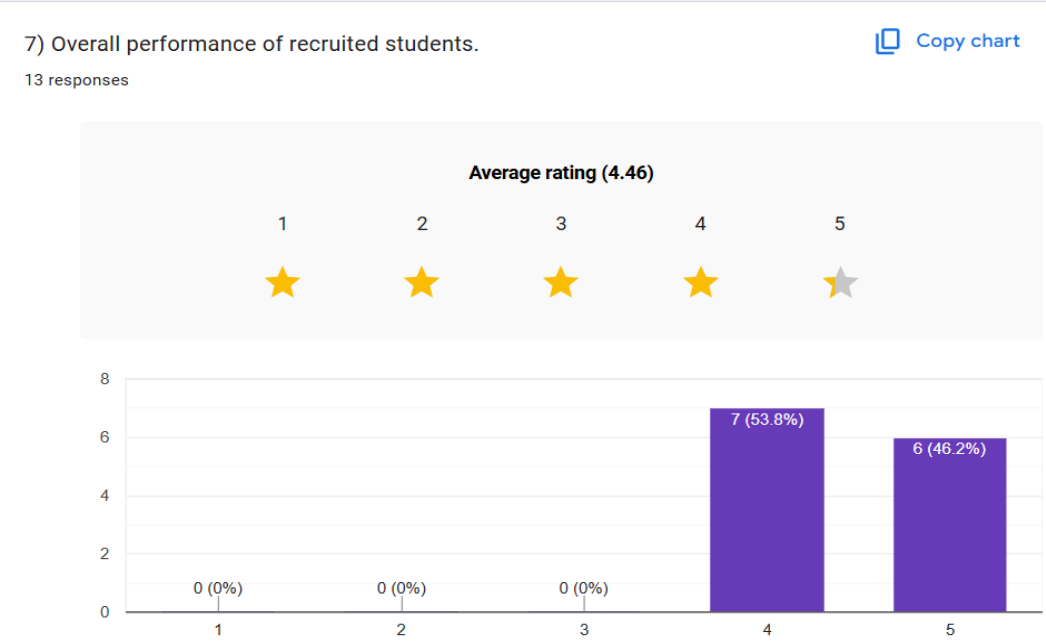
We are satisfied with the College environment and teaching. I thank the managment and proffessors in the college are excellent in Teaching. I like the way they are giving advices to the Students in each and every situations. We are feeling proud that she was making all her decisions about her future and going to build more confidence that she can do anything with out the help of others. We are very thankful to all the managment and the faculty members who had supported her in real-time experinces like projects, Trainings and internships.

Finally I thank each and every one in the college for encouraging and advising her. Now, she was very strong enough to face each and every situation in her life.

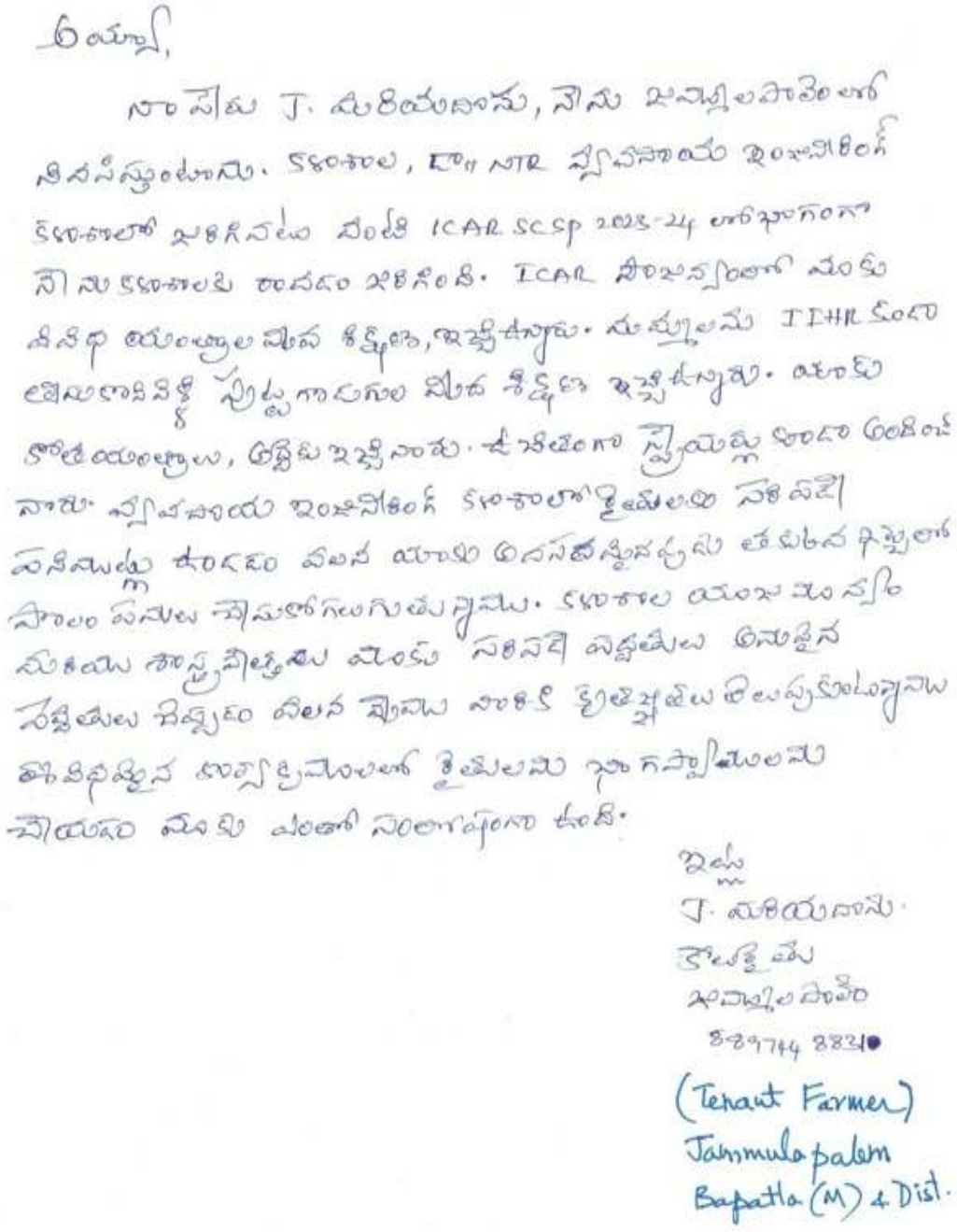
— M. [redacted]

Industries and Employers:

Employers: A feedback form was designed to collect opinions regarding technical knowledge, problem solving skills, teamwork and communication skills, professional ethics and discipline, ability to adapt to work requirements, leadership qualities, industry exposure and overall behaviour of the students from Dr. N.T.R. College of Agricultural Engineering, Bapatla. A questionnaire was circulated among the employers of the students of B. Tech. (Agricultural Engineering) and 13 companies responded on a linear scale of 1 to 5 representing poor to excellent respectively. The list of companies responded to the feedback is available with the College Office along with their corporate name, name and designation of the person filling the form and emails for further enquiry.



Farmers' feedback:



ఓయ్ నీ,

నా పేరు J. మరియమ్మ, నేను జామ్మిలపాలెంలో నివసిస్తుంటాను. కళాశాల, డా॥ NTR వ్యవసాయ ఇంజనీరింగ్ కళాశాలలో ఏర్పడినటు వంటి ICAR SCSP 2023-24 లో భాగంగా నేను కళాశాలకు రావడం జరిగింది. ICAR వారు మాలాంటి వారికి అనేక రకాల యంత్రాలు మీద శిక్షణ, ఇచ్చి ఉన్నారు. మమ్మల్ని IIHR ద్వారా బెంగుళూరు నుండి స్పీడ్ గా రుగుల మీద శిక్షణ ఇచ్చి ఉన్నారు. అలాగే కోతి యంత్రాలు, ఉచితంగా ఇచ్చినారు. ప్రతిరోజు స్నేహపూర్వకంగా అందరూ ఉండేవారు. వ్యవసాయ ఇంజనీరింగ్ కళాశాలలో ఫైలులు సరిపడే పనిముట్లు చూడటం వలన మాకు ఉదయపు స్నేహపూర్వకంగా లభించిన స్థాలం పనులు చేసుకోగలుగుతున్నాము. కళాశాల యందు మంచి నుకయిన శాస్త్రవేత్తలు మరియు సరిపడే వెచ్చదనులు అనుభవిన సలహాలు ఇవ్వడం వలన మాలాంటి కొత్తవారికి తెలుసుకోవడానికి తొవిధమైన టువ్వాక్రియలలో కైమిలను ఫాగస్పాయిలను చెయ్యడం మాకు ఎంతో సంతోషంగా ఉంది.

ఇట్లు
J. మరియమ్మ
కౌలుదారు
జామ్మిలపాలెం
8-89744 8834●

(Tenant Farmer)
Jammulapalem
Bapatla (M) & Dist.

LETTER OF APPRECIATION

To
The Associate Dean
Dr. NTR CAE Bapatla

I, Rajesh Nirmal currently working as Village Agricultural Assitant in the Agricultural Department at Returu, would like to bring to your kind notice that the farmers of Jammulapalem, Kavurivaripalem, Narasayapalem, and Bapatla villages have greatly benefited from the ICAR Sub-Plan programmes conducted by Indian Council of Agricultural Research and Dr. NTR College of Agricultural Engineering during the years 2023 and 2024. Earlier, I had the opportunity to work at Jammulapalem village and closely observed the positive impact of these programmes on the farming community.

I sincerely thank Indian Council of Agricultural Research for providing generous financial support towards uplifting the farmers through these developmental activities. The farmers expressed immense satisfaction in acquiring additional knowledge and practical skills related to repair and maintenance of agricultural equipment, as well as the operation and handling of harvesting machinery. They were also very happy to receive sprayers and insect trappers distributed under the programme.Such institutional programmes play a vital role in creating awareness among farmers regarding modern agricultural technologies and contribute significantly towards improving their standard of living.

On behalf of all the farmers who benefited from these programmes, I extend my heartfelt gratitude to Indian Council of Agricultural Research, the administration of Dr. NTR College of Agricultural Engineering, and the Agricultural Department for involving the line departments in farmer-oriented developmental activities.

Village Agriculture Assista
Rythu Bharosa Kendram
Returu (V), Kakumanu(M)
A.P - 522 113

6.4.8. Student intake and attrition in the program for last five years

Admissions into the First Year B. Tech. (Agricultural Engineering) degree programme are carried out through the Andhra Pradesh Engineering, Agriculture and Pharmacy Common Entrance Test (AP EAPCET) conducted by the Government of Andhra Pradesh. The admission process is undertaken through the centralized counselling system implemented by

the Andhra Pradesh State Council of Higher Education (APSCHE), ensuring transparency and adherence to the reservation policies prescribed by the Government of Andhra Pradesh. The approved intake capacity for the First Year B. Tech. (Agricultural Engineering) programme is 60 students per academic year. Out of these 60 seats, 60 percent (36 seats) are filled through the AP EAPCET Convener quota based on merit and reservation norms. The remaining 40 percent (24 seats) are allotted under the Farmers' Quota in accordance with the policies and reservation guidelines of the Government of Andhra Pradesh, thereby providing opportunities to students from farming backgrounds to pursue professional education in agricultural engineering.

In addition to the regular intake, 20% supernumerary seats over and above the sanctioned intake are allotted to students nominated under the ICAR quota by the Indian Council of Agricultural Research (ICAR), New Delhi. These admissions are made at the national level, enabling students from different states of the country to join the programme and contribute to a diverse academic environment within the College. The College also provides an opportunity for diploma holders to pursue higher education through the lateral entry system. Candidates who have completed a three-year Diploma in Agricultural Engineering are eligible for admission directly into the Second Year of the B. Tech. (Agricultural Engineering) programme through the Andhra Pradesh Engineering Common Entrance Test (AP ECET) conducted annually by APSCHE, Government of Andhra Pradesh. For these lateral entry admissions, an additional 20% seats over and above the regular intake are reserved as per the prescribed norms. Over the years, the College has observed a moderate increase in student attrition at the undergraduate level. The major reasons identified include health-related issues, difficulties faced by some students in adapting to the local climatic conditions, and migration of students to institutions located nearer to their hometowns through the process of counselling-based sliding or upgradation. Despite these challenges, the College continuously strives to provide a supportive academic environment, counselling services, and student mentoring mechanisms to improve student retention and ensure the successful completion of the programme.

Table 5. Percentage of attrition during the last five years (2021-22 to 2025-26)

Name of the Degree Programme	Actual students admitted in last five years						Attrition (%)					
B Tech (Ag.Engg.)	2021 - 22	2022 - 23	2023 - 24	2024 - 25	2025 - 26		2021 - 22	2022 - 23	2023 - 24	2024 - 25	2025 - 26	2025 - 26
	60	**80**	**86**	**84**	**83**		**10% (6)**	**6.25% (5)**	**1.16% (1)**	**1.19% (1)**	**1.20% (1)**	**1.20% (1)**

6.4.9. ICT Application in Curricula Delivery

The B. Tech. (Agricultural Engineering) undergraduate programme at Dr. N.T.R. College of Agricultural Engineering is supported by modern teaching and learning facilities that enhance academic delivery and create an interactive learning environment for students. The College has established well-equipped smart classrooms integrated with advanced instructional technologies such as interactive boards, multimedia projectors, audio-visual systems, internet connectivity, and other ICT-enabled teaching aids to facilitate effective classroom teaching, seminar presentations, project reviews, and academic discussions. Faculty members extensively utilize these modern teaching resources to deliver course lectures through PowerPoint presentations, multimedia modules, video demonstrations, animations, and various digital learning platforms. The integration of contemporary teaching tools and ICT has significantly improved the quality of classroom instruction by promoting better conceptual understanding, active student participation, and enhanced interaction between teachers and students.

The undergraduate programme follows learner-centric teaching methodologies that encourage analytical thinking, practical problem-solving, innovation, and experiential learning among students. In addition to regular classroom teaching, students are encouraged to actively participate in seminars, technical presentations, project discussions, and research-oriented activities using digital presentation tools and online resources. These activities help students improve their technical communication skills, presentation abilities, confidence, leadership qualities, and overall professional competence. As part of the academic curriculum, regular seminars, technical discussions, guest lectures, and student presentations are organized to expose students to recent advances, emerging technologies, and current developments in the field of agricultural engineering. Such activities provide opportunities for students to broaden their technical knowledge, strengthen professional skills, and develop awareness of industry and research trends. Dr. N.T.R. College of Agricultural Engineering continuously strives to strengthen and modernize its academic infrastructure in accordance with ICAR recommendations and contemporary educational requirements. The available teaching and learning facilities provide a highly conducive academic environment for effective teaching, practical learning, research, innovation, and knowledge dissemination. These initiatives collectively contribute towards maintaining high standards of education and

achieving academic excellence in the B. Tech. (Agricultural Engineering) undergraduate programme.

6.4.10 SSR of the B.Tech. (Agricultural Engineering) Degree

Self-study report of the B.Tech. (Agricultural Engineering) programme of Dr N.T.R. College of Agricultural Engineering, Bapatla is here with submitted.

6.4.11 Certificate

I, the Associate Dean of the Dr NTR College of Agricultural Engineering, Bapatla, Andhra Pradesh hereby certify that the information contained in Sections 6.4.1 to 6.4.9 are furnished as per the records available in the college and degree awarding university.



SELF STUDY REPORT for the Dr. N.T.R. College of Agricultural Engineering, Bapatla

6.5.1. College Administration

Dr. N.T.R. College of Agricultural Engineering, Bapatla is having Associate Dean's Office supported by Administrative Office. The Head of the Departments work under the guidance of the Associate Dean. The Associate and Assistant Professors work under the guidance of the Head of the Departments. Every department is supported by the College office run under the control of an Administrative Officer. In turn, the Administrative officer is assisted by the ministerial staff like superintendent, Junior and senior clerks. The Regional Accounts Officer stationed at Agricultural College, Bapatla will coordinate with this college to obtain funds and further processing of bills by forwarding to the University headquarter. The higher officers above the Associate Dean ie the Faculty Dean is stationed at University headquarters to coordinate with the University for the overall Development of the College.

6.5.1.1. College Dean's Office Establishment

The college was founded in the year 1983 in the premises of Agricultural College, Bapatla by elevating the then Department of Agricultural Engineering to College of Agricultural Engineering to its present campus in the year 1994. Office Staff of the College shown under Section 6.4.3 of B. Tech. (Agricultural Engineering) degree programme are responsible for all the activities of B.Tech. (Agricultural Engineering), M. Tech. (Agricultural Engineering) and Ph.D. (Agricultural Engineering) degree programmes.

The post of the Associate Dean is a sanctioned post by the Government of Andhra Pradesh through G.O. Ms. No. 30 dt 19-02-2019 of Agriculture and Cooperation Department (Agri.III). The Associate Dean position available in the College. Associate Dean is nominated as per the seniority of the Professor in Faculty of Agricultural Engineering and Technology. Normally, Associate Dean's tenure is for two years and it may be extended further based on the requirement. The present Associate Dean is Dr. M. Sardar Baig joined in the present position on 07-08-2026.

Associate Dean is having a separate office with conference table to accommodate 10 people. Associate Dean is equipped with computer, printer, internet LAN facility, surveillance CC camera network, fridge, etc. There is a separate meeting hall with audio system available adjacent to Associate Dean's chamber. Administrative Office is established separately in two different halls having full-fledged office furniture. All staff members are provided with furniture, computer, internet LAN facility, etc.

6.5.1.2. Monitoring Mechanism for Quality Assessment (on-line)

The college implements several monitoring mechanisms for improving the quality of the teaching, research and extension. Among these internal quality assessment, Student-Teacher council, Advisory system, Feedback systems, regular visits by Deans are predominantly notable. The council discusses and makes recommendation for further improvement of quality of curricular, co-curricular and extra-curricular activities.

Apart from Student-Teacher Council meetings being held regularly, all classrooms, laboratories, examination hall, all premises of the College, etc. relate to surveillance CC cameras to continuously monitor the teaching activities both in classrooms and laboratories. Video of each camera is recorded in the system.

6.5.1.3. CC/Board of Studies

There is no CC at department level and Board of Studies for the College. However, the College conducts regular meetings known as 'Staff Meetings' to take decisions for the overall improvement of the Institution and to uphold in the highest at an average frequency of one in a month. Apart from the college level meetings, the Faculty Board meeting are being held at University level twice in a year to improve the academics at every College in the faculty. Dr NTR College of Agricultural Engineering, Bapatla implements the decisions of the University without any deviation.

Planning process in respect of academic and non-academic activities at college level is done mainly through the following two Institutional bodies.

College Academic Committee:

- Responsibility to advise the Associate Dean on academic matters running of the college within the theme and framework of ANGRAU rules and guidelines.
- The committee constitute of Associate dean, Academic Matters (UG and PG), Head of the departments and officers-in-charge of the student activities

College Advisory Board (CAB)

- To advise the Associate Dean on administrative matters, co-curricular and extra-curricular activities and other matters relating to the development and running of the college within the framework of ANGRAU rules and guidelines.

- The committee constitutes of Associate Dean as chairman, Head of the departments, wardens, Officer in charge of Student Affairs, Resident Medical Officer, Librarian and Administrative Officer as secretary.

6.5.1.4. Anti Ragging Cell

Dr. N.T.R. College of Agricultural Engineering maintains a safe, disciplined, and ragging-free campus environment that promotes mutual respect, healthy interaction, and cordial relationships among students. The College is committed to ensuring a congenial atmosphere in which senior and junior students interact in a friendly, supportive, and academically oriented manner. In compliance with the guidelines and regulations issued by the University Grants Commission (UGC), Government of India, and Government of Andhra Pradesh, the College has implemented several preventive and monitoring measures to eliminate ragging from the campus and hostels. To create awareness about the serious consequences of ragging and to promote a culture of respect and harmony among students, anti-ragging awareness programmes are organized regularly. These programmes involve the participation of police officials, legal experts, local administrators, faculty members, students, media representatives, and parents. Through counselling sessions, interactive meetings, and awareness campaigns, students are educated about the legal implications, disciplinary actions, and psychological impact of ragging.

The College has constituted Anti-Ragging Committees and Anti-Ragging Squads to monitor and prevent any form of ragging within the campus and hostel premises. Faculty members are assigned anti-ragging duties on a daily rotational basis to ensure continuous supervision and interaction with the students. In addition, anti-ragging squads conduct surprise inspections and regular checks in the hostels, particularly during the night, weekends, and holidays, to ensure the safety and well-being of freshers. Special arrangements were made to provide a secure and comfortable environment for first-year students. Separate hostel accommodation is provided for freshers away from senior students to minimize unnecessary interactions during the initial period of adjustment to campus life. Dining schedules in the hostels are also staggered to ensure smooth functioning and avoid overcrowding and unwanted interaction during mealtimes. To reinforce awareness of anti-ragging regulations, posters, banners, and display boards highlighting punishments and penal provisions related to ragging are prominently exhibited in hostels, classrooms, corridors,

notice boards, and other important locations across the campus. These displays serve as constant reminders of the institution's zero-tolerance policy on ragging.

The College campus is further strengthened by CCTV surveillance systems installed in classrooms, corridors, hostels, and other identified locations to continuously monitor student movement and activities. The surveillance system acts as an effective preventive measure and helps maintain discipline and security on campus. As part of the mandatory anti-ragging compliance process, undertakings and online anti-ragging affidavits are collected from all students and their parents at the time of admission, as prescribed by the regulatory authorities. Anti-ragging registers are also maintained separately in both boys' and girls' hostels to enable students to report grievances or incidents without hesitation. Furthermore, important contact details, such as anti-ragging helpline numbers, Officer in Charge of Student Activities (OSA), hostel wardens, medical emergency services, and ambulance numbers, are prominently displayed on hostel and college notice boards for immediate assistance during emergencies. Through these comprehensive preventive measures, counselling mechanisms, monitoring systems, and awareness programmes, Dr. N.T.R. The College of Agricultural Engineering has successfully maintained a peaceful, student-friendly, and ragging-free academic environment that supports the overall well-being and holistic development of students.

To prevent and effectively address the menace of ragging within the campus, the following Anti-Ragging Committee is hereby constituted at Dr. N.T.R. College of Agricultural Engineering, Bapatla, in connection with the admission of fresh students into the B. Tech. (Agricultural Engineering) program for the academic year 2025–26. The Committee was constituted in accordance with the guidelines and regulations issued by the Honourable Supreme Court of India, the University Grants Commission (UGC), Government of India, and Acharya N.G. Ranga Agricultural University to ensure a safe, disciplined, and ragging-free environment within the college and hostel premises. The Anti-Ragging Committee shall be responsible for creating awareness among students regarding the harmful effects and legal consequences of ragging, monitoring student activities within the campus and hostels, and taking preventive measures to eliminate any form of ragging from the campus. The Committee shall also ensure that newly admitted students are provided with a secure, friendly, supportive academic atmosphere.

Table 8. Anti-Ragging Committee for the year 2025-26

Sl. No.	Name of the member	Designation	Mobile No.	Representation
1	Dr. D. D. Smith	Associate Dean	9989625205	Teaching Faculty
2	Dr. K. Krupavathi	Associate Professor, OSA and Convener	9533877666	Teaching Faculty
3	Sri. G. Ramanjeneyulu	Deputy Superintendent of Police, Bapatla (Member)	9440796165	Police Administration
4	Sri. R. Ram Babu	Circle Inspector of Police (Member) Town	9440796171	Police Administration
5	Shaik. Saleema	Mandal Revenue Officer (Member)	8309588617	Civil Administration
6	M. Vijay Kumar	Sub Inspector of Police, Bapatla Town	8978777298	Police Administration
7	Ch. Chandravathi	Sub Inspector of Police, Bapatla Town	7569137676	Police Administration
8	Sri. R. Venkateswarulu	Advocate, Bapatla	9985304129	Advocate
9	Dr. G. Ravi Babu	Professor and Head	9848572321	Teaching Faculty
10	Dr. H. V. Hema Kumar	Professor and Head	8179815632	Teaching Faculty
11	Dr. K.V.S. Rami Reddy	Professor and Head	9490845917	Teaching Faculty
12	Dr. B. Sreenivasula Reddy	Professor and Head	9618800819	Teaching Faculty
13	Dr. S. Vishnu Vardhan	Professor and Head	9440363112	Teaching Faculty
14	Dr. K. Lavanya	Assoc. Professor	9295350600	Teaching Faculty
15	Dr. D. Vijay Kumar	Assistant Professor	9848186842	Teaching Faculty
16	Dr. A. Srinivasa Rao	Assistant Professor	8374647499	Teaching Faculty
17	Dr. A. Ashok Kumar	Assistant Professor	9492686634	Teaching Faculty
18	Dr. N. Ashok Kumar	Assistant Professor		Teaching Faculty
19	Dr. B. Raj Kiran	Assistant Professor	9177345631	Teaching Faculty
20	Er. M. Karunya	Assistant Professor	8106307360	Teaching Faculty
21	Smt. V. Padma Tulasi	Administrative Officer	9494684341	Non-Teaching
22	Sri. K. Venkateswarlu	Physical Director	9010131084	Teaching Faculty
23	Mr. M. Umamaheswara Rao	Student	9550398596	VI year
24	Miss. S. Vasantha	Student	7989815216	VI year
25	Mr. K. Charan Sai	Student	9441961255	III year
26	Miss N. Snehitha	Student	7981918759	III year
27	Mr. N. Gangadhara Reddy	Student	8919570601	II year
28	Miss. K. Priyanka	Student	9182266310	II year
29	Mr. M. Chandra Reddy	Parent	8498863467	Parent of II year student (F/O M. Vara Prasad Reddy)
30	Mr. Ch. Yhoshuva	Parent	9908963206	Parent of III year student (F/O Ch. Smily Priya)

31	Mr. E. Thammaya	Parent	8074955419	Parent of III-year student (F/O E. Ganeswari)
32	Sri. V. Srinivasa Rao	Reporter, CVR News	8019945257	Media representative
33	Sri. G. Suresh Kumar	Reporter, Eenadu Daily News paper	8008573141	Media representative
34	Sri. B. Dhan Raj	Reporter, Andhra Jyothi Daily News paper	9985411240	Media representative
35	Sri. Anjaiah	Reporter, Sakshi Daily News paper	9505517849	Media representative
36	Sri. B. Sambaiah	Secretary, Adarsa Educational Society, Karlapalem	9866725646	Representative of NGO
37	Sri. B.S. Narayana Bhattu	Secretary, Red Cross Society, Bapatla.	9346911199	Representative of NGO/Voluntary Organization
38	Sri. P.C. Sai Babu	Secretary, Forum for Better Bapatla, Bapatla	9963521857	Representative of NGO/Voluntary Organization

Any incident of ragging, whether reported directly or noticed by the Committee, shall be viewed with the utmost seriousness. Strict disciplinary action will be initiated against the individuals involved in accordance with the existing anti-ragging regulations and disciplinary guidelines prescribed by the Honourable Supreme Court of India, UGC, and the University. Depending on the severity of the offence, the punishment may include suspension, expulsion from the institution, cancellation of admission, withholding of scholarships, debarring from examinations, filing of police complaints, and other disciplinary measures as deemed appropriate by the competent authorities. All students are hereby instructed to maintain discipline, mutual respect, and cordial relationships on campus and to actively cooperate in maintaining Dr. N.T.R. College of Agricultural Engineering, Bapatla, as a completely ragging-free institution.

The following **Anti Ragging Squad Committees** are here by constituted to deal with the menace of ragging in Dr. NTR College of Agricultural Engineering, Bapatla premises in view of admission of fresh students of B. Tech. (Agril. Engg.) during the academic year 2025-26. The incidence of ragging will be viewed seriously by the committee, and action against the culprits will be initiated as per the existing disciplinary guidelines issued by ***the Honorable Supreme Court of India***.

Table 9. Anti ragging squads

Squad No.	Name of the faculty	Particular
I	**Dr. G. Ravi Babu**, Professor & Head, Dept. of Soil and Water Conservation Engineering	Chairperson
	Dr. A. Srinivasa Rao, Assistant Professor, Dept. of Farm Machinery and Power Engineering	Member
	Dr. N. Ashok Kumar, Assistant Professor, Dept. of Irrigation and Drainage Engineering	Member
	Mr. M. Rahul, Third Year Student	Member
II	**Dr. H. V. Hema Kumar**, Professor & Head, Dept. of Irrigation and Drainage Engineering	Chairperson
	Dr. B. Raj Kiran, Assistant Professor, Dept. of Farm Machinery and Power Engineering	Member
	Sri G. Sesha Reddy, Teaching Associate	Member
	Dr. Kishore Kumar, Teaching Associate	Member
	Mr. K. Rajeev, Third Year Student	Member
III	**Dr. K.V.S. Rami Reddy**, Professor & Head, Dept. of Farm Machinery and Power Engineering	Chairperson
	Dr. A. Ashok Kumar, Assistant Professor, Dept. of Farm Machinery and Power Engineering	Member
	Dr. K. Rakesh, Teaching Associate	Member
	Er. K.V.S. Kartheek, Teaching Associate	Member
	Mr. R. Mohan, Second Year Student	Member
IV	**Dr. B. Sreenivasula Reddy**, Professor & Head, Dept. of Processing and Food Engineering	Chairperson
	Dr. D. Vijay Kumar, Assistant Professor, Dept. of Farm Machinery and Power Engineering	Member
	Sri. K. Venkateswarlu, Physical director	Member
	Sk. Imran, Second Year Student	Member
V	**Dr. K. Lavanya**, Associate Professor, Dept. of Processing and Food Engineering	Chairperson
	Dr. K. Krupavathi, Associate Professor, Dept. of Irrigation and Drainage Engineering	Member
	Dr. M. Karunya, Assistant Professor, Dept. of Soil and Water Conservation Engineering	Member
	Dr. B. Pravallika, Teaching Associate	Member

	Miss. N. Snehitha, Third Year Student	Member



Fig. 3. Release of anti-ragging poster by Associate Dean and Rural Sub Inspector of police, lawyer and students



Fig. 4. Associate Dean Addressing the students during anti-ragging awareness campaign



Fig. 5. Anti ragging warning boards at classrooms



Fig. 6. Anti to helpline, Associate Dean, warden phone numbers and anti-ragging poster display at corridors



Fig. 7. Conducting student awareness with visual aids on anti-ragging awareness week from 13th to 18th 2023



Fig. 8. Essay writing competitions conducted during anti-ragging awareness week



Fig. 9. Certificates distributed to winners during anti-ragging awareness week

6.5.1.5. Biological waste disposal facility

Dr. N.T.R. College of Agricultural Engineering, Bapatla, has established adequate waste disposal facility. One is the waste from the hostel mess is converted into manure through an electric machine provided in the Boys Hostel. Another one is a preparation of dry compost using the plant materials like dried leaves, etc.

6.5.1.6. Institutional Ethics Committee for Experiment on Animals

The college does not experiment on animals.

6.5.1.7. Committee for Prevention of Sexual Harassment of Women at Workplaces

The Women Protection Cell at Dr. N.T.R. The College of Agricultural Engineering, Bapatla, is committed to promoting gender equality, ensuring the safety and dignity of women, and creating a supportive and inclusive academic environment for female students and staff. The Cell functions in accordance with institutional policies and statutory guidelines to prevent discrimination, harassment, and any form of gender-based misconduct on campus. The primary objective of the Women Protection Cell is to foster a safe, respectful, and nurturing atmosphere that enables female students to pursue their academic and personal development with confidence and security. The Cell actively works towards creating awareness of women's rights, gender sensitization, personal safety, and legal provisions related to women's protection through counselling sessions, awareness programs, interactive meetings, and educational activities conducted periodically.

A designated Women's Cell Coordinator supervises and monitors matters related to the welfare and well-being of female students. The coordinator serves as a point of contact for guidance, counselling, grievance redressal, and emotional support whenever required. Female students are encouraged to approach the Cell freely to discuss their concerns in a confidential and supportive environment. The College has also provided essential infrastructure and amenities to ensure the comfort and convenience of female students. Adequate sanitary facilities are available for female students, including approximately five to six well-maintained toilets with an uninterrupted water supply for female day scholars. Special attention is paid to cleanliness, hygiene, and regular maintenance of these facilities to ensure a healthy campus environment.

To support menstrual hygiene management, sanitary pad disposal machines were installed in the girls' hostel for the safe, hygienic, and environmentally responsible disposal of used sanitary napkins. These facilities help ensure privacy, cleanliness, and convenience for the residents. In addition, the college provides medical support and emergency care facilities for female students. A dedicated female nurse is available round the clock to provide immediate medical assistance, health support, and counselling, whenever necessary. This arrangement ensures timely attention to health-related concerns and contributes to the overall safety and well-being of female students on campus. Through these welfare measures, support systems, and awareness initiatives, the Women Protection Cell of Dr. N.T.R. The College of Agricultural Engineering continuously strives to create a secure, respectful, and

empowering environment that promotes the holistic development and well-being of female students and staff.

Table 10. Women Protection Cell committees from 2021 – 2026

Year	Name of the Staff Member	Designation
2012 – 22	Dr. K. Lavanya, Chairperson	Assistant Professor
	Dr. N. Vinoda, Member	Assistant Professor
2022 – 23	Dr. K. Lavanya, Chairperson	Assistant Professor
	Er. M. Karunya, Member	Assistant Professor
2023 – 24	Dr. K. Krupavathi, Chairperson	Assistant Professor
	Dr. K. Lavanya, Member	Assistant Professor
	Smt. V. Padma Tulasi, Member	Administrative Officer
2024 – 25	Dr. K. Krupavathi, Chairperson	Associate Professor
	Er. M. Karunya, Member	Assistant Professor
	Smt. V. Padma Tulasi, Member	Administrative Officer
2025 – 26	Dr. K. Krupavathi, Chairperson	Associate Professor
	Dr. K. Lavanya, Member	Associate Professor
	Smt. V. Padma Tulasi, Member	Administrative Officer

The following are the activities done under women protection cell

1. **Awareness Programme on Health and Hygiene**

An awareness programme on health and hygiene was organized on 08 March 2023 to educate students on the importance of maintaining personal cleanliness and adopting healthy lifestyle practices. Dr. Lakshmi delivered an informative session that emphasized personal hygiene, sanitation, disease prevention, nutritional habits, and environmental cleanliness. The programme created awareness among students regarding the significance of maintaining both personal and surrounding hygiene for a healthy and productive life.

2. **Awareness Programme on Women Empowerment and Entrepreneurship**

An awareness program on Women Empowerment and Entrepreneurship was conducted on August 17, 2023, to encourage female students towards self-reliance and leadership. The program highlighted the importance of education, self-confidence, financial independence, and entrepreneurial development among women. Participants were motivated to explore small-scale and home-based business opportunities and were informed about the role of entrepreneurship in achieving economic empowerment and social upliftment of women.

3. Self-Defence and Yoga Training Programme

A self-defence and yoga training program was organized on 05 December 2023 for female students of the college. The program focused on basic yoga postures, breathing techniques, self-control practices, and self-protection methods that contribute to physical fitness and mental well-being. The sessions were designed to improve self-confidence, concentration, emotional balance, and inner strength among students, thereby enabling them to handle stressful and challenging situations effectively.

4. Awareness Programme on Women's Health Issues

An awareness program on women's health issues, such as Polycystic Ovarian Disease (PCOD) and cervical cancer, was organized on February 26, 2024. Medical experts explained the causes, symptoms, preventive measures, and management practices related to PCOD and cervical cancer. The program emphasized the importance of early diagnosis, regular medical screening, healthy lifestyle habits, and timely medical consultation. The session helped create awareness among the participants regarding women's reproductive health and the importance of preventive healthcare.

5. Cyber Safety Awareness Programme on Phone Hacking

An awareness program on cyber safety and phone hacking was conducted on September 11, 2024, to educate female students about cyber threats and online safety measures. Smt. Rajyalakshmi, a Sub-Inspector from Bapatla Police Station, addressed the students and explained the common methods used by cybercriminals to hack mobile phones and misuse personal information. The session also highlighted precautionary measures, such as identifying fake links, suspicious messages, fraudulent applications, and unsafe online practices. Students were educated on the immediate steps to be taken in the case of cyber fraud or phone compromise.

6. Awareness Programme on Yoga Postures, Mudras, and NCC Self-Defence

An awareness program on yoga postures, mudras, and NCC-based self-defence techniques related to girls' health and safety was organized on November 20, 2024. The session emphasized the importance of yoga for physical fitness, mental peace, stress management, concentration, and emotional wellbeing. In addition, NCC self-defence training highlighted the importance of discipline, courage, physical strength, and self-confidence. The

program encourages students to become fearless, independent, and capable of handling emergency situations while also promoting leadership qualities and gender equality.

7. International Women's Day Celebration at Returu Village

International Women's Day was celebrated on 08 March 2025 at Returu village with active participation from local women and girl students. The program focuses on women's empowerment, education, health awareness, and social equality. Motivational talks delivered during the event encouraged young girls and women to pursue education, career goals, and leadership opportunities with confidence and determination. The program also created awareness of women's rights, safety, and the importance of equal participation in social and economic development.

8. Awareness Programme on Women's Legal Rights

An awareness programme on "Women's Legal Rights" was conducted on June 19, 2025, to educate students about the important legal protections available for women in India. The program highlighted key legislation, including the Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act, 2013, the Protection of Women from Domestic Violence Act, 2005, and the Dowry Prohibition Act, 1961. Participants were informed of their legal rights, complaint mechanisms, and the importance of seeking legal support in cases of harassment, discrimination, or violence.

9. Awareness Programme on Women Protection Mobile Applications

An awareness program on Women Protection Mobile Applications was organized on November 28, 2025, in collaboration with the Bapatla Police She Team. Police officials explained the importance and usage of various women's safety applications designed for emergency assistance, location tracking, and quick communication during unsafe situations. Girl students were encouraged to install these applications on their mobile phones and educated about the effective use of digital safety tools for personal security and emergency response.

10. Training Programme on the Sexual Harassment of Women at Workplace Act

The women faculty members of Dr. N.T.R. College of Agricultural Engineering, namely Dr. K. Lavanya and Dr. K. Krupavathi, participated in a two-day online training programme conducted on 26 and 27 September 2025 on "The Sexual Harassment of Women

at Workplace (Prevention, Prohibition and Redressal) Act, 2013." The program provided a detailed understanding of legal provisions, institutional responsibilities, preventive measures, grievance redressal mechanisms, and workplace safety practices aimed at ensuring a safe and respectful environment for women.

11. International Women's Day Celebrations

As per the guidelines of the Government of Andhra Pradesh and Acharya N. G. Ranga Agricultural University, International Women's Day was celebrated on 08 March 2026 at Dr. N.T.R. College of Agricultural Engineering, Bapatla, India. Dr. K. Rani, CEO, Kshetra Natural Farmer Producer Company, Bapatla, attended the program as the Chief Guest. In her address, she motivated students to focus on self-employment, entrepreneurship, and innovation, instead of depending solely on conventional job opportunities. As part of the celebrations, literary competitions such as essay writing on the topic "Importance of Women Education" and elocution on "Role of Women in Nation Building" were organized for female students. Prizes and certificates were distributed to the winners by the Associate Dean and the Chief Guest, encouraging active participation and promoting awareness of women's empowerment and leadership.

Some glimpses of the Women Protection Cell activities



Fig. 10. Dr. K. Rani, CEO, Kshetra Natural Farmer Producer Company, Bapatla, giving message to students on the eve of "International Women's Day" on 08-03-2026



Fig. 11. Women's day celebrations at Returu village on 08-03-2025 with local women and girl students



Fig. 12. Awareness programme on different Women Protection Apps on 28-11-2025



Fig. 13. Awareness programme and training for girl students on NCC



Fig. 14. Awareness programme on women protection apps by the staff of local Bapatla police station



Fig. 15. Awareness programme on health and hygiene by Dr. Lakshmi

6.5.2. Faculty

6.5.2.1. Faculty Strength

There is sufficient faculty available to teach all the courses of the B Tech (Agricultural Engineering) degree. The college has sanctioned 31 faculty positions. Apart from 16 filled posts of the College, there are six Teaching Associates working in the College, a Scientist of AICRP on Farm Implements Machinery Scheme, and support from Agricultural College to cater to the needs of the academic requirement. The College meets the number of faculty members recommended by the Indian Council of Agricultural Research, New Delhi.

Table 11. Sanctioned and In-Position Faculty at Dr. N.T.R. CAE, Bapatla

SI. No.	Position	Sanctioned Faculty	Faculty in Place	Vacant Positions
1	Associate Dean	1	1	0
2	Professor	5	5+2*	0
3	Associate Professor	8	2+2*	4
4	Assistant Professor	17	8+16*	0
	Total	**31**	**36 (16 + 20*)**	**4**

**Staff from Agricultural College/Research/Extension and Contractual faculty*

6.5.2.2. Faculty Profile (Department wise)

Dr. N.T.R. The College of Agricultural Engineering, Bapatla, possesses qualified and experienced faculty members across various departments to effectively carry out teaching, research, extension, student mentoring, and academic administration. The College has faculty members in different cadres, namely Assistant Professor, Associate Professor, and Professor, in accordance with the sanctioned positions approved by the University. Almost all faculty members are Doctoral degree holders from national premier Universities/Institutes like Indian Institute of Technology (IIT), Kharagpur; Indian Agricultural Research Institute (IARI), New Delhi; Tami Nadu Agricultural University (TNAU), Coimbatore; University of Agricultural Sciences (UAS), Raichur; Acharya N. G. Ranga Agricultural University (ANGRAU), Guntur; University of Nebraska-Lincoln (UNL), USA; etc.

The department-wise sanctioned strength and faculty in position are presented below:

Table 12. Department wise Sanctioned and In-Position Faculty at Dr NTR CAE, Bapatla

Sl. No.	Department	Assistant Professor		Associate Professor		Professor		Total	
	Faculty / Department	**SS**	**IP**	**SS**	**IP**	**SS**	**IP**	**SS**	**IP**
1	Associate Dean	-	-	-	-	1	1	**1**	**1**
2	Farm Machinery and Power Engineering	5	5+1*	2	-	1	1	**8**	**7**
3	Processing and Food Engineering	3	3*	2	1+1*	1	2	**6**	**7**
4	Irrigation and Drainage Engineering	2	2+1*	1	1	1	1	**4**	**5**
5	Soil and Water Conservation Engineering	2	1+2*	1	-	1	1	**4**	**4**

6	Renewable Energy Engineering	1	1*	-	-	0	1	**1**	**2**
7	Basic Engineering and Applied Sciences	4	7*	2	1*	1	1	**7**	**9**
8	Physical Education	-	1*	-	-	-	-	-	**1**
	Total number of posts	**17**	**24**	**8**	**4**	**6**	**8**	**31**	**36**

Staff from Agricultural College/Research/Extension and Contractual faculty
SS - Sanctioned Strength
IP - In Position

The College has a competent pool of faculty members with expertise in various disciplines of Agricultural Engineering and Applied Sciences. The existing faculty strength is effectively supporting the academic requirements of the B. Tech. (Agricultural Engineering) undergraduate program through classroom teaching, laboratory instruction, student advisory services, research guidance, experiential learning programs, and extension activities. In addition to regular faculty positions, the College is also utilizing faculty support from Agricultural Colleges, Research Stations, Extension Wings, and contractual appointments wherever necessary to ensure uninterrupted academic activities and effective curriculum delivery. The availability of experienced faculty members across departments has enabled the College to maintain quality teaching standards in accordance with the ICAR recommendations and university norms. Overall, the present faculty profile is adequate and sufficient to meet the academic, practical, research, and student mentoring requirements of the College for the effective implementation of the B.Tech. (Agricultural Engineering) program.

6.5.2.3. Credentials of the Faculty

The following the details of the qualifications of the faculty:

B.Tech. (Agril. Engg.) Under Graduate:

S. No.	Name of the Teaching Staff	Designation	Highest qualification obtained	Experience, years
1.	Dr. G. Ravi Babu	Professor & Head, Dept. of SWCE	Ph.D.	26
2.	Dr. K.V.S. Rami Reddy	Professor & Head, Dept. of FMPE	Ph.D.	25
3.	Dr. B. Sreenivasula Reddy	Professor & Head, Dept. of PFE	Ph.D.	19
4.	Dr. S. Vishnuvardhan	Professor & Head, Dept. of REE	Ph.D.	18
5.	Dr. K. Lavanya	Associate Professor & Head, Dept. of BEAS	Ph.D.	18

6.	Dr. K. Krupavathi	Associate Professor & Head, Dept. of IDE	Ph.D.	18
7.	Dr. D. Vijay Kumar	Assistant Professor	Ph.D.	12
8.	Dr. B. Raj Kiran,	Assistant Professor	Ph.D.	12
9.	Dr. A. Srinivasa Rao	Assistant Professor	Ph.D.	12
10.	Dr. N. Ashok Kumar	Assistant Professor	Ph.D.	12
11.	Dr. A. Ashok Kumar	Assistant Professor	Ph.D.	12
12.	Er. M. Ratna Raju	Assistant Professor	M. Tech.	9
13.	Er. M. Karunya	Assistant Professor	M. Tech.	9
14.	Er. N. Kishore	Assistant Professor	M. Tech.	9
15.	Dr. G. Veera Prasad	Assistant Professor	Ph.D.	12
16.	Dr. V. Vasu Deva Rao	Senior Scientist	Ph.D.	12
17.	Er. K. Rakesh	Teaching Associate	M. Tech.	10
18.	Er. N. Kishore	Teaching Associate	M. Tech.	15
19.	Sri K. Venkateswarlu	Physical Director	M.P.Ed.	13
20.	Er. K.V.S. Karthik	Teaching Associate	M. Tech.	9
21.	Sri G. Sesha Reddy	Teaching Associate	M. Sc. (Mathematics)	22
22.	Dr. B. Pravallika	Teaching Associate	Ph.D.	2
Agricultural College				
22.	Dr. M. Latha	Professor & Head, Dept. of SSAC	Ph.D.	
23.	Dr. N. Ratna Kumari	Asst. Professor, Dept, of Horticulture	Ph.D.	
24.	Dr. B. Suresh Kumar	Teaching Associate, Dept. of Agronomy	Ph.D.	
25.	Smt. M. Swapna	Assistant Professor, Dept. of SSAC	M.Sc.	

Post Graduate M. Tech. (Agril. Engg.)

Soil and Water Conservation Engineering

S. No.	Name of the Teaching Staff	Designation	Highest qualification obtained	Experience, years
1.	Dr. G. Ravi Babu	Professor & Head, Dept. of SWCE	Ph.D.	26
2.	Dr. K. Krupavathi	Associate Professor & Head Dept. of IDE	Ph.D.	18
3.	Dr. N. Ashok Kumar	Assistant Professor	Ph.D.	12
4.	Dr. A. Ashok Kumar	Assistant Professor	Ph.D.	12
5.	Er. Ch. Ratna Raju	Assistant Professor	M. Tech.	9
6.	Er. M. Karunya	Assistant Professor	M. Tech.	9
Agricultural College				
7.	Dr. Lal Ahmad Mohammad	Professor, Dept. of Genetics	Ph.D.	
8.	Dr. T. Srinivasa Rao	Associate Professor, Library Information Sciences	Ph.D.	
9.	Dr. D. Ramesh	Assistant Professor, Dept. of Statistics & Computer Applications	Ph.D.	

10.	Dr. K. Arjuna Rao	Teaching Associate	Ph.D.	
11.	Dr. BV. Viswa Teja	Teaching Associate	Ph.D.	

Processing and Food Engineering

S. No.	Name of the Teaching Staff	Designation	Highest qualification obtained	Experience, years
1.	Dr. B. Sreenivasula Reddy	Professor & Head, Dept. of PFE	Ph.D.	19
2.	Dr. S. Vishnuvardhan	Professor & Head, Dept. of REE	Ph.D.	18
3.	Dr. K. Lavanya	Associate Professor & Head Dept. of BEAS	Ph.D.	18
4.	Dr. V. Vasu Deva Rao	Senior Scientist	Ph.D.	12
5.	Dr. G. Veera Prasad	Assistant Professor	Ph.D.	12
6.	Dr. B. Raj Kiran	Assistant Professor	Ph.D.	12
Agricultural College				
7.	Dr. K. Kiran Prakash	Professor & Head, Dept. of Statistics & Computer Applications	Ph.D.	
8.	Dr. M. Latha	Professor & Head, Dept. of SSAC	Ph.D.	
9.	Dr. Lal Ahmad Mohammad	Professor & Head, Dept. of Genetics	Ph.D.	
10.	Dr. T. Prasanth Kumar	Associate Professor, Dept. of Agricultural Extension	Ph.D.	
11.	Dr. T. Srinivasa Rao	Associate Professor, Library Information Sciences	Ph.D.	
12.	Dr. P. Lavanya Kumari	Assistant Professor, Dept. of Statistics & Computer Applications	Ph.D.	
13.	Dr. K. Arjuna Rao	Teaching Associate	Ph.D.	

Farm Machinery and Power Engineering

S. No.	Name of the Teaching Staff	Designation	Highest qualification obtained	Experience, years
1.	Dr. K.V.S. Rami Reddy	Professor & Head, Dept. of FMPE	Ph.D.	25
2.	Dr. B. Raj Kiran,	Assistant Professor	Ph.D.	12
3.	Dr. A. Srinivasa Rao	Assistant Professor	Ph.D.	12
4.	Dr. D. Vijay Kumar	Assistant Professor	Ph.D.	12
5.	Er. N. Kishore	Assistant Professor	M. Tech.	9
6.	Dr. K. Lavanya	Associate Professor & Head Dept. of BEAS	Ph.D.	18
7.	Dr. A. Ashok Kumar	Assistant Professor	Ph.D.	12
8.	Dr. G. Veera Prasad	Assistant Professor	Ph.D.	12

h.D. (Agril. Engg.)

Soil and Water Conservation Engineering

S. No.	Name of the Teaching Staff	Designation	Highest qualification obtained	Experience, years
1.	Dr. G. Ravi Babu	Professor & Head, Dept. of SWCE	Ph.D.	26
2.	Dr. K. Krupavathi	Associate Professor & Head Dept. of IDE	Ph.D.	18
3.	Dr. N. Ashok Kumar	Assistant Professor	Ph.D.	12
4.	Er. Ch. Ratna Raju	Assistant Professor	M. Tech.	9
5.	Er. M. Karunya	Assistant Professor	M. Tech.	9

Processing and Food Engineering

S. No.	Name of the Teaching Staff	Designation	Highest qualification obtained	Experience, years
1.	Dr. B. Sreenivasula Reddy	Professor & Head, Dept. of PFE	Ph.D.	19
2.	Dr. S. Vishnuvardhan	Professor & Head, Dept. of REE	Ph.D.	18
3.	Dr. L. Edukondalu	Principal Scientist	Ph.D.	20
4.	Dr. K. Lavanya	Associate Professor & Head Dept. of BEAS	Ph.D.	18
5.	Dr. D. Vijay Kumar	Assistant Professor	Ph.D.	12
6.	Sri. G. Sesha Reddy	Teaching Assistant	M. Sc. (Mathematics)	22

Farm Machinery and Power Engineering

S. No.	Name of the Teaching Staff	Designation	Highest qualification obtained	Experience, years
1.	Dr. K.V.S. Rami Reddy	Professor & Head, Dept. of FMPE	Ph.D.	25
2.	Dr. B. Raj Kiran	Assistant Professor	Ph.D.	12
3.	Dr. A. Srinivasa Rao	Assistant Professor	Ph.D.	12
4.	Dr. D. Vijay Kumar	Assistant Professor	Ph.D.	12
5.	Dr. K. Lavanya	Associate Professor & Head Dept. of BEAS	Ph.D.	18
6.	Dr. A. Ashok Kumar	Assistant Professor	Ph.D.	12

6.5.2.4. Technical and Supporting Staff

The College is sanctioned with 10 permanent positions as technical and supporting staff, apart from 26 positions working on a time scale/outsourcing/contract basis. The technical and supporting staff of 36 is sufficient to carry out the daily required turnout, which is much more than the number of positions recommended by the Indian Council of Agricultural Research, New Delhi.

Table 13. Sanctioned and In-Position Office Staff at Dr. N.T.R. CAE, Bapatla

Sl. No.	Sanctioned Post / Category	Sanctioned Strength	In-Position	Vacant Positions	Office Staff Recommended by ICAR
1	Administrative Officer	1	1	-	1
2	Superintendent	1	1	-	20 Clerks, steno, Messenger
3	Senior Assistant	2	2*	-	
4	JACT (Junior Asst cum Typist)	2	1+7*	-	
5	Record Assistant	1	1	-	
6	Office Subordinate	2	2	-	
7	Office supporting staff	-	8*	-	-
8	Messenger/Peon	-	10*		8
9.	Watchman	1			
	Total	**10**	**36**	**-**	**29**

** Staff recruited on Time scale/Outsourcing/Contract*

The College has a sanctioned strength of 16 permanent positions as technical support to laboratories, whereas a total of 41 personnel is working on a time scale/outsourcing/contract basis. A Laboratory staff of 41 is sufficient to carry out the daily required turnover, which is equal to the number of positions recommended by the Indian Council of Agricultural Research, New Delhi.

Table 14. Sanctioned and In-Position Laboratory Staff at Dr. N.T.R. CAE, Bapatla

Sl. No.	Sanctioned Post / Category	Sanctioned Strength	In-Position	Vacant Positions	Laboratory Staff Recommended by ICAR
1	Workshop Supervisor	1	1*	-	27 Workshop Staff, Laboratory Staff, Driver, Technicians
2	Technician (Mech. Gr.I)	3	5*	-	
3	Technician (Mech.Gr.II)	2	2*	-	
4	Draughtsman Grade III	1	1*	-	
5	Carpenter Grade II	1	1*	-	
6	Welder	1	1*	-	
7	Blacksmith	1	1*	-	
8	Turner	1	1*	-	
9	Driver	1	1+3*	-	
10	Workshop Helper	2	2*	-	

11	Luskar	2	2*	-	
12	Watch & Ward Staff	0	8*	-	
12	Laboratory Assistant	-	8*	-	14
13	Computer operator	-	3*	-	
	Total	**16**	**41**	**-**	**41**

**Positions filled with timescales, outsourcing, or contractual employees.*

6.5.3. Learning Resources

The learning resources provided in the college are sufficient and meet the expectations for learning defined by ICAR curricula.

6.5.3.1. College library (digital)

The fully air-conditioned digital library is provided with 32 computers. These provides highly interactive learning environment. A high speed internet is connected in the Library. There are also other facilities like photocopying, computer printing and internet facility in college library which helps students in their academic activities. The practical classes of "Computer Programming in C" AUTOCAD and "Computational methods through C Programming", Remote sensing, solid works, Ansys, software's English Grammar Lab Library and Information services for U.G. students used to conduct in computer centre. Computers having high end configuration to run software like CeRA, CAB Abstracts, KrishiKosh, EBSCO, Indiastat, CRC Netbase e-books, CAB e-books, Elsevier e-books, CMIE, EPWRF India Times Series, EZ Proxy, DELNET, OPAC, etc. for the improvement of academic and research standards.

6.5.3.2. Laboratories, Instructional Farm, Workshops, Dairy plant, Veterinary clinic, Hatchery, Ponds, etc.

The college is well-equipped with a wide range of advanced machinery and modern equipment across its various laboratories, which significantly enhance the quality of teaching, research, and practical training. These facilities are designed to provide students with hands-on experience using contemporary tools and technologies that are widely used in the field of agricultural engineering and allied disciplines. Some of the advanced machinery available in the different laboratories includes precision instruments for soil and water analysis, modern irrigation systems such as drip and sprinkler setups, and hydraulic testing equipment used in Irrigation and Drainage Engineering laboratories. In the Farm Machinery and Power Engineering

laboratories, students are exposed to advanced tractors, power tillers, seed drills, planters, harvesters, and other farm implements. These machines are often integrated with latest technologies such as sensors and automation systems to demonstrate modern mechanized farming practices.

The Soil and Water Conservation Engineering laboratories are equipped with apparatus for studying soil erosion, sediment transport, watershed management, and land development techniques. Instruments such as rainfall simulators, runoff measuring devices, and soil moisture sensors help students understand field conditions and conservation strategies in a scientific manner. In addition, the workshops house sophisticated fabrication and manufacturing equipment, including lathe machines, milling machines, drilling machines, welding units, and CNC-based systems. These facilities enable students to gain practical skills in machining, fabrication, equipment maintenance, and prototype development. The availability of such advanced machinery not only supports the academic curriculum but also encourages research, innovation, and skill development among students. It prepares them to adapt to modern agricultural practices and emerging technologies, thereby making them industry-ready professionals. Each department is having different laboratories to conduct the practical classes regularly. Each class is made into 2 batches (Batch A and Batch B) for the purpose of conducting practical separately. Accordingly, the class Timetable is set. Each department of the College is having dedicated laboratories.

Department-wise laboratories are furnished hereunder:

A. **Department of Farm Machinery and Power Engineering**

2. Farm Power Laboratory
3. Farm Machinery Laboratory
4. Workshop
5. Soil Dynamics Laboratory
6. Ergonomics Laboratory
7. Field Lab

B. **Department of Processing and Food Engineering**

1. Instrumentation Laboratory
2. Process Engineering Laboratory
3. Food Analysis Laboratory

4. Food Microbiology Laboratory
5. Environmental Control Laboratory
6. Extrusion Processing Laboratory

C. **Department of Irrigation and Drainage Engineering**

1. Field Irrigation Laboratory
2. Field Solar Pump Laboratory
3. Wells and Pumps Laboratory

D. **Department of Soil and Water Conservation Engineering**

1. Watershed
2. Soil and Water Conservation Laboratory
3. Field Lab

E. **Department of Renewable Energy Engineering**

1. Renewable Energy Laboratory

F. **Department of Basic Engineering and Applied Sciences**

1. Workshop with equipment of all manufacturing processes
2. Thermodynamics Laboratory
3. Central Computer Centre
4. Soil Mechanics Laboratory
5. Engineering Drawing Laboratory
6. Theory of Machines Laboratory
7. Strength of Materials Laboratory
8. Engineering Physics Laboratory
9. Applied Electronics Laboratory
10. Electrical Engineering Laboratory
11. Hydraulics and Fluid Mechanics Laboratory
12. Surveying and Levelling Laboratory

The Agricultural College, Bapatla possesses a well-established and fully functional college farm that plays a vital role in supporting academic instruction, research, and practical training for students. The farm serves as a living laboratory where theoretical knowledge gained in classrooms is effectively integrated with hands-on field experience. The Faculty of Agriculture actively utilizes this farm to conduct various courses related to crop production, agronomy, horticulture, and allied agricultural sciences. Students are provided with

opportunities to engage in real-time farming operations such as land preparation, sowing, irrigation management, nutrient application, pest control, and harvesting practices. This experiential learning approach enhances their practical skills, decision-making abilities, and overall understanding of agricultural systems.

In addition to routine instructional activities, the college farm is extensively used for conducting field experiments and research trials. Apart from the dedicated Irrigation Field Laboratory, the available agricultural land within the campus is efficiently utilized by multiple departments, including Farm Machinery and Power Engineering, Irrigation and Drainage Engineering, and Soil and Water Conservation Engineering. These departments carry out experiments on farm mechanization, irrigation methods, drainage systems, soil conservation techniques, watershed management, and resource optimization under actual field conditions. Such interdisciplinary use of land ensures optimal utilization of resources while promoting collaborative research and innovation.

The institution is also equipped with a highly advanced and well-maintained workshop that supports the practical needs of all engineering departments. The workshop is furnished with ultra-modern machinery, tools, and equipment required for fabrication, repair, maintenance, and testing of agricultural implements and machinery. It provides students with hands-on training in machining operations, welding, fitting, and assembly, thereby strengthening their technical competencies. Overall, the combination of a well-managed college farm, dedicated laboratories, and state-of-the-art workshop facilities creates a comprehensive learning environment. This integrated infrastructure not only fulfills the curriculum requirements but also prepares students to meet real-world challenges in agriculture and agricultural engineering with confidence and competence.



Fig. 16. Dr. M.V Rao workshop Laborartory



Fig. 17. Farm Machinery Lab



Fig. 18. Process Engineering Laboratory



Fig. 19. Food Analysis Laboratory



Fig. 20. Fluid Mechanics Laboratory



Fig. 21. Engineering Workshop Laboratory



Fig. 22. Irrigation Field Laboratory



Fig. 23. Watershed

Some of the advanced machinery available in the various laboratories is listed below.



Fig. 24. Refractive Window Drier



Fig. 25. Extruder



Fig. 26. Millet Processing Equipment



Fig. 27. Tray/Cup sealing Machine



Fig. 28. Continious band sealinhg machine





Fig. 29. Fish cutting machine & Tray Dryer



Fig. 30. Computer Lab



Fig. 31. Multicrop combine harvester & Wear Testing Apparatus



Fig. 32. Hollow Fiber Ultra-Filtration & Soil Moisture Probe





Fig. 33. Unibloc Moisture Analyzer & Digital planimeter



Fig. 34. Round soil bin



Fig. 35. Advanced CNS Lathe and Milling machines



Fig. 36. Rectangular Soil bin

6.5.3.3. Student READY/In-Plant Training/Internship/Experimental Learning Programmes

B.Tech. (Agricultural Engineering) degree program is supported by the following hands-on training courses:

1. SRDY 281 Skill Development Training – I (4 weeks)
2. SRDY 381 Skill Development Training – II (4 weeks)
3. SRDY 481 Industrial Attachment/Internship (10 weeks)
4. SRDY 482 Experiential Learning (10 weeks)
5. SRDY 483 Educational Tour (2 weeks)
6. SRDY 484 Project Work and Report Writing (20 weeks)

SRDY 281 – Skill Development Training is a vital component of the B. Tech. (Agricultural Engineering) curriculum, aimed at equipping second-year students with practical exposure and industry-oriented training. The program is conducted in small-scale agricultural equipment manufacturing units, esteemed tractor service centers, and agricultural machinery repair workshops to bridge the gap between theoretical knowledge and practical field applications. At the second-year level, students gain foundational engineering concepts through classroom and laboratory instruction; however, practical exposure to real-time agricultural machinery operations, manufacturing processes, and maintenance practices is equally crucial for enhancing technical competency. In this context, the Skill Development Training program offers students hands-on learning opportunities in areas such as fabrication, assembly, machining, welding, material handling, quality control, testing procedures, and cost-effective manufacturing techniques related to agricultural machinery and farm equipment. Training in tractor repair and service workshops enables students to acquire a

practical understanding of agricultural machinery systems, engine components, transmission systems, hydraulics, electrical systems, and preventive maintenance practices. Students are introduced to troubleshooting methods, repair procedures, diagnostics, servicing techniques, and operational safety measures associated with tractors and farm machinery commonly used in agricultural operations.

Practical exposure significantly enhances students' technical knowledge, analytical abilities, problem-solving skills, and confidence. It also familiarizes them with industrial work culture, safety standards, teamwork, professional ethics, and customer-oriented service approaches. Furthermore, the training program aids students in understanding workshop management practices, spare part handling, inventory management, equipment servicing, and the economic aspects of machinery maintenance and repair services. The Skill Development Training program also fosters entrepreneurial aptitude among students by exposing them to the operational aspects of small-scale manufacturing units and machinery service enterprises. Students gain valuable insights into self-employment opportunities in the agricultural machinery manufacturing, repair, servicing, and custom hiring sectors, which are increasingly important in the current era of agricultural mechanization. Given the growing demand for skilled technical manpower in the agricultural engineering sector and the rapid expansion of farm mechanization in rural areas, industry-oriented training is highly relevant and justified. The program ultimately improves employability, enhances practical competency, encourages entrepreneurship, and contributes to the efficient operation and maintenance of agricultural machinery systems, thereby supporting agricultural productivity and rural development in the region.

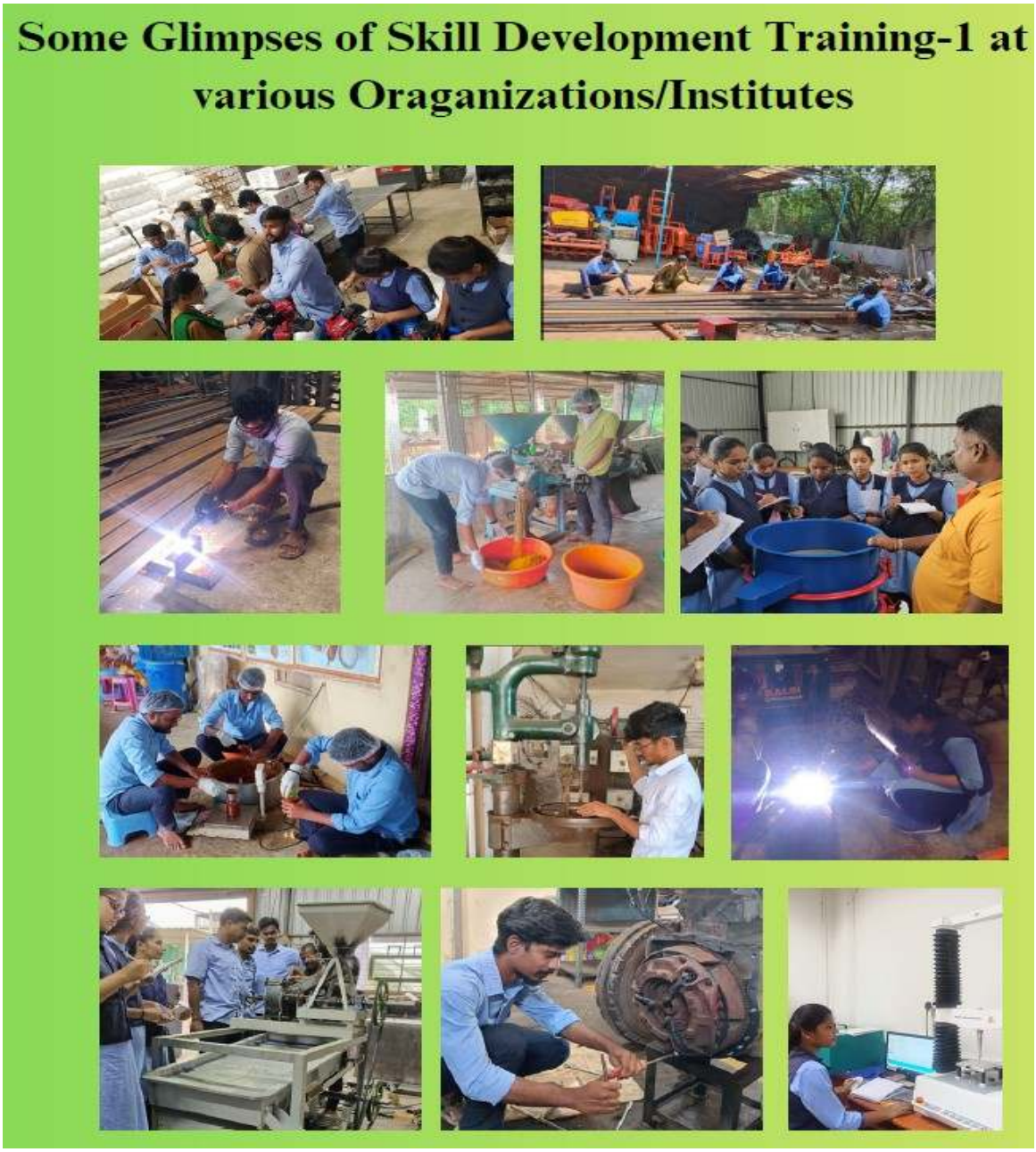
Completion of the Skill Development Training program is followed by a systematic and well-structured evaluation procedure to assess the learning outcomes and practical competencies acquired by students. This comprehensive evaluation process integrates academic assessments, industrial feedback, and reflective learning components. Students are assessed based on their practical performance, technical skills acquired during training, understanding of machinery and tools, work discipline, communication skills, and ability to adapt to industrial environments. The assessment process may include practical demonstrations, viva voce examinations, and the submission of a detailed training report describing the activities undertaken, observations made, technical knowledge gained, and overall learning experiences during the training period. Supervisors and experts from the

respective training institutions or workshops may also provide feedback regarding students' punctuality, discipline, technical aptitude, work efficiency, and professional behavior. The final evaluation is conducted by the faculty members of the College based on the training report, presentation, industrial feedback, and viva voce examination.

The details of students who underwent SRDY 281 – Skill Development Training-I at various institutes, industries and workshops are presented below.

Table 15. Details of Students Attended Skill Development Training–I at Different Institutes

Name of the Institution /company/ Organization	2021-22	2022-23	2023-24	2024-25	2025-26
Farm Machinery Training & Testing Institute, Garladinne.	74	--	--	--	10
Mahindra and Mahindra, Zahirabad	--	21	--	--	--
CSIR-Central Mechanical Engineering Research Institute, Gill Road, Ludhiana	--	18	--	--	--
ICAR- Indian Institute of Horticultural Research, Bangaluru	--	08	--	--	--
FIM, Bapatla	--	12	--	--	--
Sridevi Rice mill, Bapatla	--	11	--	--	--
Viswakarma Industries, Hyderabad	--	--	10	10	10
Gousiya Engineering works, Nandyal	--	--	11		--
V.R. Agro Equipments, Allagadda	--	--	15	13	10
K S R Agricultural Implements, Proddutur	--	--	15	09	--
V.V.C. Farm Equipment Pvt Ltd (John Deere), Guntur, AP	--	--	10	--	--
Navya Engineering works, Gudivada	--	--	05	--	--
Sri Padmavathi Automotives, Guntur	--	--	05	--	--
Sri Lakshmi Agencies, Ongole	--	--	05	--	--
Sri Srinivasa Auto Engineering Works, Gudiwada	--	--	03	04	06
Vedhik Enterprises, Karnataka	--	--	--	13	10
Nabhitha Engineering Private Limited, Hyderabad	--	--	--	10	10
Mech O Tech LLP, Hyderabad	--	--	--	10	--
Tractor Farm Equipment Limited, Guntur	--	--	--	05	--
Tractor Farm Equipment Limited, Nellore	--	--	--	05	--
Jayarama Motors Pvt., Ltd., Mahaboobnagar, Telangana	--	--	--	05	--
CATIA, ANSYS, Solid works, Dr, NTR CAE, Bapatla	--	--	--	--	28
Total	**74**	**70**	**79**	**84**	**84**



SRDY 381 - For third-year B.Tech (Agricultural Engineering) students, skill development training at esteemed Indian Council of Agricultural Research institutes and Farm Machinery Testing & Training Institutes (FMTTIs) is essential at this advanced stage of their studies. By this point, students have acquired foundational theoretical knowledge in subjects such as farm machinery, power engineering, soil and water engineering, and post-harvest technology. However, structured exposure to leading research and testing institutions enables them to translate this knowledge into practical skills. The necessity for such training lies in familiarizing students with sophisticated agricultural machinery, testing protocols, standardization processes, and performance evaluation methods employed at national-level

institutes. Facilities such as the Central Farm Machinery Training and Testing Institute and other FMTTIs provide practical experience in machinery calibration, testing under diverse field conditions, safety compliance, and adherence to BIS standards. Exposure to ICAR institutes also acquaints students with ongoing research, innovation, and emerging technologies such as precision farming, automation, and smart farm equipment.

The rationale for this training is founded on the enhancement of employability, technical proficiency, and research aptitude. It equips students for careers in government sectors, research organizations, agri-machinery industries, and entrepreneurship. The training also cultivates analytical and reporting skills essential for machinery evaluation and project work. The benefits of such training include improved problem-solving abilities, a better understanding of real-world farm mechanization challenges, and preparedness for industry or advanced studies. Ultimately, it contributes to producing skilled agricultural engineers capable of supporting sustainable and mechanized agriculture in India. Students will be assessed based on practical performance, technical skills acquired, and understanding of tools, machinery, and processes. This assessment may include hands-on tests, viva voce, and the submission of a training report detailing activities performed, observations, and learning outcomes. Supervisory institutions may provide performance feedback on discipline, skill level, and adaptability. The student details who had undergone skill development training - II at various institutes were listed below:

Table 16. Details of Students Attended Skill Development Training–II at Different Institutes

Name of the Institution /company/ Organization	2021-22	2022-23	2023-24	2024-25	2025-26
Nano Scientific Research Centre Python (Software) Hyderabad	77	--	--	--	--
Mahindra and Mahindra, Zahirabad	--	15	--	--	--
ICAR- Central Institute of Agricultural Engineering, Regional Centre, Coimbatore	--	15	08	09	10
SRFMT&TI, Tractor Nagar, Garladinne, Anantapur	--	15	16	10	--
ICAR- Indian Institute of Horticultural Research, Bangaluru	--	15	10	06	--
ICAR-IISWC, Research Centre, Ballari-583104, Karnataka	--	14	08	08	10
Indian Institute of Technology Design & Manufacturing (IIITDM) Jabalpur	--	--	13	--	--
ICAR-Indian Institute of Oil Palm Research, Pedavegi	--	--	05	--	05

ICAR-Central Institute of Post-Harvest Engineering & Technology (CIPHET) P.O. PAU LUDHIANA	--	--	07	--	10
ICAR-Central Plantation Crops Research Institute, Kudlu P.O., Kasaragod, Kerala	--	--	04	07	10
NRFMTTI, Hisar	--	--	--	06	
IISWC, Ooty	--	--	--	08	10
IISWC, Vasad	--	--	--	09	
IISWC, Dehradun	--	--	--	08	06
CIAE, Bhopal	--	--	--	--	05
IISWC Koraput	--	--	--	--	10
Onion And Garlic Research Pune	--	--	--	--	09
Dr. YSRHU Horticultural Research Station Pandirimamidi	--	--	--	--	01
Total	**77**	**74**	**71**	**71**	**86**



SRDY 481 - Internship or industrial attachment for final-year B. Tech (Agricultural Engineering) students is a critical component of professional preparation, especially when undertaken in process industries, dairy plants, irrigation companies, and machinery manufacturing units. At this stage, students are expected to transition from academic learning to industry readiness, and internships provide the necessary platform to achieve this. The need for such training arises from the requirement to integrate multidisciplinary knowledge

covering farm machinery, food processing, irrigation systems, and energy applications with real industrial operations. Exposure to facilities such as dairy plants Drip/Sprinkler irrigation companies, hydroponic cultivation, farm as well as process machinery manufacturers allows students to understand production workflows, quality assurance, supply chain management, and industrial safety practices. It also provides hands-on experience in equipment handling, system design, maintenance, and troubleshooting under real working conditions.

The justification lies in enhancing employability and bridging the academia–industry gap. Internships cultivate practical skills, professional discipline, teamwork, and communication abilities that are often not fully developed in classroom settings. Students gain exposure to industry standards, documentation practices, and performance expectations, making them job ready. The uses of such internships are multifaceted. They strengthen technical competency, improve decision-making and problem-solving skills, and provide clarity on career interests. Importantly, internship experience significantly improves employment prospects. Recruiters often prefer candidates with prior industry exposure, as it reduces training time and ensures immediate productivity. Internships also offer networking opportunities, references, and sometimes pre-placement offers (PPOs), giving students a competitive advantage in campus placements and off-campus recruitment. Overall, industrial internships are indispensable for shaping competent agricultural engineers capable of contributing effectively to modern, technology-driven agriculture and allied industries.

Evaluation after Internship / Industrial Attachment

Assessment shall focus on both technical and professional competencies. Key components include:

- **Internship Report** documenting work experience, projects handled, and problem-solving approaches
- **Industry Supervisor Evaluation** covering punctuality, teamwork, communication, and technical ability
- **Seminar/Presentation** where students present their learning to a faculty panel
- **Logbook/Work Diary** maintained throughout the internship period Organizations

SRDY 482 - Study tours for final-year B. Tech (Agricultural Engineering) students are an essential academic activity that complements classroom learning with real-world exposure.

At the final stage of their program, students are expected to have a comprehensive understanding of engineering principles, and study tours help them visualize how these concepts are applied across diverse agricultural and industrial settings. The need for study tours arises from the limitation of classroom-based instruction in demonstrating large-scale operations, modern technologies, and integrated systems. Visits to advanced institutions such as Indian Council of Agricultural Research institutes, agro-processing industries, irrigation projects, and farm machinery manufacturing units enable students to observe practical applications of mechanization, precision farming, post-harvest processing, and water management systems. These tours expose students to recent innovations, automation, and sustainable agricultural practices that are shaping the future of the sector.

Study tours also serve as an important platform for experiential learning, allowing students to interact with scientists, engineers, and industry professionals. Such interactions provide insights into current challenges, industry expectations, and emerging career opportunities. They help students connect theoretical knowledge with practical scenarios, thereby strengthening conceptual clarity and analytical skills. Furthermore, study tours enhance soft skills such as teamwork, communication, adaptability, and professional behaviour. They broaden students' perspectives by exposing them to different agro-climatic regions, farming systems, and industrial practices. This is particularly important for agricultural engineers who must work in diverse field conditions. Overall, study tours are necessary for producing well-rounded, industry-ready graduates by providing holistic exposure, fostering innovation, and preparing students for professional roles in agriculture and allied sectors.

Assessment of student's performance shall be based on their observation, analysis, and knowledge integration. Students are evaluated through:

- **Tour Report** summarizing visits, technologies observed, and key learnings
- **Group Discussion or Viva** to assess understanding of field exposure
- **Assignments/Case Studies** based on industries or institutes visited
- **Participation and Conduct** during the tour

Table 17. Year wise details of SRDY 481 Skill Development Training-I (Student READY)

Name of the Institution /company/ Organization	2021-22	2022-23	2023-24	2024-25	2025-26
ICAR- Central Institute of Agricultural Engineering, Regional Centre, Coimbatore	20	--	--	--	--
JK Seeds, Hyderabad	06	--	--	--	--
Watershed Support Services and Activates Network (WASSAN), Hyderabad	04	--	--	--	07
Dhan Foundation, Hyderabad	16	--	07	04	06
Central Mechanical Engineering Research Institute (CMERI), Ludhianina	12	--	--	--	--
Tractor and Farm Equipment Limited (TAFE), Vijayawada	19	--	--	--	--
Corteva Agri Science,Telangana	--	04	--	03	--
DAAT-C (Dara Akruthi Agriculture Training Centre), Hyderabad	--	04	--	--	---
Aerospace Engineer Exodrone Systems Pvt. Ltd. Telangana	--	04	--	--	04
Foundation for Ecological Security Madanapalle	--	06	04	04	04
Kerala Agro Machinery Corporation Ltd. (A Government of Kerala Undertaking)	--	07	--	--	--
Kisankraft, Podalakuru,Nellore	--	05	--	--	--
Hyfunfoods Pvt. Ltd., Ahemadabad	--	09	--	--	--
Nava Kisan Agri Farms, Bangalore	--	08	--	--	--
Telangana State Remote Sensing Applications Centre (TRAC), Telangana	--	26	--	--	--
Jaya Sree Rice Mill, Bapatla	--	01	--	--	----
MaLaxmi Polymers Pvt.Ltd, Thukkuguda, Hyderabad	--	--	06	--	--
IIT, Tirupati	--	--	03	03	--
Haritage Food, Tirupati	--	--	05	06	01
Indus Coffee Pvt. Ltd, Nellore	--	--	03	--	--
Sri Vijaya Visakha Milk Producers Company Limited, Ankapalle Visakhapatnam, AP	--	--	05	06	10
Visakha Dairy, Amudalavalasa- Andhra Pradesh	--	--	06	--	--
Godrej Agrovet Limited, Ch. Pthepalli W.G. (District)-AP	--	--	05	--	--
Vijaya Dairy, Vijayawada	--	--	05	--	--
CCL Products (India) Limited, Duggirala – A.P.	--	--	05	--	--
Sangam Dairy Vadlamudi	--	--	02	--	04
Eid Parry India Limited, Srikakulam	--	--	--	06	--
Touchstone Foundation Iskon, Bangalore	--	--	--	06	02
Andhra Pradesh State Seeds Development	--	--	--	06	12

Corporation ltd. Kurnool					
Heritage Foods Limited, Bayyavaram	--	--	--	01	--
SIDS Farm Private Limited, Hyderabad	--	--	--	06	--
Nabhitha Engineering Private Limited, Hyderabad	--	--	--	06	--
Samunatifinance, Srikakulam and Warangal	--	--	--	02	--
Search (NGO) Bapatla	--	--	--	03	--
Mahindra Irrigation - Guntur	--	--	--	03	
Simply Fresh Private Limited, Hyderabad	--	--	--	07	--
Sprouted Jungle Pvt Ltd., Nagpur, Maharashtra	--	--	--	--	08
Biofuel Circle Pvt Ltd., Kakinada	--	--	--	--	07
SEARCH (Markapur)	--	--	--	--	06
Strato sky drones, Tanku West Godavari	--	--	--	--	01
Thulasi Seeds, Guntur	--	--	--		01
Signet Industries Ltd, Kadapa	--	--	--	--	03
Signet Industries Ltd, Hyderabad	--	--	--	--	02
Akshyakalpa Pvt Ltd, Bangalore	--	--	--	--	03
Total	**77**	**74**	**56**	**72**	**81**

SRDY 483 - Experiential Learning Programme module (ELMs) for final-year B. Tech. (Agricultural Engineering) students are a vital component of outcome-based education, designed to provide hands-on, enterprise-oriented learning. These modules covering hydroponics equipment manufacturing, hydroponics crop production, farm machinery production units, grain and spice processing, and tertiary food product development enable students to function in real production and business environments rather than simulated academic settings. The need for ELMs arises from the increasing demand for industry-ready graduates who possess not only technical knowledge but also practical skills, entrepreneurial mindset, and managerial competence. Final-year students must be capable of independently handling production systems, understanding market dynamics, and managing resources efficiently. Exposure to modern systems like hydroponics and precision agriculture aligns with advancements promoted by institutions such as Indian Council of Agricultural Research. Similarly, modules on processing and value addition reflect the growing importance of agri-business and post-harvest management in India's economy.

The justification lies in bridging the gap between theory and practice. Traditional classroom teaching cannot fully impart skills such as equipment fabrication, quality control, product development, cost estimation, and enterprise management. ELMs simulate real-life business environments where students learn by doing, decision-making, and problem-solving under actual constraints.

Uses and Learning Outcomes

- **Technical Skill Development:** Hands-on experience in fabrication, processing, and system operation
- **Entrepreneurial Skills:** Understanding business planning, costing, marketing, and profit analysis
- **Interdisciplinary Integration:** Application of knowledge from farm machinery, food engineering, irrigation, and agronomy
- **Innovation and Creativity:** Development of new products (e.g., value-added food products, hydroponic systems)
- **Teamwork and Leadership:** Managing group-based enterprises or production units

Students working in modules such as farm machinery production or spice processing units gain practical exposure like industrial environments, enhancing their confidence and competence.

Role in Employment and Recruitment

Experiential learning significantly improves employability. Recruiters prefer candidates who can contribute immediately without extensive training. Experience gained through ELMs demonstrates:

- Practical exposure to real-world operations
- Ability to handle machinery, production systems, and quality standards
- Problem-solving and decision-making skills
- Entrepreneurial and managerial capabilities

Companies such as Mahindra & Mahindra, Jain Irrigation Systems, and food processing industries actively seek candidates with hands-on experience. ELM participation can lead to internships, industry connections, and even pre-placement offers (PPOs).

Profit Sharing Model (Student: Mentor: Department = 75 : 15 : 10)

To strengthen motivation and ensure sustainability of ELMs, a structured profit-sharing mechanism can be adopted:

- **Students (75%):** Major share allocated to students to encourage ownership, accountability, and entrepreneurial thinking

- **Mentor (15%):** Faculty/technical supervisor share as incentive for guidance, mentoring, and technical support
- **Department (10%):** Institutional share for maintenance of infrastructure, raw materials, and future module development

This model promotes active participation, financial literacy, and a sense of enterprise among students.

Evaluation Criteria

A comprehensive evaluation system should be adopted to assess student performance in ELMs:

1. **Enterprise/Project Report (20%)**: Detailed documentation of processes, production, cost analysis, and outcomes
2. **Practical Performance (50%)**: Skill in handling equipment, quality of output, and efficiency
3. **Viva Voce / Presentation (10%)**: Understanding of concepts, decision-making ability, and communication skills
4. **Continuous Assessment (10%)**: Attendance, participation, teamwork, and discipline
5. **Financial and Business Analysis (10%)**: Cost-benefit analysis, profitability, and marketing strategies
6. **Innovation and Problem-Solving (Bonus/Additional Weightage)**: Creativity in product or process improvement

Overall, experiential learning modules are indispensable for transforming students into competent agricultural engineers, entrepreneurs, and industry-ready professionals capable of contributing effectively to modern agriculture and agri-based industries. The profits generated during different academic years through ELPM units is listed below:

Sl. No.	Name of the ELP Unit	Profit Generated, Rs.				
		2021-22	2022-23	2023-24	2024-25	2025-26
1	Hydroponics -I	-	-	15798	55192	46,263
2	Hydroponics- II	-	-	16200	55028	47342
3	ANGRAU Farm Implements Production -I	6370	2560	10800	23020	11,002
4	ANGRAU Farm Implements Production -II	2120	-	22955	30718.50	20,815
5	Grain Processing	416	4560	6317	17422	20,057
6	Spice Processing	4200	12690	16409	31,827	25,899

7	Tertiary Food Products	-	-	14,604	37,433	37,250
	Total	**13,106**	**19,810**	**1,03,083**	**2,50,640**	**2,08,628**

Some of the glimpses of ELP Products are



Fig. 37. Gud Makhana



Fig. 38. Instant Ragi Dosa Mix



Fig. 39. Instant Foxtail and Little millet Dosa mix



Fig. 40. Ragi malt



Fig. 41. Finger Millet cookies



Fig. 42. Multi millet atta



Fig. 43. Vertical hydroponic unit



Fig. 44. Hydroponics unit



Fig. 45. Leafy vegetable cultivation



Fig. 46. Cultivation practice with hydroponics



Fig. 47. Hydroponic unit construction



Fig. 48. Selling of value added products



Fig. 49. Ragi mixture



Fig. 50. Black gram Laddu



Fig. 51. Fabrication of cultivator frame



Fig. 52. Levelling blade attachments

SRDY 484 - A final-year B. Tech (Agricultural Engineering) project is the capstone of undergraduate study, where students integrate knowledge, skills, and practical exposure to solve real-world agricultural problems or develop innovative technologies. It reflects technical competence, research ability, and readiness for professional careers or higher studies. Through the final-year project, students shall

1. Apply theoretical concepts to practical engineering problems
2. Develop problem-solving, design, and analytical skills
3. Encouraged for innovation and research orientation
4. Prepare students for industry, entrepreneurship, or higher education
5. Align with national priorities in mechanization, food processing, and sustainability,

A standard B. Tech project includes:

1. Problem Identification & Objectives
2. Literature Review
3. Design/Methodology
4. Fabrication/Experimentation
5. Data Collection & Analysis
6. Results & Discussion
7. Conclusion & Future Scope

Evaluation Criteria

- **Project Report (30–40%)** – Technical content, clarity, and documentation
- **Innovation & Design (20–25%)** – Novelty and practical relevance
- **Execution/Experimentation (15–20%)** – Implementation quality

- **Presentation & Viva (15–20%)** – Communication and understanding
- **Teamwork & Regularity (10–15%)**

Students will prepare a Project Report and present their work in front of the Evaluation Committee and then, Viva-Voce Examination will be held for evaluation. A well-executed final-year project is not just an academic requirement but a professional launchpad. It transforms students into competent engineers capable of addressing real agricultural challenges, contributing to mechanization, sustainability, and agri-business development. The following were the projects carried out by the students during 2021-22 to 2025-26.

S. No.	Year	Title of the B. Tech Project	Student team	Project guide
1.	2021	Performance Evaluation of an indirect type Solar Dryer for Drying of Pineapple and Jackfruit	T. Yashwanth Krishna, BE-17-64 K. Naveen BE-17-33 P. Venkata Sukesh BE-17-57 M. Venu Gopal BE-17-58 Gulshan Kumar BE-17-66 B. Hari Prasad BE -16-23	Dr. M. Madhava
2.		Optimization of Rasam and Sambar Powder ingredients using linear Programming Techniques	P. Baby Amrutha BE-17-06 D. Nagur Vali, BE-17-31 G. Sunny BE-17-49 G. Vivek BE-17-62 Yogesh Singh Dosad BE-17-70 K. Jagadeesh BE-17-77	Dr. B. Sreenivasula Reddy
3.		Effect of Climate change on Crop Water Requirement in Visakhapatnam District of Andhra Pradesh using Cropwat 8.0 Software	S. Archana BE -17-05 J. Swathi BE -17-50 R. Ranjith Kumar BED 18-06 E. V. Sai Kumari BED 18-08	Dr. R. Ganesh Babu
4.		Studies on Skimming well Technology for Kitchen Gardens of Coastal Sandy Areas	Mahesh Pamisetty BE-17-22 N. Nageswara Rao BE-17-30 Naveen Rachamalla BE-17-34 Satish Kongara BE-17-45 Samanwayi Madina BE D17-83	Dr. H.V. Hema Kumar
5.		Measurement of Physical properties of Black gram and Green gram Seed for the Evaluation of vertical plate metering Mechanism	K. Ramesh BE-17-39 S. Pavan Kumar BE-17-36 V. Vani Neelima Devi BE-17-53 P. Divya Nagaraju BE -16-17	D. Vijay Kumar
6.		Study on operational parameters for different nozzles suitable for organic farming using Spray patternator	B. D. Venkata Ramana BE-17-10 K. Koteswara Rao BE-17-20 Y. Manjusha BE-17-23 N. Mikhilya BE-17-24	G. Veera Prasad

7.		Effect of Ultrasound on drying characteristics of Beetroot Slices	D. L. Ramakumari BE-17-02 J. Durgasyamala BE-17-11 M. Nataraj BE-17-32 M. Sasipavan BE-17-44 B. Harshika BED 17-76	N. Vinoda
8.	2021	Effect of extraction temperature and roasting time on quality characteristics of Sesame Oil	N. Ganesh BE-17-12 B. Haritha BE-17-13 L. Janaki Ramani BE-17-16 M. Nagaraju BE-17-28 M.Nagarjuna BE-17-29	K. Lavanya
9.		Study on Wear characteristics of Material used for Shovels	M. Surendra Kumar BE-17-18 SK. Mohammad Ali BE-17-25 CH. Ramya BE-17-40 M. Sudhakar BE-17-47 P.Prasad BED 18-05	B. Raj Kiran
10.		Making Karela Biscuits for Good Health	N. Anil Kumar BE-17-03 P. Anjali BE -17-04 N. S. Sree Harshitha BE-17-46 CH. Tarun Teja Reddy BE-17-51 T. Vineela BE-17-60 V. Ratna Kumari BE-17-80	Dr. D. Bhaskar Rao
11.		Investigation and Assessment of physical and chemical properties of Soils on infiltration rate at Dr. NTR CAE College Farm, Bapatla	K. Bhargavi BE-17-07 V. Kalyani BE-17-17 K. Kiran Kumar BE-17-19 S. Kumar Sahoo BE-17-68 Ch. V. Lakshmi, BED 18-09	K.N. Raja Kumar
12.		Comparative study on different Harvesting machines for paddy crop	M. Haritha BE-17-14 K. Murali Krishna BE-17-26 K.N.D. Sai Mythri BE -17-27 Rama Kumar J, BE-17-38 K. Sagarika BE-17-41 K. Sudhir Kumar BE-17-69	Dr. B. Hari Babu
13.		Modified Digital circular flume aided with ultrasonic sensor for measurement of flow discharge	N. K. Chaudhary BE-17-09 V.V.B. Reddy, BE-17-55 M. V. Ramprakash BE-17-56 A. Vyshnavi BE-17-63 Sanjukta Bera BE-17-67	Dr. K. Krupavathi
14.	2022	Design, fabrication and performance Evaluation of Tribolium Cartaneum separator from RICE.	K. Akash BE-18-01 K. Akhil sai BE-18-02 D. Vasu, BE-18-54 L. Divya Sai Teja BE-18-12 S. Vijay Kumar BE17-59 Y. Vani Raj BE17-54	Dr. B. Sreenivasula Reddy
15.		Evaluation of water use Efficiency of Paddy fields in Bapatla Region.	S. Poojith BE18-28 A. Lakshmi Sailaja BE18-19 K.A. K. Sailesh BE18-04 M. Daya Sagar BE18-11 B. Vamsi Krishna BE18-53	Dr. H.V. Hema Kumar

			Y. Lokesh BE17-21	
16.		Drying Kinetics and Quality Attributer of Vacuum Dried Shrimps.	G.C.N.V.B. Lakshmi BED-19-02 N. Sai Sumanth BE18-37 N. Vyshnavi Vineela BE18-61 G. Naga Tata Rao BE-18-24 M. Gopala Krishna BE18-13 T. Uday Kumar Be18-51	Dr. M. Madhava
17.	2022	Development and Evalaution of Seed Metering Mechanism for Manually Operated Single Row Jowar Planter.	D. Anil Kumar BE18-07 M. Srieeja BE18-43 D. Swathi Kiran BE18-48 M. Venka Praven BE18-58 M. Samba siva Reddy BE17-43 K.Vinod Kumar BE17-61	Dr. K.V.S. Rami Reddy
18.		Development of Reciprocating Type Trunk Shaker for Fruit Harvesting.	M. Bharath Reddy BE18-08 P, Nandini BE18-25 Shashwat Maurya BE18-67 J. L. Veera Sai BED19-06 D. Vimsi Krishmi BE17-52	Dr. B. Raj Kiran
19.		Impact of Irrigation Levels Under Mulch on Water Use Efficiency for Broad Bean Crop.	Niraj Kumar BE18-65 T. Praven Kumar BE18-31 T.S.S. Brahma Naidu BE18-44 T. Chranteja BE18-10 T. Hari rama Krishna BE18-15 B. Anand Babu BE18-03	Dr. K.N. Raja Kumar
20.		Studies on the Effect of Dragon Fruit Peel and Tamarind Leaves on Storage of Chicken for Chicken Nuggets.	K.V.L. Supraja BE18-57 P. Shyam Kumar, BED19-08 R. Sireesha BE18-39 V. Sahithi BE18-34 S. Somanandini- BE18-41 K. Tanu Sri BE18-50	Dr. D. Bhaskara Rao
21.		Development and Evaluation of Manually Operated Wheel Sprayer for row crops.	V. Lalindra Babu BE18-20 K. Naga Jyothi BE18-23 Shaik Sajidha BE 18-38 Priyanka Kumari BE18-68 Shaik Nijama MED19-45 B. Chandra Lekha BE17-08	Dr. B. Hari Babu
22.		Development of a Full-Face Shield with Fresh Air Supplier for Power Sprayer.	K. Yamini BE18-62 C.H.V.N. Srisha BE18-60 Satyam Purohit BE18-66 J. Issaac Sindhiya BE18-17 Devkaran Patidar BE18-64 R. P. Kumar Naik, BED 17-78	Er. D. Vijay Kumar
23.		Laboratory Evaluation of Battery Power Remote Controlled Sprayer for Field Crops.	P. Anantha Rami Reddy BE18-05 G. Anatha Varma BE18-06 Y. Poojitha BE-18-29 K. Pujitha BE18-32 G. Sivaji BE18-40 J. Veeranjaneyulu BE18-56	Er. G. Veera Prasad
24.		Studies on Foam Mat	K. Lokesh BE18-21	Dr. K. Lavanya

		Drying of TOMATO.	SK. Md. Junaid BE18-22 D. Pranav BE18-30 P. Srivani BE18-46 P. Uday Sri Sai Ram BE18-52 B. Adarsha BE17-01	
25.		Development of IoT based integrated soil moisture cum nutrient sensor for Geo-tagging soil properties in Bapatla Mandal.	B. Harika BE18-16 L. Navya Tejaswini BE18-26 N. Revathi BE18-33 B. Sai Kumar BE18-35 G. Sri Vidya BE18-45 Srijan Upadhya BE18-70	Dr K. Krupavathi
26.	2022	Monitoring Crop Growth Through Remote Sensing Indes Using Google Earth Engine.	P.G.V.S. Naga Durga BE18-14 G. Yuva Raj BE18-63 G. Sai Mounika BE18-36 P. Navya BE18-27 P. Swathi BE-18-49 R. Sai Brahma Reddy BE17-42	Dr. A. Mani
27.		Effect of Drip Fertigation Levels on Growth and Yield of OKRA (Abelmoschus esculents L.Moench)	J. Lakashmi Madhvi BE18-18 B. Veera BabuBE18-55 S. Harsha BED19-04 Y.S. Phnimanikanti BE18-47 B.V. Rama Krishna BE18-59 G. Lolesh BED19-07	Dr. R. Ganesh Babu
28.	2023	Development of Pendulum Impact Cutter and Measurement of Cutting Force and Energy for Plant Stalks	G. Mounika BE19-27 G. Sailikitha BE19- 44 R. Sunitha Bai BE 19-55 D. Surendra BE19-56 Ch. Vennela BE19-66	Dr. B. Hari Babu
29.		Studies on Shelf-life Enhancement of Pomegranate Arils	CH. Pavithra BE19-34 A. Prasuna BE 19-36 R. Vamsi BE19-60 G. Mohan Kumar BED20-03 P. Rahul BE19-39	B. Sreenivasula Reddy
30.		Studies on Ultrasonic Assisted Osmo-Convective Drying of Onion Slices	A. Chandini BE19-04 M. Komali Srilatha BE19-19 S. Maheswari BE19-23 T. Saikrishna BE19-43 S. H. Reddy BED20-08	Dr. S. Vishnu Vardhan
31.		Effect of Transplanting Dates on Growth, Yield &Length of Crop Cycle of Basil Crop in Hydroponics	N. Naveen BE19-31 K. Ravi Chandra BE19-41 K. Venkata Sai, BE19-64 T. Surendra, BED20-05 T. Sri Munnika, BED20-011	Dr. K. Krupavathi
32.		Modification and Performance of Evaluation of Drum Seeder for Rice Fallows	D. Jhansi Lakshmi BE19-15 Sk. Mastanbi BE19-24 Sk. Mohammad Moin BE19-26 L. SravaniBE19-52 P.R.L. BhavaniBED20-04	Er. D. Vijay Kumar
33.		Studies on Drying	A. Lakshmi PavaniBE19-20	Dr. K. Lavanya

		Characteristics of Carrotslices Using Freeze-Dryer	V.N. Deekshitha BE19-256 L. Rupa, BED20-10 Eliya Ramavath BED20-07 D.DeviPrasannaBE19-06	
34.	2023	Study of Sprinkler Irrigation System Under Different Irrigation Levels for Groundnut Crop (Kadiri Lepakshi) In Costal Sandy Soil	M. Chandini Priya BE19-05 M. Jahnavi BE19-14 D.R.S. Kumar Reddy BE19-40 P.S.P. Srinivas Be19-45 Tejas Shrivas BE19-58	Dr G. Ravi Babu
35.		Effect of Microwave Drying on Some Physical and Chemical Properties of Moringa Leaf	N. Prasanthi BE19-35 D. Radhika BE19-38 T. Saicharantej BE19-42 K. Seshu Babu BE19-48 R. Varun BE19-63 T. Venkatasai BE19-65	Dr. B. John Wesley
36.		Estimation of Soil Loss Using Rusle and Remote Sensing with Goodle Earth Engine for Sub Region of Lower Krishna River Basin	P. Dharani BE19-07 D. Likhitha BE19-22 A. Suresh BE 19-57 P. Vamsi Krishna BE19-59 O. Devi Vineela BED20-01	Er. M. Karunya
37.		Effect of Deficit Irrigation on Cluster Bean Under Drip Irrigation System	P. Jithendra Varma BE19-16 D. Pavan Teja BE19-33 D. Varshita Reddy BE19-62 K. Usha Sree BED20-06 K. Navya Sri BED20-09	Dr. R. Ganesh Babu
38.		Real Time Irrigation Scheduling Using Automated Class A Pan Evaporimeter for Okra Crop	P. Bhargavi Jyothi BE19-02 K. Bhargavi BE19-03 D. Jyothi BE19-17 K. Sravani BE19-53 G. Harshitha Reddy BED20-02	Er Ch. Ratna Raju
39.		Effect of Microwave Pretreatment on Dehulling Characteristics of Pigeon Pea	M. Hari Krishna BE19-09 P. Meghana BE19-25 M. Naga Charan BE19-30 A. Sofiya BE19-50 M. Yuvaraju BE19-67	Dr M. Madhava
40.		"Development of Hand Operated Two Row Seed Dibbler for Mazie"	M. Gopal Krishna BE19-08 B. Jahnavi BE19-13 D. Indu Madhavi BE19-11 M. Sai Sri BE19-46 B. Vinay Kumar, BED 17-85	Dr. B. RajKiran
41.		Development of Seed Drill Suitable For Robot Prime Mower	K. Havilah BE19-10 S. Keerthi BE 19-18 T. Lavanya BE19-21 J. Padmaja BE19-32 V. SainathBED 17-82	Dr. K.V.S. Rami Reddy
42.		Development of Punch Planter Suitable For Robotic Prime Mover"	K. Murali BE 19-28 S. Satya BE19-47 K. Sonika BE19-51	Er. G. Veera Prasad

			B. Srihari BE19-54 G. Vani Sree BE19-61	
43.	2024	Fabrication of Loose oil Palm fruit collector	R. Maruthi Naik BE/2020-22 V. Sai Deepika BE/2020-30 N. Siddu BE/2020-37 B. Durga Prasad BED/2021-01 G. Sirisha BED/2021-10	Dr. S. Vishnu Vardhan
44.		Development of Io T based NPK Sensor	K. Chandrisha BE/20-06 B. Sowmya Satya Sri BE/20-40 P. Venkatesh Varma BE/20-58 G. Jnaneshwar BED/18-02	Dr. K. V. S. Rami Reddy
45.		Performance evaluation of single row self-propelled Maize harvester	Jogi Jessy BE/2020-011 Chintapalli Jhansi BE/20-12 Suman Kumar Saha BE/20-48 M. Narsimha Reddy BE/20-63	Dr. D. Vijay Kumar
46.		Development of micro controller based automatic spot granular fertilizer application for row Crops	J. Ravve Teja BE/20-28 R. Supraja Suma BE/20-49 Ch. Vinni BE/20-60 P. Vyshnavi BE/20-61	Dr. A. Srinivasa Rao
47.		Performance evaluation of Accelerated Ageing chamber for Paddy	T. Harshitha BE/20-10 G. Rohith BE/20-29 P Sai Tejeswi BE/20-32 A. Sri Ram Rahul BE/20-47	Dr. V. Vasudeva Rao
48.		Development of anthocyanin based PH Sensitive strip to Detect food freshness	M. Mounika BED/21-07 G. Uday Kiran BE/20-55 Y. Vaishna Bhargavi BE/20-56 SK. Sameer BE/20-33	Dr. M. Madhava
49.		Development of Electronic punch planter attached to Robotic Mini Tractor	P. Kusala BE/20-15 Raunak Kumar BE/20-27 M. S. Vyshnavi BE/20-36 K. Sravani BE/20-41 T. Tulasiram BE/20-54	Dr. G. Veera Prasad
50.		Study of impact of Climate change on summer Groundnut(TAG-24) in coastal Sandy Soils	S. Manindra BE/20-21 D. Sri Lakshmi BE/20-44 M. Kavya Devi BE/20-64 T. Gunasri BE/D21-03)	Dr, G. Ravi Babu
51.		Induced Moisture stress on plant Growth parameter on Brinjal Plant by using deep learning algorithm	Ch. Swetha BE/20-52 T. V. M. Sudheer BE/20-57 Yashwanth B. BE/20-62 K. Haripriya BED/21-04	Dr. K. Krupavathi
52.		Assessment of landuse classes and optimization of water resources for major crop of Guntur District	S. Anusha BE/20-02 B. Aravind BE/20-03 D. Sai Srinivasa Rao BE/20-31 T. Srija BE/20-45 K. Swetha BE/20-53	Dr. N. Ashok Kumar
53.		Design and Development of integrated grain storage environment monitoring	M. Ganesh BE/20-08 K. Lakshmi BE/20-017 S. Premanvitha BE/20-26	Dr. B. Sreenivasula Reddy

		system	K. Srinivasa Rao BE/20-46	
54.		Development of Multi-Linear Regression(MLR) model for soil moisture estimation using satellite data	P. Anitha Swarupa BE/20-01 M. Aravind BE/20-04 Ch. Kranthi Kiran BE/20-14 G. Navya BE/20-25	Er. Ch. Ratna Raju
55.	2024	Development of hand operated dibbler with electronic metering unit	V. Lavanya BE/20-19 S. Sameera BE/20-34 A. Santhi Priya BE/20-35 U. Sravani BE/20-42 V. Venkatesh BE/20-59	Dr. B. Raj Kiran
56.		Studies on effect of ultrasonication treatment on physic-chemical properties and shelf life of button Mushroom	K. Madhuri BE/20-09 V. Sushma Raj BE/20-50 R. Gopalswami Naik BED/21-02 N. Manikanta BED/2018-04	Dr. K. Lavanya
57.		Rose petal and Apple jam : Preparation, Physio-chemical and sensory analysis	B. Naga Chandrika BE/20-23 J. Siri Chandana BE/20-38 G. Vaddikasulu BED/21-07 SK. Parvin BED/21-08	Dr. B. Sreenivasula Reddy
58.		Performance evaluation of PTO operated Mobile Rice Mill	T. Karthik Reddy BE/20-13 V. Lakshmi Narayana BE/20-18 Narsi Meena BE/20-24 Swarppa K. BE/20-51 P. Pavan Sreehitha BED/21-09	Dr. D. D. Smith
59.	2025	Studies on Engonomical Characteristics of Farm workers	P. Jaya BED/2022-006 Anshuman Ghosh BE/2021-004 T.L. Tirupathamma BE/2021-028 B. Neeraja Blessy BE/2021-035 SK. Nagur Mastan, BE/2021-034	Dr. D. Vijay Kumar
60.		Formulation of Jamun blended with Ginter RTS beverage for enhanced shelf life	M. C. Preethi BE/2021-012 R. Chandana BE/2021-013 V. Padimini BE/2021-036 G. Rajini BED/2022-014 P. Yaswanth BED/2022-025	Dr. D. D. Smith
61.		Comparative study of Drip Irrigation system and Furrow Irrigation system for Tomato Crop (SOLANUM LYCOPERSICUM) in Light sandy soil	T. Jahnavi Lakshmi Phani BE/2021-023 N. Maneesha BE/2021-030 Prakash Kumar BE/2021-038 G. S. B. Sankar BE/2021-043 A. R. S. Kumar BED/2022-013)	Dr. N. Ashok Kumar
62.		Comparative Study of Micro Sprinkler Irrigatin and Flood Irrigation for Leafy vegetables cultivation in Sandy Soil	S. Chandini BE/2021-014 B. Hitha Varshini BE/2021-021 J. M. N. L. Sai BE/2021-033 P. Susan Raja BE/2021-05 B. Preethi BED/2022-012	M. Karunya
63.		Effect of Edible coatings on quality and shelf of Sapota (MANILKARA ZAPOTA.	B. Dellieswari BED/2022-03 T. Santhoshini BED/2022-018 Ch. Sarathsai BE/2021-046	Dr. K. Lavanya

		L) cv. Kali Patti	P. Anusha BE/2021-005 V. Mahesh Babu BE/2021-020	
64.		Design and Development of solar photovoltaic sonar Sensor Based Bird scarer system	Y. Sireesha BE/2021-48 G. Sujatha BE/2021-50 CH. T. Lokesh BE/2021-056 S. Sujitha BE/2021-051 SK. Reshma BED/2022-15	Dr. A. Srinivasa Rao
65.	2025	Development of solar operated sprayer with Integrated wheel Hoe weeder system	SK. Abhisha BE/2021-003 S. Aruna BE/2021-006 V. Bhavya Sai BE/2021-010 N. Hima Bindu BED/2022-005 P. Sravani BED/2022-019	Dr. K.V.S. Rami Reddy
66.		Development of Tractor operated sensor-based site specific Chemical Application	Sk. Imrana BE/2021-022 A. Sivateja BE/2021-047 G. Tejaswini BE/2021-055 T. Vasavi BE/2021-058 A. Muskan BE/2022-011	Dr. A. Ashok Kumar
67.		Microwave assisted-high Temperature short time pulse Milling	K. V. Haripriya BED/2022-023 G. Vineetha BED/2022-024 M. Samhitha BE/2021-044 K. Supraja BED/2022-021 K. Karunya BE/2021-026	Dr. B. Sreenivasula Reddy
68.		Preparation of Drainage Proposal for Dr. NTR College of Agricultural Engineering	V. Chakravathi BE/2021-011 T. Harshaswi BE/2021-018 B. Lavanya BED/2022-008 M. Sai Druvitha BED/2022-016 M. Sai Lakshmi BED/2022-017	Dr. H.V. Hema Kumar
69.		Development of Groundnut Digger Shaker blade	Y. Avinash BE/2021-007 G. Manikanta NE/2021-031 Rimjhim Kumari BE/2021-041 K. V. Narayana BE/2021-059 B. Srikanya BED/2022-020	Dr. B. Raj Kiran
70.		Gravity-Fed Automated Irrigation systems for potted plants	Ch. Ajay BE/2021-001 N. Bhavani BE/2021-08 Y. Sandeep BE/2021-045 M. Uday BE/2021-057 A. Maruthi BED/2022-010	Dr. K. Krupavathi
71.		Studies on the Effect of carrier Material on masking of Bitterness of sweet orange juice Through freeze-drying	N. Harsha Vardhan BE/2021-019 D. Pavani BE/2021-037 M. Ranga Naik BE/2021-040 R. Gopi Naik BED/2022-004 B. Tharun Kumar BED/2022-022	Dr. S. Vishnu Vardhan
72.		Study of Growth and Yield of Cucumber in Different soilless Hydroponics system	A. Kusuma BED/2022-007 M. Mahesh Babu BED/2022-009 K. J. S. Sugnan BE/2021-024 Y. Aarthi Sri BE/2021-002	Dr. G. Ravi Babu

6.5.3.4. Curricula Delivery through IT (smart classroom/interactive board, etc.)

The B.Tech. (Agricultural Engineering) undergraduate programme at Dr. N.T.R. College of Agricultural Engineering is supported by modern teaching and learning facilities that enhance academic delivery and create an interactive learning environment for students. The College has established well-equipped smart classrooms integrated with advanced instructional technologies such as interactive boards, multimedia projectors, audio-visual systems, internet connectivity, and other ICT-enabled teaching aids to facilitate effective classroom teaching, seminar presentations, project reviews, and academic discussions. Faculty members extensively utilize these modern teaching resources to deliver course lectures through PowerPoint presentations, multimedia modules, video demonstrations, animations, and various digital learning platforms. The integration of contemporary teaching tools and ICT has significantly improved the quality of classroom instruction by promoting better conceptual understanding, active student participation, and enhanced interaction between teachers and students.

The undergraduate programme follows learner-centric teaching methodologies that encourage analytical thinking, practical problem-solving, innovation, and experiential learning among students. In addition to regular classroom teaching, students are encouraged to actively participate in seminars, technical presentations, project discussions, and research-oriented activities using digital presentation tools and online resources. These activities help students improve their technical communication skills, presentation abilities, confidence, leadership qualities, and overall professional competence. As part of the academic curriculum, regular seminars, technical discussions, guest lectures, and student presentations are organized to expose students to recent advances, emerging technologies, and current developments in the field of agricultural engineering. Such activities provide opportunities for students to broaden their technical knowledge, strengthen professional skills, and develop awareness of industry and research trends. Dr. N.T.R. College of Agricultural Engineering continuously strives to strengthen and modernize its academic infrastructure in accordance with ICAR recommendations and contemporary educational requirements. The available teaching and learning facilities provide a highly conducive academic environment for effective teaching, practical learning, research, innovation, and knowledge dissemination. These initiatives collectively contribute towards maintaining high standards of education and achieving academic excellence in the B.Tech. (Agricultural Engineering) undergraduate programme.



Fig. 53. Podium, LCD Projectors and Interactive board in the class room

6.5.4. Student Development

6.5.4.1. Student Intake and Attrition

Admissions into the First Year B. Tech. (Agricultural Engineering) degree programme are carried out through the Andhra Pradesh Engineering, Agriculture and Pharmacy Common Entrance Test (AP EAPCET) conducted by the Government of Andhra Pradesh. The admission process is undertaken through the centralized counselling system implemented by the Andhra Pradesh State Council of Higher Education (APSCHE), ensuring transparency and adherence to the reservation policies prescribed by the Government of Andhra Pradesh. The approved intake capacity for the First Year B. Tech. (Agricultural Engineering) programme is 60 students per academic year. Out of these 60 seats, 60 percent (36 seats) are filled through the AP EAPCET Convener quota based on merit and reservation norms. The remaining 40 percent (24 seats) are allotted under the Farmers' Quota in accordance with the policies and reservation guidelines of the Government of Andhra Pradesh, thereby providing opportunities to students from farming backgrounds to pursue professional education in agricultural engineering.

In addition to the regular intake, 20 percent supernumerary seats over and above the sanctioned intake are allotted to students nominated under the ICAR quota by the Indian Council of Agricultural Research (ICAR), New Delhi. These admissions are made at the

national level, enabling students from different states of the country to join the programme and contribute to a diverse academic environment within the College. The College also provides an opportunity for diploma holders to pursue higher education through the lateral entry system. Candidates who have completed a three-year Diploma in Agricultural Engineering are eligible for admission directly into the Second Year of the B. Tech. (Agricultural Engineering) programme through the Andhra Pradesh Engineering Common Entrance Test (AP ECET) conducted annually by APSCHE, Government of Andhra Pradesh. For these lateral entry admissions, an additional 20 percent seats over and above the regular intake are reserved as per the prescribed norms. Over the years, the College has observed a moderate increase in student attrition at the undergraduate level. The major reasons identified include health-related issues, difficulties faced by some students in adapting to the local climatic conditions, and migration of students to institutions located nearer to their hometowns through the process of counselling-based sliding or upgradation. Despite these challenges, the College continuously strives to provide a supportive academic environment, counselling services, and student mentoring mechanisms to improve student retention and ensure the successful completion of the programme.

This multi-channel admission system ensures merit-based selection, inclusivity, opportunities for students from agricultural backgrounds, national-level participation through ICAR, and academic progression pathways for diploma holders. The admission process thereby contributes to maintaining academic standards, accessibility, and diversity within the B.Tech. (Agricultural Engineering) programme.

Table 18. Attrition percentage during the last five years

Name of the Degree Programme	Actual students admitted in last five years						Attrition (%)					
B Tech (Ag.Engg.)	2021 - 22	2022 - 23	2023 - 24	2024 - 25	2025 - 26		2021 - 22	2022 - 23	2023 - 24	2024 - 25	2025 - 26	2025 - 26
	60	**80**	**86**	**84**	**83**		**10% (6)**	**6.25% (5)**	**1.16% (1)**	**1.19% (1)**	**1.20% (1)**	**1.20% (1)**

6.5.4.2. Average Number of Students in Theory and Practical Classes

Dr. N.T.R. College of Agricultural Engineering, Bapatla, follows an effective batch system for conducting theory and practical classes to ensure quality teaching, better faculty – student interaction, and efficient utilization of academic and laboratory facilities. For the B.Tech. (Agricultural Engineering) undergraduate programme, students of each academic year are divided into two batches for both theory and practical classes. This system facilitates effective classroom management, improved interaction, and better understanding of technical concepts. In practical and laboratory courses, the batch-wise division enables students to gain adequate hands-on experience with machinery, instruments, tools, and experimental setups without overcrowding. On an average, each batch consists of approximately 40-45 students depending on the admission strength of the respective academic year.

Table 19. Average Number of Students in Theory and Practical Classes in different programmes

S. No.	Name of the Degree Programme	Batch of students in theory class	Batch of students in practical class
1.	B. Tech (Agricultural Engineering)	Max 87	Max 40 students per batch
2.	M. Tech (Farm Machinery and Power Engineering)	1	1
3.	M. Tech (Processing and Food Engineering)	1	1
4.	M. Tech (Soil and Water Conservation Engineering)	1	1
5.	PhD (Farm Machinery and Power Engineering)	1	1
6.	PhD (Processing and Food Engineering)	1	1
7.	PhD (Soil and Water Conservation Engineering)	1	1

For the Postgraduate (M.Tech.) and Doctoral (Ph.D.) programmes, theory and practical classes are conducted in a single batch due to the comparatively lower student intake. The smaller class size enables close interaction between faculty and students, individualized guidance, better discussion-based learning, and effective research supervision. The average strength in PG and Ph.D. classes generally ranges from 2-6 students depending on the specialization and year of admission. The existing classroom infrastructure, laboratories, instructional facilities, and faculty strength are adequate to effectively handle the student strength at undergraduate, postgraduate, and doctoral levels in accordance with University and ICAR norms.

6.5.4.3. Admission Process

Admissions into the First Year B. Tech. (Agricultural Engineering) degree programme are carried out through the Andhra Pradesh Engineering, Agriculture and Pharmacy Common Entrance Test (AP EAPCET) conducted by the Government of Andhra Pradesh. The admission process is undertaken through the centralized counselling system implemented by the Andhra Pradesh State Council of Higher Education (APSCHE), ensuring transparency and adherence to the reservation policies prescribed by the Government of Andhra Pradesh. The approved intake capacity for the First Year B. Tech. (Agricultural Engineering) programme is 60 students per academic year. Out of these 60 seats, 60 percent (36 seats) are filled through the AP EAPCET Convener quota based on merit and reservation norms. The remaining 40 percent (24 seats) are allotted under the Farmers' Quota in accordance with the policies and reservation guidelines of the Government of Andhra Pradesh, thereby providing opportunities to students from farming backgrounds to pursue professional education in agricultural engineering.

In addition to the regular intake, 20 percent supernumerary seats over and above the sanctioned intake are allotted to students nominated under the ICAR quota by the Indian Council of Agricultural Research (ICAR), New Delhi. These admissions are made at the national level, enabling students from different states of the country to join the programme and contribute to a diverse academic environment within the College. The College also provides an opportunity for diploma holders to pursue higher education through the lateral entry system. Candidates who have completed a three-year Diploma in Agricultural Engineering are eligible for admission directly into the Second Year of the B. Tech. (Agricultural Engineering) programme through the Andhra Pradesh Engineering Common Entrance Test (AP ECET) conducted annually by APSCHE, Government of Andhra Pradesh. For these lateral entry admissions, an additional 20 percent seats over and above the regular intake are reserved as per the prescribed norms. Over the years, the College has observed a moderate increase in student attrition at the undergraduate level. The major reasons identified include health-related issues, difficulties faced by some students in adapting to the local climatic conditions, and migration of students to institutions located nearer to their hometowns through the process of counselling-based sliding or upgradation. Despite these challenges, the College continuously strives to provide a supportive academic environment, counselling services, and student mentoring mechanisms to improve student retention and ensure the successful completion of the programme.

6.5.4.4. Conduct of Practical and Hands on Training

Dr. N.T.R. College of Agricultural Engineering, Bapatla, places strong emphasis on practical learning and hands-on training as an integral part of the teaching–learning process in the B.Tech. (Agricultural Engineering), M.Tech., and Ph.D. programmes. The practical courses are designed to provide students with experiential learning opportunities and to strengthen their technical knowledge, analytical ability, and professional skills in various disciplines of agricultural engineering. Practical classes are conducted regularly in well-equipped laboratories, workshops, instructional farms, and field units as per the academic curriculum and timetable prescribed by the University. Students are provided hands-on exposure in operating agricultural machinery, farm equipment, laboratory instruments, survey tools, irrigation systems, food processing equipment, renewable energy devices, and soil and water conservation structures. The practical sessions are aimed at developing technical competency, operational skills, problem-solving ability, and confidence among students. For undergraduate programmes, students are divided into batches to ensure effective conduct of practical classes and optimum utilization of laboratory facilities and equipment. Faculty members and supporting technical staff closely supervise the practical sessions and provide necessary demonstrations, guidance, and safety instructions before the commencement of experiments and field operations.

Hands-on training is also imparted through workshops, field demonstrations, fabrication activities, machinery operation and maintenance exercises, industrial visits, and Student READY programmes. Students are trained in various practical aspects such as machinery calibration, equipment servicing, welding, fabrication, processing operations, irrigation management, renewable energy applications, surveying techniques, and precision farming technologies. In addition to regular laboratory practicals, students are encouraged to participate in experiential learning activities, mini-projects, seminars, model development, and research-oriented assignments to enhance their practical understanding and innovation skills. Training programmes conducted in industries, research institutes, and agricultural organizations further expose students to real-time field conditions, industrial practices, and emerging technologies in agricultural engineering. The College ensures that all practical and hands-on training activities are conducted in accordance with ICAR recommendations and University guidelines. The available laboratories, instructional facilities, workshop infrastructure, and experienced faculty members adequately support the effective conduct of

practical classes and skill-oriented training programmes, thereby contributing to the overall professional development and employability of students.

6.5.4.5. Examination and Evaluation Process

There shall be one mid-semester examination conducted for each course during the semester. The examination shall be organized by the course teacher after completion of approximately 50% of the instructional (working) days in the semester. The schedule of the mid-semester examinations shall be prepared and notified by the Controller of Examinations (CoE) Section, ANGRAU. The duration of the mid-semester examination shall be one and a half (1½) hours, and the maximum marks allotted shall be 50 marks. The question paper for the mid-semester examination shall be prepared centrally and transmitted by the COE Section, ANGRAU to the Office of Academic Matters (OAM) official email ID of the respective college. The question paper shall be sent one hour prior to the commencement of the examination in a password-protected format. The password for accessing the question paper shall be communicated separately to the Associate Dean concerned.

The duplication (printing) of the question papers shall be carried out strictly at the Examination Printing Cell of the college. The process shall be conducted under continuous surveillance using CCTV cameras to ensure confidentiality and security.

The printing process shall take place in the presence of:

- The Course Teacher
- The Officer In-charge of Academic Matters (OAM)
- Staff deputed by the Associate Dean

A confidential register/logbook shall be maintained at the Examination Printing Cell to record all relevant details, including:

- Date and time of printing
- Name of the course
- Number of copies printed
- Names and signatures of personnel involved

All personnel involved in the process shall ensure strict confidentiality, integrity, and security of the examination materials at every stage. The course teacher shall evaluate the mid-semester answer scripts and return them to the students in the classroom within seven days after the completion of the examination. The teacher shall obtain the students' signatures as acknowledgment, duly verified by the course teacher.

Practical Examination:

The practical examinations shall be conducted by the respective course teacher, who is responsible for the overall planning, execution, and evaluation of the examination. The practical examination timetable shall be prepared well in advance by the OAM office and communicated to all students and course in-charges through official channels to ensure proper coordination and preparedness. Each practical examination shall be conducted for duration of 2 hours and 30 minutes. The evaluation shall be carried out for a total of 50 marks, which includes different components such as practical performance, viva voce, identification of specimens/equipment, and a written test, wherever applicable.

The course teacher shall ensure that all necessary equipment, materials, and laboratory facilities are made available prior to the commencement of the examination. Proper instructions regarding the examination procedure, safety precautions, and evaluation criteria shall be clearly explained to the students before the start of the examination. During the examination, strict discipline shall be maintained, and the performance of each student shall be assessed individually and fairly based on predefined criteria. The marks awarded shall be recorded systematically and submitted to the OAM office within the stipulated time.

Semester Final Examination Procedure:

The semester final examinations shall be conducted at the end of each semester for all registered courses as per the academic schedule notified by the Controller of Examinations (COE), ANGRAU. The duration of the semester final examination for the theory component of each course shall be two and a half (2½) hours. The maximum marks allotted for the semester final theory examination shall be 100 marks. The question papers for the semester final examinations shall be prepared centrally and transmitted by the COE Section, ANGRAU to the Office of Academic Matters (OAM) official email ID of the respective college. The question papers shall be sent one hour prior to the commencement of the

examination in a password-protected format. The password for accessing the question paper shall be communicated separately to the Associate Dean concerned, ensuring confidentiality and controlled access.

The duplication (printing) of question papers, including both mid-semester and semester final theory examinations, shall be carried out exclusively at the Examination Printing Cell of the college. This process shall be conducted under strict security and continuous CCTV surveillance (using two cameras) to prevent any breach of confidentiality. The printing and handling of question papers shall be performed in the presence of:

- The Officer In-charge of Academic Matters (OAM)
- Staff members deputed by the Associate Dean

A confidential register/logbook shall be maintained at the Examination Printing Cell to document all activities related to question paper handling, including:

- Date and time of receipt of question paper
- Date and time of printing
- Course title and code
- Number of copies printed
- Names and signatures of authorized personnel involved

All officials and staff involved in the process shall strictly adhere to confidentiality protocols, ensuring the security, integrity, and secrecy of examination materials at all stages.



Fig. 54. Printing and duplication of Mid and Semester Final Theory Question Papers



Fig. 55. Question papers printing under CC camera surveillance

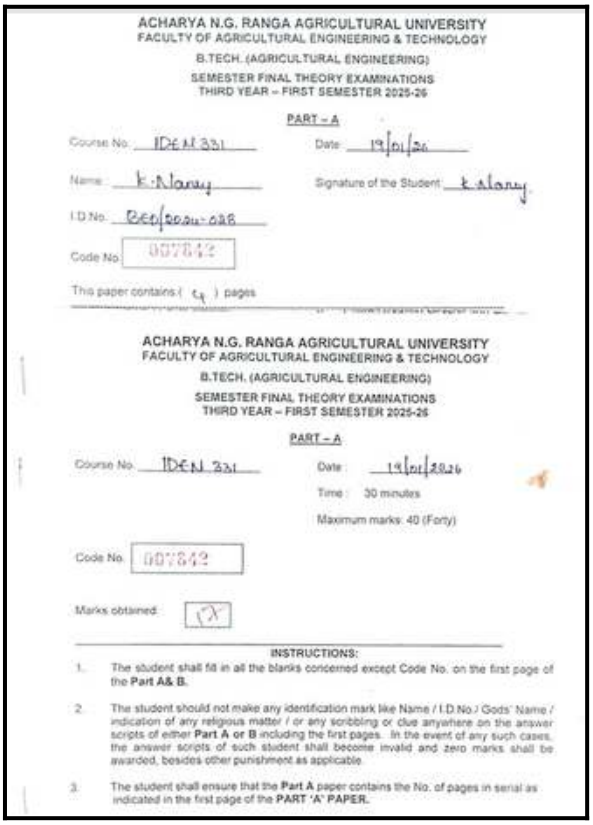
Evaluation Procedure:

In B.Tech. (Agricultural Engineering), the spot valuation procedure shall be followed. The Associate Dean shall send the answer scripts of each course, year-wise, to the nominated evaluation centre, where they will be jumbled and coded. The spot evaluation centre will be decided by the Controller of Examinations, ANGRAU, and the faculty for evaluation will be nominated by the Dean of Agricultural Engineering & Technology, ANGRAU. The spot evaluation is generally conducted over a period of three days. At the evaluation centre, the answer scripts from both colleges Dr. NTR College of Agricultural Engineering, Bapatla, and the College of Agricultural Engineering, Madakasira are combined course-wise and jumbled thoroughly (thrice) in the presence of the Spot Valuation Committee, chaired by the Associate Dean and the OAM, along with other nominated faculty members.

Once the jumbling process is completed, all answer scripts are coded with identical numbers on the first page for both Part-A and Part-B. The portion containing the students' details is removed so that the evaluator cannot identify the individual student's name or other details, thereby ensuring unbiased evaluation. The detached coded slips are packed in a separate cover in the presence of the OAM and are kept securely in the Associate Dean's chamber.

After completion of the evaluation, the evaluator submits the award list indicating the code number and marks obtained in Part-A and Part-B, after verification by another faculty

member. Upon receipt of all award lists from the concerned nominated course teachers, the award lists are decoded using the coded slips. The decoded marks award list, along with the students' ID numbers, is then sent to the respective colleges by the Associate Dean of the evaluation centre. Upon receipt of the award lists from the Associate Dean, the OAM office will distribute them to the respective course teachers for the preparation of GP reports. The GP reports will be collected within seven days from the date of distribution of the award lists.



ACHARYA N.G. RANGA AGRICULTURAL UNIVERSITY
FACULTY OF AGRICULTURAL ENGINEERING & TECHNOLOGY
B.TECH. (AGRICULTURAL ENGINEERING)
SEMESTER FINAL THEORY EXAMINATIONS
THIRD YEAR – FIRST SEMESTER 2025-26

PART – A

Course No. IDEN 321 Date 19/01/26

Name K.Nanny Signature of the Student K.Nanny

I.D No. BEG/2020-038

Code No. 007842

This paper contains (04) pages

ACHARYA N.G. RANGA AGRICULTURAL UNIVERSITY
FACULTY OF AGRICULTURAL ENGINEERING & TECHNOLOGY
B.TECH. (AGRICULTURAL ENGINEERING)
SEMESTER FINAL THEORY EXAMINATIONS
THIRD YEAR – FIRST SEMESTER 2025-26

PART – A

Course No. IDEN 321 Date 19/01/2026

Time : 30 minutes

Maximum marks: 40 (Forty)

Code No. 007842

Marks obtained 17

INSTRUCTIONS:

1. The student shall fill in all the blanks concerned except Code No. on the first page of the **Part A& B.**
2. The student should not make any identification mark like Name / I.D No./ Gods' Name / indication of any religious matter / or any scribbling or clue anywhere on the answer scripts of either **Part A or B** including the first pages. In the event of any such cases, the answer scripts of such student shall become invalid and zero marks shall be awarded, besides other punishment as applicable.
3. The student shall ensure that the **Part A** paper contains the No. of pages in serial as indicated in the first page of the **PART 'A' PAPER.**

Fig. 56. Coding of the answer scripts after jumbling



This paper contains 04 pages
Please verify

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FACULTY OF AGRICULTURAL ENGINEERING & TECHNOLOGY
B.TECH. (AGRICULTURAL ENGINEERING)
SEMESTER FINAL THEORY EXAMINATIONS
THIRD YEAR – FIRST SEMESTER 2025-26

IDEN 331 SPRINKLER AND MICROIRRIGATION SYSTEMS Marks: 40 Time: 30 min

PART – A

I. Underline the correct answer 30 × 0.5 = 15M

1. Micro-irrigation is also called
   a. Nano-irrigation b. Petite irrigation
   c. Localized irrigation d. Flood irrigation
2. In solid set sprinkler irrigation system, laterals are positioned during
   a. Early of crop season b. Mid of crop season

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SEMESTER FINAL THEORY EXAMINATIONS
THIRD YEAR – FIRST SEMESTER 2025-26

PART – A

Course No. IDEN - 331 Date : 19/01/2026
Time : 30 minutes
Maximum marks: 40 (Forty)

Code No. 007041

Marks obtained 17

INSTRUCTIONS:

1. The student shall fill in all the blanks concerned except Code No. on the first page of the Part A& B.
2. The student should not make any identification mark like Name / I.D.No./ Gods' Name / indication of any religious matter / or any scribbling or clue anywhere on the answer scripts of either Part A or B including the first pages. In the event of any such cases, the answer scripts of such student shall become invalid and zero marks shall be awarded, besides other punishment as applicable.
3. The student shall ensure that the Part A paper contains the No. of pages in serial as indicated in the first page of the PART 'A' PAPER.

ACHARYA N.G. RANGA AGRICULTURAL UNIVERSITY
FACULTY OF AGRICULTURAL ENGINEERING & TECHNOLOGY
B.TECH. (AGRICULTURAL ENGINEERING)
SEMESTER FINAL THEORY EXAMINATIONS
THIRD YEAR – FIRST SEMESTER 2025-26

PART – A

Course No. IDEN - 331 Date: 19/01/2026
Name : A. Hemalatha Signature of the Student: A. Hemalatha
I.D.No. MED/2021-031
Code No. 007041
This paper contains (04) pages

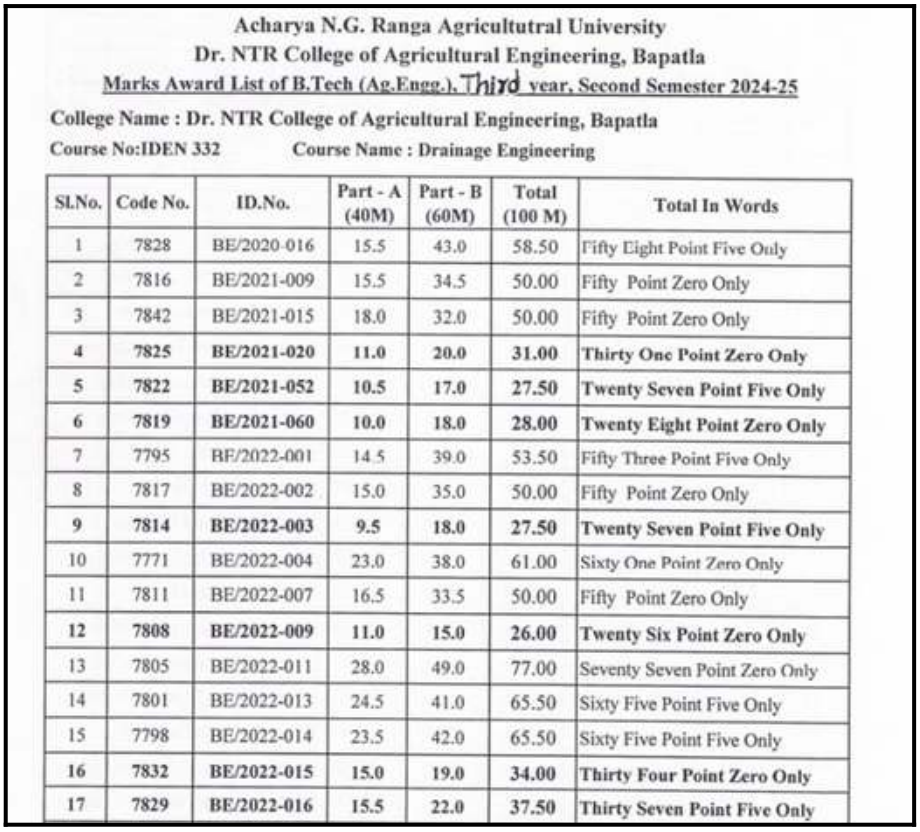
Fig. 57. Separation of the coded slip having students' details

Acharya N.G. Ranga Agricultutral University
Dr. NTR College of Agricultural Engineering, Bapatla
Marks Award List of B.Tech (Ag.Engg.), Third year Second Semester 2024-25

College Name : Dr. NTR College of Agricultural Engineering, Bapatla
Course No:IDEN 332 Course Name : Drainage Engineering

Sl.No.	Code No.	ID.No.	Part - A (40M)	Part - B (60M)	Total (100 M)	Total In Words
1	7697		17.0	40.5	57.50	Fifty Seven Point Five Only
2	7698		19.5	38.0	57.50	Fifty Seven Point Five Only
3	7699		22.5	49.0	71.50	Seventy One Point Five Only
4	7700		20.5	45.0	65.50	Sixty Five Point Five Only
5	7701		23.5	53.0	76.50	Seventy Six Point Five Only
6	7702		29.0	55.0	84.00	Eighty Four Point Zero Only
7	7703		31.0	58.0	89.00	Eighty Nine Point Zero Only
8	7704		25.5	39.0	64.50	Sixty Four Point Five Only
9	7705		21.0	44.0	65.00	Sixty Five Point Zero Only
10	7706		33.0	53.0	86.00	Eighty Six Point Zero Only
11	7707		11.5	38.5	50.00	Fifty Point Zero Only
12	7708		28.0	59.0	87.00	Eighty Seven Point Zero Only
13	7709		21.0	43.0	64.00	Sixty Four Point Zero Only
14	7710		25.5	43.0	68.50	Sixty Eight Point Five Only
15	7711		20.5	41.0	61.50	Sixty One Point Five Only
16	7712		19.0	40.0	59.00	Fifty Nine Point Zero Only

Fig. 58. Marks obtained list submitted by the evaluator with Codes



Acharya N.G. Ranga Agricultutral University
Dr. NTR College of Agricultural Engineering, Bapatla
Marks Award List of B.Tech (Ag.Engg.), Third year, Second Semester 2024-25

College Name : Dr. NTR College of Agricultural Engineering, Bapatla
Course No:IDEN 332 Course Name : Drainage Engineering

Sl.No.	Code No.	ID.No.	Part - A (40M)	Part - B (60M)	Total (100 M)	Total In Words
1	7828	BE/2020-016	15.5	43.0	58.50	Fifty Eight Point Five Only
2	7816	BE/2021-009	15.5	34.5	50.00	Fifty Point Zero Only
3	7842	BE/2021-015	18.0	32.0	50.00	Fifty Point Zero Only
4	7825	BE/2021-020	11.0	20.0	31.00	Thirty One Point Zero Only
5	7822	BE/2021-052	10.5	17.0	27.50	Twenty Seven Point Five Only
6	7819	BE/2021-060	10.0	18.0	28.00	Twenty Eight Point Zero Only
7	7795	BE/2022-001	14.5	39.0	53.50	Fifty Three Point Five Only
8	7817	BE/2022-002	15.0	35.0	50.00	Fifty Point Zero Only
9	7814	BE/2022-003	9.5	18.0	27.50	Twenty Seven Point Five Only
10	7771	BE/2022-004	23.0	38.0	61.00	Sixty One Point Zero Only
11	7811	BE/2022-007	16.5	33.5	50.00	Fifty Point Zero Only
12	7808	BE/2022-009	11.0	15.0	26.00	Twenty Six Point Zero Only
13	7805	BE/2022-011	28.0	49.0	77.00	Seventy Seven Point Zero Only
14	7801	BE/2022-013	24.5	41.0	65.50	Sixty Five Point Five Only
15	7798	BE/2022-014	23.5	42.0	65.50	Sixty Five Point Five Only
16	7832	BE/2022-015	15.0	19.0	34.00	Thirty Four Point Zero Only
17	7829	BE/2022-016	15.5	22.0	37.50	Thirty Seven Point Five Only

Fig. 59. Decoded Award list with student details

After the completion of the preparation of GP reports, all course in-charges shall compile and verify the results carefully to ensure accuracy. The results shall include detailed information such as the grade points obtained by each student and their status (Pass/Fail), which shall be clearly indicated in the remarks column. The verified results shall then be displayed on the designated academic notice board for the information of students. Care shall be taken to ensure that the displayed results are clear, complete, and easily understandable. Course in-charges shall also ensure that confidentiality and data accuracy are maintained during the preparation and display of results. Any discrepancies, if identified by students, shall be promptly addressed and rectified following due verification. A copy of the finalized results shall be submitted to the OAM office for official record and further processing. The entire process of result preparation and display shall be completed within the stipulated timeline prescribed by the academic authorities.

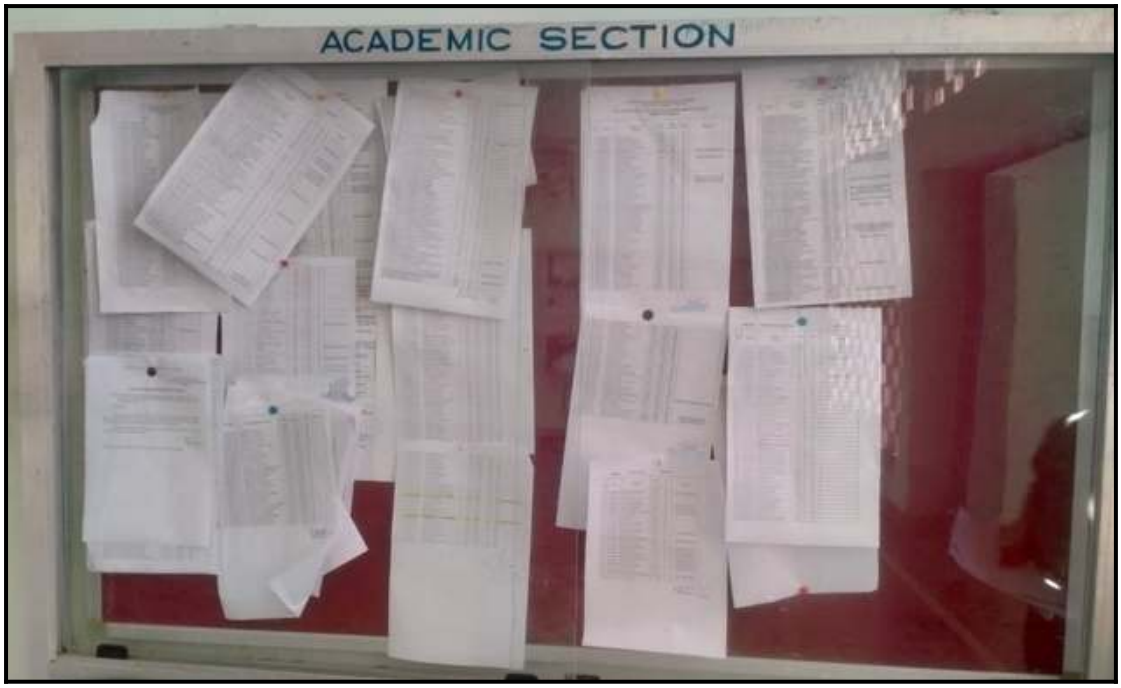


Fig. 60. Academic Notice Board Displaying Examination Results and Timetables

6.5.4.6. NCC/NSS/RVC Units

NCC was initiated for the first time in Dr. NTR College of Agricultural Engineering during the academic year 2025-26 on 24.10.2025. Students were enrolled as NCC cadets under 22 Andhra Battalion, Tenali Unit located at Tenali, Andhra Pradesh. Second year students of the college studying in the academic year 2024-25 has the opportunity in participating in the NCC selections as they have sufficient residence time to compete NCC-B and C certification trainings. 55 students showed interest in joining NCC and all were screened based on the initial screening, physical fitness test and interview.

18 students in which 10 boy cadets (Senior Division) and 8 girl cadets (Senior Wing) were finally enrolled as NCC cadets and additionally 4 students are under reserve during the First enrolment year i.e., 2025-26.



Fig. 61. Associate Dean, Dr. D. D. Smith Addressing the students before enrolment



Fig. 62. Lt. Colonel. Udaya Kumar, 22 (A) Tenali Unit addressing the students

Initially the students were screened based on the age, height and eye sight. Out of 54 students, 25 students were selected for the physical test. Initial screening includes the height of 152 cm and 160 cm for girls, and 157 cm to 168 cm for boys. Height may varies based on the additional credential that the student having, such as past NCC participation, involvement in extracurricular activities.



Fig. 63. Dr. D.D. Smith, Associate Dean Dr. NTR CAE, Bapatla inspecting the initial screening test



Fig. 64. Dr. D.D. Smith, Associate Dean Dr. NTR CAE, Bapatla, interacting with Lt. Colonel. Udaya Kumar 22(A) Tenali Unit

Physical test includes 1.2 km in 6 min, 25 pushups, 25 shuttle runs in 1 min. The screen students were further interviewed by the **Lt. Colonel. Udaya Kumar, 22 (A) Tenali Unit.**



Fig. 65. Students participated in running physical test



Fig. 66. Students participated in push ups physical test



Fig. 67. Students participated in push ups physical test



Fig. 68. Students participated in short running test



Fig. 69. Students inspection by 22(A) Tenali NCC staff for the physical test

Interview was conducted for all the selected students and willing ness to join and aim of joining the NCC unit was ascertained. All the 21 cadets shown their willingness. Out of 21 cadets 18 cadets were selected and 3 cadets were made reserve.



Fig. 70. Lt. Colonel. Udaya Kumar, 22 (A) Tenali Unit explaining the procedure and interviewing the cadets



Fig. 71. Associate Dean, Dr. D.D. Smith and Lt. Colonel. Udaya Kumar, 22 (A) Tenali Unit, examiners from NCC unit and cadets honouring the National and NCC Flags



Fig. 72. Associate Dean, and Lt. Colonel. Udaya Kumar, 22 (A) Tenali Unit, examiners from NCC unit with final Enrolled NCC Cadets

List of the cadets selected are from the second-year students for the year 2025-26 are as mentioned in the Table 20. 18 students from s.no.1 to 18 are enrolled as NCC cadets. Remaining 4 students from s.no.19 to 22 are in reserve. There is a change to register the reserve cadets if the regular cadets cancel the registration.

Table 20. List of the cadets selected for the NCC

S.NO.	REG.NO	NAME OF CADET	ID NO.
1	AP2025SDIA0570659	KOLLATI BALAJI	BE/2024 -001
2	AP2025SDIA0570658	MADAKA BHARATH KUMAR	BE/2024 -002
3	AP2025SDIA0570683	DONDAPATI SUSEEL RANA	BE/2024 -028
4	AP2025SDIA0570665	PANDI VENKATA SUBBAIAH	BE/2024 -032
5	AP2025SWIA0570649	BHAWNA CHOUDHARY	BE/2024 -036
6	AP2025SWIA0570650	KONDRA PRANAYA	BE/2024 -037
7	AP2025SWIA0570655	ARIGELA DEVI AMRUTHA	BED/2025-005
8	AP2025SDIA0570664	MIRAPAKAYALA GOPI CHANDU	BED/2025-008

9	AP2025SDIA0570661	SHAIK IMRAN	BED/2025-009
10	AP2025SWIA0570652	LAKE. LAVANYA	BED/2025-017
11	AP2025SWIA0570651	DUDDU NALINI	BED/2025-022
12	AP2025SWIA0570653	NELLI POOJA SRI	BED/2025-023
13	AP2025SDIA0570660	SIRRA PRINCE	BED/2025-024
14	AP2025SDIA0570654	ELLAPU SAI SUDHARSHINI	BED/2025-031
15	AP2025SDIA0570656	KOYYA SANDYA	BED/2025-033
16	AP2025SDIA0570663	CHILUMUNTHA SRINIVASA RAO	BED/2025-035
17	AP2025SDIA0570662	KUNDETI MANIKANTA	
18	AP2025SDIA0570663	AMAN BARWAD	
19	-	KAGEPOGU JESSI SAJEEVAN	RESERVE
20	-	A. RAHUL	RESERVE
21	-	M. VARA PRASAD REDDY	RESERVE
22	-	G. RUPASRI	RESERVE

All the above students with Reg. No. in the above table will receive a uniform allowance of Rs. 4650/- toward the purchase of the uniform and shoes for the regular drills and NCC camps. **Classes/ drill attended by Cadets:** Cadets are attending 2 classes per month by the PI s from the unit and the remaining classes were practised by the cadets



Fig. 73. Drill by the cadets



Fig. 74. NCC Cadets attending the theory class



Fig. 75. NCC Cadets attending the practical drill class

NCC cadets of 16 numbers (8 SD +8 SW) of Dr. NTR College of Agricultural Engineering Bapatla participated in simulator firing at Koneru Lakshmaiah (KL) University at Vaddeswaram, Guntur (Dst), Andhra Pradesh. Initially the cadets were introduced about the firing positions and handling of rifles.



Fig. 76. NCC Cadets at KL University for practise on simulator firing



Fig. 77. Cadets firing by Kneeling position



Fig. 78. Cadets firing by Lying position



Fig. 79. Cadets (SW) firing on the simulator by lying position

NSS Activities

National Service Scheme (NSS) is an Indian government-sponsored public service program conducted by the Department of Youth Affairs and Sports of the Government of India. The NSS programme is an integral part of course curriculum of B.Tech. (Agricultural Engineering). The NSS programme comprises of regular activities (120 hours) and Annual Special Camp (120 hours). The regular activities are being conducted on campus and off campus throughout the year. The Annual Special Camps are being conducted in a rural village for 7 days and the NSS volunteers must stay in the village during the special camp. All the NSS volunteers who have served NSS for 3 years and have performed 240 hours of work under NSS are being issued a certificate from the university with the signature of the Hon'ble Vice-Chancellor and the Programme Coordinator (Dean of Student Affairs) at University level.

Regular NSS activities like Swachh Bharat, plantation, social awareness, door to door surveys, rallies, etc. are conducted in and around Bapatla and in adopted villages. The details of NSS volunteers are shown in the below table. Two NSS units are allocated to this college and are in full functioning mode from first year of the course. NSS aimed at developing students' personality through community service was achieved at this college in undergraduate students. NSS special camp was organized every year at two places by two NSS units. The NSS volunteers actively participated in cleaning the premises of college campus every week as per the schedule. Further, the following activities are being conducted under NSS programme as furnished below. NSS volunteers learned many skills through NSS activities; some important are as follows:

- Enabled students to acquire specialized competence in communication skills
- Improved awareness on local crops related problems and needs of the farmers
- Acquired leadership qualities
- Students improved in attitude, behavioural aspects

Table 21. Details of NSS units during last five years

Sl. No.	Year	No. of NSS volunteers	No. of NSS units allotted	Status of function
1	2020-21	287	2 Units	Functioned
2	2021-22	285	2 Units	Functioned
3	2022-23	225	2 Units	Functioned
4	2023-24	250	2 Units	Functioned

5	2024-25	288	2 Units	Functioned
6	2025-26	285	2 Units	Functioned

Table 22. Category-wise No. of NSS volunteers

Sl. No.	Year	SC	ST	OBC	General	Total
1	2020-21	39	20	145	103	287
2	2021-22	39	26	150	70	285
3	2022-23	31	19	131	44	225
4	2023-24	47	24	122	57	250
5	2024-25	65	27	142	54	288
6	2025-26	62	25	158	40	285

Table 23. Details of NSS special camps

Sl. No.	Year	Place of NSS special camp held	No of NSS units participated	Period
1	2020-21	Ch.J.Palem (V), Bapatla (M), Guntur Dist.	2 Units	30-03-2021 to 31-03-2021 (2 days)
2	2021-22	Chintayapalem (V) & Thurpupalem (V), Karlapalem (M), Bapatla Dist.	2 Units	21-03-2022 to 27-03-2022 (7 days)
3	2022-23	--	--	--
5	2023-24	Kavuru (V) & Yeminenivaripalem (V), Cherukupalli (M), Bapatla Dist.	2 Units	09-03-2024 to 15-03-2024 (7 days)
5	2024-25	Returu (V) & Appapuram (V), Kakumanu (M), Guntur Dist.	2 Units	04-03-2025 to 10-03-2025 (7 days)
6	2025-26	Kothapalem (V) & Cherukuru (V), Parchur (M), Bapatla Dist.	2 Units	11-02-2026 to 17-02-2026 (7 days)

Table 24. Brief activities year-wise under NSS programme on campus and off-campus

2020-21

S.No.	Activity	Date
1.	International yoga day (online)	21-06-2020
2.	Andhra Pradesh state farmers day	08-07-2020
3.	Independence day	15-08-2020
4.	NSS foundation day and plantation of saplings	24-09-2020
5.	150th birth anniversary of Mahatma Gandhi	02-10-2020
6.	National women farmers day	15-10-2025
7.	National unity day	31-10-2025
8.	Andhra Pradesh state formation day	01-11-2020
9.	Indian Constitution day	26-11-2020

10.	Death anniversary of Dr. B.R. Ambedkar	06-12-2020
11.	Republic day	26-01-2021
12.	125th Birth Anniversary of Netaji Subhash Chandrabose	23-01-2021
13.	Anti-ragging awareness programme	04-03-2021
14.	World water day	22-03-2021
15.	NSS special camp	30-03-2021 to 31-03-2021

2021-22

1	Birth anniversary of Dr B.R. Ambedkar	14-04-2021
2	International day of yoga (online classes on yoga)	18-06-2021; 19-06-2021; 21-06-2021
3	Andhra Pradesh state farmers day	08-07-2021
4	Swatch Bharat (College campus cleaning)	22-08-2021
5	Covid vaccination	23-09-2021
6	NSS Foundation Day function	24-09-2021
7	Gandhi Jayanthi	02-10-2021
8	National Unity Day	31-10-2021
9	Indian Constitution day	26-11-2021
10	Agricultural Education Day	03-12-2021
11	Death Anniversary of Dr B.R. Ambedkar	06-12-2021
12	Republic day	26-01-2022
13	Sri Amara Jeevi Potti Srirumulu Jayanthi	23-03-2022
14	International Women's Day	08-03-2022
15	NSS special camp	21-03-2022 to 27-03-2022

2022-23

1	Birth anniversary of Dr. B.R. Ambedkar	14-04-2022
2	International Yoga Day	21-06-2022
3	Independence day	15-08-2022
4	NSS Foundation Day function and Blood donation camp	24-09-2022
5	Gandhi Jayanthi	02-10-2022
6	National Unity Day	31-10-2022
7	Indian Constitution day	26-11-2022
8	Death anniversary of Dr. B.R. Ambedkar	06-12-2022
9	Bamboo plantation	12-01-2023
10	Republic day	26-01-2023

2023-24

1	Birth anniversary of Dr. B R Ambedkar	14-04-2023
2	World Environment day	05-06-2023
3	Awareness programme on Global warming	17-06-2023
4	International Yoga Day	21-06-2023
5	Seminar on water conservation and its techniques	24-06-2023
6	Cleaning of local water bodies	26-06-2023
7	Seminar on Organic farming and its benefits	28-06-2023

8	Rally to create awareness about ecofriendly ways to reduce waste at home	28-06-2023
9	Meri Maati Mera Des programme (rally and plantation)	11-08-2023
10	National Sports Day	29-08-2023
11	NSS Foundation Day function and Blood donation camp	24-09-2023
12	National unity day	31-10-2023
13	Constitution day	26-11-2023
14	World water day	22-03-2024
15.	NSS special camp	09-03-2024 to 15-03-2024

2024-25

1	International Yoga Day	21-06-2024
2	Teak and forest tree saplings plantation under the programme on “Ek Ped Maa Ke Abhiyon”	08-07-2024; 13-09-2024; 18-10-2024
3	Awareness programme on ill effects of Tobacco under “ Tobacco Campaign 2.0”	31-08-2024
4	Campaign on “Swachhata Hi Sewa”	17.09.2024 to 01.10.2024
5	NSS Foundation Day	24-09-2024
6	Gandhi Jayanthi	02-10-2024
7	National Constitution Day	26-11-2024
8	International AIDS Day	01-12-2024
9	Republic Day celebrations	26-01-2025
10	Swachha Andhra Programme	15-02-2025; 15-03-2025
11	Rally on “Avoiding Plastic in Daily Life”	22-02-2025
12	Voters Awareness Programme in view of Graduate MLC elections	24-02-2025
13	NSS special camp	04-03-2025 to 10-03-2025

2025-26

1	Birth anniversary of Dr. B R Ambedkar	14-04-2023
2	Swachh Andhra Programme	19-04-2025; 17-05-2025; 21-06-2025; 19-07-2025, 23-08-2025; 27-09-2025; 18-10-2025, 15-11-2025; 21-12-2025
3	World Environment Day	05-06-2025
4	11th International Yoga Day	21-06-2025
5	Independence day	15-08-2025
6	57th NSS Foundation Day	24-09-2025
7	Blood donation camp	24-09-2025

8	Campus cleaning	22-11-2025; 29-11-2025; 03-01-2026; 28-02-2026; 28-03-2026
9	Death anniversary of Dr. B.R. Ambedkar	06-12-2025
10	Indian Constitution day	26-11-2025
11	National Agricultural Education Day – Rythanna Mikosam Programme	03-12-2025
12	Republic day	26-01-2026
13	NSS special camp	11-02-2026 to 17-02-2026

Some glimpses of NSS Activities during 2021 to 2026



Fig. 80. Blood Donation Camp



Fig. 81. International Yoga Day



Fig. 82. Plantation programme



Fig. 83. Rally on Swachhata Hi Sewa



Fig. 84. Campus cleaning



Fig. 85. Health camp under NSS SC



Fig. 86. World Environment Day



Fig. 87. Swachh Andhra Programme







Fig. 88. Demostration of latest agricultural machinery/tecnologies to farmers under NSS special camp during 2024-25 and 2025-26

6.5.4.7. Language Laboratory

The English Language Lab is proposed to be conducted in the same space as computer Lab is conducted with software like k-Van/Waldon/Globerena for communicative skills, writing skills, presentation skills and inter personal skills. However, right now this software is not available in College. In near future, the college will procure and improve the value of the institution and students also.

The computer Lab is facilitated in a well-lit, spacious and air conditioned with a floor space of 1800 sq .ft. (60 X 30 ft) and false roofing to reduce the cooling load. The Lab is provided with 30+ numbers of desktop systems connected with LAN. The ICT course is being conducted in this lab besides other courses of computers like C language, autocad, Ansys, etc.

6.5.4.8. Cultural Centre

The College has established a dedicated Cultural Centre to promote holistic student development through cultural, literary, artistic and creative activities. The Centre provides infrastructure and institutional support for students to explore their talents, preserve cultural heritage, develop leadership and communication skills, and participate in university, state and national-level cultural events. Student clubs operating under the Centre provide sustained opportunities for engagement throughout the academic year. The activities and achievements of the Centre are systematically documented as part of the institutional student-development initiatives. The Centre will facilitate regular cultural activities and provide opportunities for

students to participate in **music, dance, drama, literary activities, visual arts, traditional arts, photography, film-making and other creative pursuits**.

The Centre will also serve as a platform for preserving and promoting the rich **Indian, Telugu and regional cultural heritage**, while encouraging students to develop confidence, communication skills, teamwork, creativity and leadership abilities.

Objectives

The Cultural Centre will aim to:

- Provide a dedicated space for **cultural and creative activities**.
- Identify and nurture the talents of students in various art forms.
- Promote **Indian, Telugu and regional cultural heritage**.
- Encourage participation in inter-collegiate, university and national-level cultural competitions.
- Develop students' **communication, confidence, teamwork and leadership skills**.
- Organize cultural festivals, literary events, exhibitions and competitions.
- Provide opportunities for students to learn traditional and contemporary art forms.
- Encourage faculty–student interaction through cultural activities.
- Promote inclusiveness and participation of students from diverse backgrounds.
- Contribute to the **holistic development and personality development of students**.

Governance Structure

A **Cultural Centre Committee** may be constituted with:

Chairperson: Associate Dean

Coordinator: Officer In-Charge of Student Affairs, Physical Director

Members: Faculty representatives and student representatives

Proposed Activities

Performing Arts

- Classical and folk dance
- Vocal and instrumental music
- Theatre and drama
- Mime and skits

- Folk arts
- Cultural performances

Literary Activities

- Essay writing
- Poetry
- Elocution
- Debate
- Quiz
- Creative writing
- Telugu literary activities

Visual & Creative Arts

- Painting
- Drawing
- Photography
- Poster making
- Rangoli
- Sculpture and handicrafts
- Digital art

Cultural Events

- Annual cultural festival
- Traditional day
- Telugu language and cultural programmes
- National and international day celebrations
- Heritage exhibitions
- Student talent shows
- Inter-department cultural competitions
- University-level cultural competitions






WORLD WATER DAY







6.5.4.9. Personality Development

Considerable efforts are being invested every year for the personality development of the students. The following is the table showing the details of the lectures conducted to improve the personality and communication. A three day on campus 'Personality Development & Soft Skills Training' was conducted at Dr. N.T.R. College of Agricultural Engineering, Bapatla during 29th - 31st August 2023. A total of 73 students were selected. Radha Shankarnarayanan Smart Series, Bangalore acted as resource persons for the programme. The day wise sessions in the three-day programme were as given below:

1. Goals -Set and achieve (Day-1)
2. Time Management (Day-1)
3. Listening Skills/ Communication Skills (Day-2)
4. Presentation Skills (Day-2)
5. Effective Teamwork (Day-3)
6. Problem Solving and Decision Making (Day-3)

A three day on campus 'Personality Development & Soft Skills Training' was conducted at Dr. N.T.R. College of Agricultural Engineering, Bapatla during 27-29, June 2022. A total of 72 students were selected. Mr Kushal Raut, Director - communicare Training and content Solutions and Mr Sanjeev Rane acted as resource person for the programme.

A three day on campus 'Personality Development & Soft Skills Training' was conducted at Dr Ntr College of Agricultural Engineering, Bapatla during 4th, 6th and 7th November 2023. A total of 75 students were participated in the training program. Mr Sanjeev

Rane, Director - communicare Training and content Solutions and Mr. Utkarsh acted as resource persons for the programme. A three day on campus ‘Personality Development & Soft Skills Training’ was conducted at Dr NTR College of Agricultural Engineering, Bapatla during 28, 30 and 31 January 2023. A total of 81 students were attended. Mrs. Radha Shankarnarayanan Smart Series, Bangalore acted as resource persons for the programme.

Dr. N.T.R. College of Agricultural Engineering, Bapatla, gives significant importance to the overall personality development of students in addition to academic and technical education. Personality development programmes are conducted regularly with the objective of enhancing students’ confidence, communication skills, leadership qualities, professional ethics, interpersonal relationships, and employability skills required for successful careers and responsible citizenship. The College organizes personality development classes, soft skill training programmes, motivational sessions, career guidance lectures, and interactive workshops for undergraduate, postgraduate, and doctoral students. These programmes are conducted by experienced faculty members, subject experts, industry professionals, psychologists, trainers, alumni, and invited resource persons from reputed organizations and institutions. The personality development sessions focus on various aspects such as:

- Communication and presentation skills
- Public speaking and stage confidence
- Leadership and team-building qualities
- Time management and goal setting
- Professional ethics and workplace behaviour
- Stress management and emotional well-being
- Interview skills and group discussion techniques
- Resume preparation and career planning
- Positive attitude and self-motivation
- Decision-making and problem-solving abilities

Students are also encouraged to participate in seminars, debates, group discussions, technical presentations, cultural activities, NSS, NCC, sports, and extension programmes to improve their confidence, leadership abilities, and social interaction skills. Training is also provided for competitive examinations, placement interviews, and entrepreneurship development to prepare students for professional careers and higher education opportunities. Special programmes on yoga, meditation, self-discipline, mental wellness, women

empowerment, cyber awareness, and self-defense are also organized periodically to promote physical fitness, mental strength, and personal safety among students. The College believes that personality development is an essential component of higher education and plays a vital role in shaping students into competent professionals and responsible individuals. These initiatives help students improve their self-confidence, communication ability, professional readiness, and overall personality, thereby enhancing their employability and contribution to society.

Few glimpses of Students personality development activities



Fig. 89. Students' personality development trainings

6.5.5. Physical facilities

6.5.5.1. Hostels

Boys Hostel Facilities

The boy students of Dr. N.T.R. College of Agricultural Engineering, Bapatla, are accommodated in two hostels namely, **Dr. K.L. Rao Hall of Residence** and the **New Boys Hostel**. Together, these hostels provide a safe, comfortable, and student-friendly residential environment that supports the academic and personal well-being of undergraduate,

postgraduate, and doctoral students. The two hostels collectively consist of 59 rooms with an accommodation capacity of approximately 177 students. At present, around 165 students belonging to UG, PG, and Ph.D. programmes are residing in the hostels. Most of the hostel rooms are spacious enough to comfortably accommodate four students. Each student is provided with an individual cot along with essential furniture such as cupboards, reading tables, and chairs to facilitate comfortable living and academic activities. The boys' hostels are equipped with adequate dining and kitchen facilities to ensure proper boarding arrangements for students. The hostel mess contains large dining tables, sufficient seating arrangements, purified drinking water facilities, refrigerator, freezer, ceiling fans, and an LED television with a sound system for student recreation. An RO water purification plant and water coolers are also installed to provide safe and hygienic drinking water.

To support academic activities and digital learning, Wi-Fi connectivity is provided in the hostels, enabling students to access online learning resources, academic materials, and digital platforms. Reading rooms are also available within the hostel premises, where English and Telugu newspapers, magazines, and other reading materials are provided regularly for the benefit of students. The hostels are equipped with sufficient bathrooms and toilet facilities, which are maintained hygienically on a regular basis. Special attention has been given to accessibility and inclusiveness by providing disabled-friendly restrooms and ramp facilities for differently abled students in the New Boys Hostel. For the welfare and convenience of students, additional facilities such as guest rooms for visiting parents and sick rooms for isolation and emergency medical care are also available within the hostel premises. First-aid kits are maintained in both hostels to provide immediate medical assistance whenever required.

The College also encourages physical fitness, sports, and recreation among students. A gymnasium with essential exercise equipment is available in the boys' hostel. Facilities for indoor games such as table tennis, chess, and carroms are provided for recreation and relaxation. Outdoor sports facilities including basketball, volleyball, badminton, cricket, and football grounds with running tracks are available adjacent to the hostels for regular sports and physical activities. To ensure the safety and security of students, CCTV cameras have been installed at strategic locations inside and outside the hostel premises. Continuous monitoring is carried out to regulate the movement of students and outsiders and to maintain discipline and security within the hostels. A senior male faculty member is designated as the

Hostel Warden to supervise the routine administration, boarding and lodging arrangements, discipline, welfare, and overall well-being of the hostel students. The Warden closely monitors hostel activities and provides guidance and support to students whenever necessary. Overall, the boys' hostels at Dr. N.T.R. College of Agricultural Engineering provide adequate residential, academic, recreational, and safety facilities, thereby creating a comfortable and conducive environment for student life and learning.



Fig. 90. KL Rao Hall of residence (Old Boys hostel), Dr. NTR CAE, Bapatla



Fig. 91. New Boys hostel, Dr. NTR CAE, Bapatla

Girls Hostels

Dr. N.T.R. College of Agricultural Engineering, Bapatla, provides hostel accommodation facilities for girl students of B.Tech. (Agricultural Engineering), M.Tech., Ph.D., and PG Diploma programmes within the Girls Hostel campus. The hostel environment is designed to ensure safety, comfort, discipline, and a conducive atmosphere for academic learning and personal well-being. At present, nearly 160 girl students are accommodated in

the Girls Hostel, which consists of five buildings, namely the Main Building (Ground Floor and First Floor), NABARD Hostel Block, and three Old Staff Quarters. The hostel accommodation is arranged systematically to ensure convenience, security, and proper supervision of students belonging to different academic programmes and years of study. The third-year B.Tech. students are accommodated on the ground floor of the main hostel block, while second-year B.Tech. students and M.Tech. second-year students are accommodated on the first floor. Final-year students are housed in the three old staff quarters consisting of 13 rooms. First-year B.Tech. students of the 2025–26 admitted batch, along with M.Tech., PG Diploma, and Ph.D. students, are accommodated separately in the NABARD hostel block to ensure a comfortable and secure residential environment for newly admitted students and research scholars.

The College ensures continuous monitoring and welfare support for hostel residents through regular visits by the Warden, Additional Warden, and Chief Warden. Girl students are permitted weekly outings for a limited duration as per hostel regulations to ensure both student welfare and discipline. The students are provided with hygienic and nutritious food prepared under proper supervision. The hostel mess is managed efficiently with active involvement of student representatives, and the menu is revised periodically based on student requirements and feedback. The entire Girls Hostel complex is secured with compound walls, gates, LED lighting systems, CCTV surveillance cameras, and night security personnel to ensure the safety and security of students. The hostel administration is supported by caretakers, housekeeping staff, mess workers, gardeners, and security personnel for smooth functioning and maintenance of the hostel facilities. Caretakers are available during daytime hours for monitoring and assistance, while dedicated women staff members are deployed during night hours for hostel security and student safety.

Most hostel rooms are spacious and comfortably accommodate four students. Each room is provided with individual cots, cupboards, reading tables, chairs, ceiling fans, and LED lighting facilities to support both comfortable living and academic activities. The Girls Hostel complex is equipped with adequate dining and kitchen facilities. The main mess hall contains stainless steel dining tables with sufficient seating capacity, while an additional mini dining hall is also available. The mess facilities include a water cooler, refrigerator, roti-making machine, dosa preparation unit, mixer grinder, hot water facilities for cleaning utensils, water dispensers, inverter backup, UPS system, fire extinguisher, and vacuum

cleaner for efficient kitchen and dining management. To support digital learning and academic access, Wi-Fi connectivity is provided throughout the Girls Hostel complex. The hostels also contain sufficient bathrooms and toilets, which are maintained hygienically on a regular basis. Hot water facilities are provided through solar water heaters and electric geysers. Additional facilities such as washing areas, cloth drying spaces, and washing machines are also provided for the convenience of students. Ramp facilities are available to ensure accessibility for differently abled students.

The hostel also includes a reading room where English and Telugu newspapers, magazines, and other reading materials are made available regularly. For hygiene and welfare purposes, electrical sanitary napkin disposal units are installed in the hostel premises. To encourage physical fitness and recreation, an open gymnasium with essential exercise equipment is available within the hostel campus. Indoor games facilities such as table tennis, chess, and carroms are also provided. Outdoor sports facilities including courts for shuttle badminton, ball badminton, Tennikoit, and volleyball are available within the hostel premises for recreational and sports activities. CCTV cameras are installed at strategic locations inside and outside the hostel buildings for continuous monitoring of student movement and campus security. First-aid kits are also maintained in the hostel to provide immediate medical assistance whenever necessary. A senior female faculty member is designated as the Hostel Warden to supervise the routine administration, boarding, lodging, discipline, and welfare of the girl students. An Additional Warden also assists in hostel management and student support activities whenever required. Overall, the Girls Hostel facilities at Dr. N.T.R. College of Agricultural Engineering provide a secure, hygienic, comfortable, and academically supportive residential environment for girl students pursuing higher education in agricultural engineering.



Fig. 92. NABARD Building



Fig. 93. Staff quarters block



Fig. 94. Main Girls Hostel Building



Fig. 95. Water Cooler



Fig. 96. Hot water geyser



Fig. 97. Clothes washing machine



Fig. 98. Sanitary napkin destroyer



Fig. 99. Open gym for girl students



Fig. 100. TT board at girls' hostel



Fig. 101. Volleyball court at girls' hostel



Fig. 102. Shuttle badminton court



Fig. 103. Tennikoit court at girls' hostel



Fig. 104. Solar water heating system



Fig. 105. Ball badminton court



Fig. 106. Associate Dean interaction with first year girl students in Mini mess hall



Fig. 107. Associate Dean interaction with second year girl students in Mess Hall

6.5.5.2. Examination Hall

Dr. N.T.R. College of Agricultural Engineering, Bapatla, is equipped with two well-furnished and adequately equipped examination halls for the smooth conduct of mid-semester and final theory examinations. These halls are designed to provide a disciplined, comfortable, and transparent examination environment in accordance with university norms and examination regulations. The first examination hall has a seating capacity of approximately 70 students and measures about 8.9 meters in width and 21.24 meters in depth. The hall is also provided with a stage measuring 5.7 meters in width (including steps) and 2.5 meters in depth. The stage facilitates effective supervision and monitoring of students during examinations by invigilators and examination staff. The hall is well-ventilated and adequately illuminated to ensure a comfortable environment for students during examinations. It is equipped with 17 ceiling fans, one table fan, 11 windows, and 25 lighting fixtures to provide proper air circulation and sufficient lighting throughout the examination period. The

examination hall has three main doors to facilitate smooth entry and exit of students and examination personnel.

To ensure transparency, security, and proper monitoring during examinations, the hall is equipped with four CCTV cameras installed at strategic locations. The CCTV surveillance system continuously monitors the activities of students, invigilators, and examination staff during the conduct of examinations. The cameras are also utilized during the packing and sealing of answer scripts to maintain confidentiality and prevent any possibility of mishandling or malpractice. Additional facilities provided in the examination hall include a prominently placed wall clock for clear visibility of time to students and invigilators, and an electrical ringing bell system used for issuing time alerts and examination warnings. The hall also displays a flex banner containing examination rules, regulations, and guidelines related to unfair means and disciplinary procedures. The banner, measuring approximately 4 × 5 feet, serves as a constant reminder to students regarding examination discipline and ethical conduct. Overall, the examination hall facilities available at Dr. N.T.R. College of Agricultural Engineering ensure the fair, transparent, and efficient conduct of university examinations in a disciplined academic environment.

Examination Hall – II

In addition to the main examination hall, Dr. N.T.R. College of Agricultural Engineering, Bapatla, is provided with an additional examination hall-cum-classroom facility for conducting examinations and academic activities efficiently. The hall has a seating capacity of approximately 50 students and is utilized for mid-semester examinations, practical examinations, class tests, and regular instructional purposes whenever required. The hall measures approximately 5.8 meters in width and 17.1 meters in depth and is designed to provide a comfortable and disciplined examination environment. It is well-ventilated and adequately illuminated with 13 windows, 12 ceiling fans, and 14 lighting fixtures to ensure proper air circulation and sufficient lighting during examinations and classroom sessions. The hall is provided with a main entrance and exit door for smooth movement of students and invigilation staff. To maintain transparency and discipline during examinations, the hall is equipped with two CCTV cameras installed at strategic locations for continuous monitoring of student and staff activities. The surveillance system helps ensure fair conduct of examinations and adherence to examination regulations.

A flex banner displaying examination rules, regulations, and guidelines related to unfair means and disciplinary procedures is prominently installed in the hall. The banner, measuring approximately 4 × 5 feet, serves as an important source of awareness for students regarding ethical examination practices and institutional examination policies. The available facilities in the additional examination hall-cum-classroom support the effective and smooth conduct of examinations and academic activities while maintaining proper discipline, transparency, and student comfort.

Examination Question Paper Handling and Evaluation Procedures

Dr. N.T.R. College of Agricultural Engineering, Bapatla, follows a secure, transparent, and systematic procedure for the conduct of examinations and handling of confidential examination materials in accordance with university norms and examination guidelines. The question papers for both mid-semester and final theory examinations are transmitted to the official email ID of the Officer In-charge of Academic Matters (OAM) approximately one hour before the commencement of the examination. To ensure confidentiality and security, the question papers are password protected. The password required to access the question papers is communicated separately to the Associate Dean through a secure channel. Printing and duplication of question papers are carried out exclusively in the designated Examination Printing Cell under strict confidentiality. The printing process is continuously monitored through CCTV surveillance using two strategically installed cameras. The entire process is conducted in the presence of the OAM and other authorized staff members deputed by the Associate Dean. A separate register is maintained in the Examination Printing Cell to record all relevant details related to the receipt, printing, handling, and distribution of question papers.

To maintain transparency and accuracy, the sealed question paper bundle is opened only five minutes before the commencement of the examination in the presence of the Course In-charge, Head of the Department, and OAM. This procedure enables the concerned Course In-charge to verify the correctness and completeness of the question paper before distribution to students. After completion of the examination, the answer scripts are carefully packed in

secure covers, duly sealed and signed by the invigilators in the presence of CCTV surveillance cameras. This procedure ensures transparency, accountability, and protection against any possibility of mishandling or tampering of answer scripts during collection and submission. The College also provides examination support facilities for differently abled students in accordance with University and Government guidelines. Students having 60 percent or more physical disability are permitted to avail the facility of a scribe upon submission of a formal request and approval by the competent authority. Through these systematic procedures and monitoring mechanisms, the College ensures the fair, confidential, and transparent conduct of examinations while maintaining high standards of academic integrity and accountability.



Fig. 108. Main Entrance of the Examination Hall – I



Fig. 109. A flexi shows displaying rules and regulations and location of CCTV Camera



Fig. 110. Students Appearing for the SFT Examination in the Examination Hall



Fig. 111. Main entrance of Examination Hall – II



Fig. 112. A flexi shows displaying rules and regulations and location of CCTV Camera



Fig. 113. Full view of Examination Hall – II

6.5.5.3. Sports and Recreation Facilities

Dr. NTR College of Agricultural Engineering, Bapatla creates various facilities for the students for their physical and mental health. The college is providing sports and games, gymnasium, health clinic facilities which are presented broadly as below. The college is having a playground of area 90 x 150 m and with a running track of 350 m. A well compacted cricket pitch for laying the cricket mat provision is also available.



Fig. 114. Running track



Fig. 115. Ground facility



Fig. 116. Cricket ground

College is having two spacious volleyball courts adjacent to the boy's hostel and college ground. Ground boy is appointed to take care of regular maintenance of the volleyball and other courts for both girls and boys. Girl's hostel is also provided with one volleyball court. From the recent permission from the university, the volleyball court was elevated and compacted to avoid water logging. The space above the compound walls of the girl's hostel was closed with the green mat to restrict the movement of the balls. Apart from the volleyball courts, the Girls hostel is also having the ball badminton court, outdoor shuttle court, tennikoit court, table tennis, carroms and outdoor gymnasia facility.



Fig. 117. Volleyball court – I



Fig. 118. Volleyball court – II

College ground is also provided with two ball badminton courts, one adjacent to the 1st year boys students hostel and another in the main ground. On the other side of the spacious ground, football posts are installed to play the football. Repairs were carried out for the basketball ground, which was established at the adjacent to the main ground.



Fig. 119. Ball Badminton court



Fig. 120. Football court



Fig. 121. Basketball court

Carroms and table tennis boards are provided both in new boy's hostel and KL Rao Hall of residence. The facility is utilized by all the students such as UG, PG and PhD of the college.



Fig. 122. Carron board facility



Fig. 123. Table tennis facility

Gymnasium Facility:

The college is provided with two indoor and outdoor gymnasiums for boys and one outdoor gym for girl students. Indoor gym is facilitated with power rack, squad rack, bench press, dumbbell set, pushup bars, weight plates, abdominal bench, butterfly and leg curl equipment. Student representatives are made involved in facilitating the needs of the gym. Open gymnasium is located in between the ambient plant space and consists of tipple twister, air walker, sky walker, chest press, sit-up board double and push bar. Similar equipment is also provided for girl students also.



Fig. 124. In door gymnasium



Fig. 125. Open gymnasium



Fig. 126. Sports and games kit

6.5.5.4. Auditorium

Dr. N.T.R. College of Agricultural Engineering, Bapatla, is provided with an open auditorium facility within the campus to conduct various academic, cultural, extracurricular, and student welfare activities. The open auditorium serves as a common platform for organizing student gatherings, seminars, awareness programmes, extension activities, sports-related events, cultural celebrations, orientation programmes, and other institutional functions. The auditorium is designed in an open-air structure with a raised stage and seating space that facilitates the conduct of events involving large student participation. The open environment provides adequate space and flexibility for conducting both formal and informal programmes in a comfortable manner. The stage area is utilized for delivering lectures, student presentations, cultural performances, competitions, and public address programmes during college events and celebrations. The open auditorium is regularly used for organizing student induction programmes, International Women's Day celebrations, anti-ragging awareness programmes, yoga sessions, NSS and NCC activities, technical events, guest lectures, and extension education programmes. It also serves as a venue for interaction between students, faculty members, invited experts, alumni, and officials from government and non-government organizations.

The facility contributes significantly to the overall personality development of students by encouraging participation in co-curricular and extracurricular activities,

leadership development, teamwork, communication skills, and cultural engagement. The open auditorium also supports the College's efforts in promoting community interaction, awareness campaigns, and student-centred activities in a spacious and accessible environment. Overall, the open auditorium serves as an important infrastructure facility of the College, providing a suitable venue for academic enrichment, student engagement, and institutional activities in a disciplined and vibrant campus atmosphere.



Fig. 127. Open auditorium

Reading room facility

Dr. N.T.R. College of Agricultural Engineering, Bapatla, provides dedicated reading room facilities to support the academic, competitive examination, and self-learning needs of students. The reading rooms are designed to offer a peaceful, comfortable, and study-oriented environment that encourages independent learning, concentration, and academic enrichment. Separate reading room facilities are available in the College and hostel premises for the convenience of undergraduate, postgraduate, and doctoral students. These reading rooms are well-ventilated, adequately illuminated, and furnished with sufficient seating arrangements, reading tables, chairs, ceiling fans, and uninterrupted power supply to ensure a comfortable study atmosphere. The reading rooms are regularly supplied with English and Telugu daily newspapers, competitive examination magazines, technical magazines, and other informative publications to enhance the general awareness, communication skills, and current affairs

knowledge of students. Students utilize these facilities for regular study, preparation for competitive examinations, assignments, seminars, project work, and academic discussions.

Wi-Fi connectivity and access to digital learning resources further support students in utilizing online academic materials, e-books, journals, and educational platforms for self-learning and research purposes. In addition, television facilities are also provided in the reading room/hostel common areas for student recreation and entertainment. The television facility enables students to view educational programmes, agricultural news, current affairs, sports events, and other informative channels, thereby contributing to both knowledge enhancement and relaxation during leisure hours. The quiet and disciplined environment maintained in the reading rooms enables students to effectively utilize their study hours beyond regular classroom schedules. The reading room facilities play an important role in promoting reading habits, self-study culture, academic discipline, and continuous learning among students. Overall, these facilities contribute significantly towards improving academic performance, intellectual development, personality enhancement, and professional preparedness of the students of Dr. N.T.R. College of Agricultural Engineering, Bapatla.



Fig. 128. Reading room in hostel



Fig. 129. Television for entertainment facility

Health Clinic:

Dr. N.T.R. College of Agricultural Engineering, Bapatla, provides health care facilities within the campus through a dedicated Health Clinic established for the welfare of students, faculty, and staff members. The clinic functions as a primary healthcare support centre to address routine medical needs, provide first aid, and handle emergency health situations within the College premises. A qualified Medical Doctor is appointed on a contractual basis to provide medical consultation and treatment during scheduled college hours and whenever emergency situations arise. The health clinic ensures timely medical attention and basic healthcare services for students and staff members residing within the campus. To support medical services efficiently, both male and female staff nurses are available at the clinic for monitoring the health condition of students, administering prescribed medicines and injections, and assisting the doctor during consultations and emergencies. The clinic is also provided with bed facilities for observation, first aid, and temporary care of patients requiring rest and monitoring.

A systematic record maintenance mechanism is followed in the clinic. Details of medical consultations, prescriptions issued, and medicines dispensed are entered in a health register for future reference and monitoring of the health condition of individual students during subsequent visits. The Health Clinic is adequately equipped with essential medicines, emergency drugs, disposables, injections, digital and analog health monitoring instruments, oxygen cylinders, stretchers, wheelchairs, and refrigeration facilities for safe storage of

injectable medicines and medical supplies. These facilities enable the clinic to effectively manage common illnesses, minor injuries, and emergency medical situations within the campus. The expenditure for medical facilities, medicines, and healthcare equipment is supported through ICAR Student Amenities funds and NUF (Non-University Fund) resources of the College. The College continuously strives to strengthen the healthcare support system to ensure the safety, well-being, and healthy academic environment for all students and staff members.



Fig. 130. Bed facility in the health clinic

Dr. N.T.R. College of Agricultural Engineering, Bapatla, provides adequate sports infrastructure and physical education facilities to promote physical fitness, recreation, discipline, and overall personality development among students. The College encourages active participation of students in indoor games, outdoor sports, intercollegiate competitions, and extracurricular activities. The Physical Director's room serves as the central facility for the storage, maintenance, and distribution of sports and games equipment. All sports materials and equipment are systematically maintained, and separate issue registers are kept for recording the issue and return of sports materials to students. Student sports representatives actively assist in maintaining the equipment in safe and usable condition and play an important role in the effective management of sports activities within the College. Student representatives are also involved in the procurement process of sports equipment to

ensure that the facilities meet the requirements and interests of students. Sports materials and equipment are generally purchased twice in an academic year, particularly during College Day celebrations and intercollegiate sports events. The expenditure for sports equipment and related activities is met through the Non-University Fund (NUF) resources of the College.

In addition to the sports facilities available for all students on the campus, special attention has been given to providing sports and fitness facilities for girl students within the Girls Hostel premises. Adequate safety and security measures, including CCTV surveillance, watch and ward arrangements, and continuous monitoring, are maintained around the girls' hostel to ensure a secure environment for students participating in sports and recreational activities. An open gymnasium facility is also available within the Girls Hostel premises to encourage regular physical exercise and fitness among girl students. The open gym is equipped with various fitness equipment such as a twister, air walker, sky walker, chest press, sit-up board, double bar, and push-up bar to support physical well-being and healthy lifestyle practices. Similar fitness facilities are also available for other students within the campus.

Outdoor recreational facilities such as Tennikoit courts are provided within the hostel premises for the convenience and recreation of girl students. These facilities encourage active participation in sports, improve physical fitness, and contribute to the mental well-being and overall development of students. Overall, the College provides a supportive environment for sports, fitness, and recreational activities, thereby promoting a healthy and balanced student life along with academic excellence.



Fig. 131. Outdoor gym facility for girl's students



Fig. 132. Tennicoit court for students in hostel premises

Table tennis provides a dedicated indoor space which is highly beneficial for students, especially during extreme weather conditions. The facility includes standard table tennis tables with proper lighting, quality rackets and balls, safe and well-ventilated playing area. An outdoor badminton court exclusively or preferentially accessible to girl students encourages active participation in sports. The facility includes properly marked court with good surface levelling, availability of shuttlecocks and rackets, safe and secure environment. Badminton enhances agility, stamina, and teamwork, and serves as an excellent stress-reliever.



Fig. 133. Table tennis board facilitated in girls' hostel



Fig. 134. Outdoor shuttle court in girls hostel premises

Outdoor volleyball facility for girl students is essential to encourage physical fitness, teamwork, and active participation in sports activities on campus. A net protection facility was provided for girl's students to avoid wrong placement of ball during playing. Net and supporting poles were procured from the ICAR student amenities grants.



Fig. 135. Volleyball court facility for girl students in hostel premises



Fig. 136. Net protection for volleyball court facilitated from ICAR grants

6.5.6. Research Facilities

6.5.6.1. Postgraduate laboratories and Equipment

Dr. N.T.R. College of Agricultural Engineering, Bapatla, has established well-equipped postgraduate laboratories and advanced research facilities to support M.Tech. and Ph.D. programmes in various disciplines of Agricultural Engineering. The laboratories are designed to provide quality practical training, advanced experimentation, research exposure, and skill development for postgraduate and doctoral students. These facilities enable students and researchers to undertake innovative research activities, thesis work, product development, testing, and analysis in accordance with ICAR recommendations and contemporary research requirements.

The postgraduate laboratories are equipped with modern instruments, machinery, analytical equipment, software tools, and experimental setups relevant to the specialized fields of Farm Machinery and Power Engineering, Processing and Food Engineering, Soil and Water Conservation Engineering, and Irrigation and Drainage Engineering. The laboratories support both academic instruction and research activities related to agricultural mechanization, food processing, renewable energy, soil and water management, precision agriculture, irrigation engineering, and protected cultivation technologies.

Research Facilities in Farm Machinery and Power Engineering

The Department of Farm Machinery and Power Engineering possesses advanced laboratories including:

- Farm Power Laboratory
- Farm Machinery Laboratory
- Advanced Workshop
- Soil Dynamics Laboratory
- Ergonomics Laboratory
- Field Laboratory

These laboratories are equipped with modern agricultural machinery, tractor systems, CNC machines, soil bins, sensors, load cells, precision instruments, ergonomics equipment, and workshop facilities required for teaching, testing, fabrication, and research. Advanced facilities such as CNC lathe machines, universal milling machines, soil dynamics instrumentation systems, data loggers, and precision agriculture equipment support research activities related to farm mechanization, machinery design, automation, ergonomics, and soil-machine interaction studies.

Research Facilities in Processing and Food Engineering

The Department of Processing and Food Engineering has specialized postgraduate laboratories including:

- Instrumentation Laboratory
- Process Engineering Laboratory
- Food Analysis Laboratory
- Food Microbiology Laboratory
- Environmental Control Laboratory
- Extrusion Processing Laboratory

The laboratories are equipped with advanced food processing and analytical equipment such as tray dryers, freeze dryers, microwave ovens, IoT-enabled refractive window dryers, UV spectrophotometers, colorimeters, viscometers, protein analyzers, sonicators, thermal imaging cameras, centrifuges, laminar airflow chambers, microbiological

equipment, rice processing units, packaging machines, and pilot-scale twin screw extruders. These facilities support research in food processing, drying technologies, product development, quality analysis, food safety, storage engineering, and value addition of agricultural produce.

Research Facilities in Soil and Water Conservation Engineering

The Departments of Soil and Water Conservation Engineering and Irrigation and Drainage Engineering are equipped with laboratories and field facilities such as:

- Soil and Water Conservation Laboratory
- Watershed Laboratory
- Hydroponics Laboratory
- Surveying and Levelling Laboratory
- Hydraulics Laboratory
- Irrigation Field Laboratory
- Wells and Pumps Laboratory
- Micro Irrigation Demonstration Units

These facilities include watershed models, farm ponds, hydroponics and aeroponics systems, total stations, GPS instruments, digital planimeters, soil moisture sensors, flumes, drip irrigation systems, fertigation units, weather stations, hydraulic flumes, pumps, and water measurement devices. Advanced software facilities such as AutoCAD, MATLAB, SURFER, CROPWAT, MODFLOW, HYDRUS, DRAINMOD, GIS, and Remote Sensing applications are also available for research and data analysis.

Computer and ICT Facilities for Research

The College also provides postgraduate computer laboratories equipped with high-speed internet connectivity, advanced software packages, LCD projectors, smart classrooms, and virtual classroom facilities. These ICT-enabled facilities support research data analysis, modeling, simulation studies, thesis preparation, seminar presentations, and online academic interactions.

Research-Oriented Learning Environment

The postgraduate laboratories and research facilities are effectively utilized for:

- Practical classes and laboratory instruction

- Thesis research work of M.Tech. and Ph.D. students
- Product development and prototype fabrication
- Testing and evaluation of machinery and equipment
- Advanced analytical studies and experimentation
- Training programmes, workshops, and demonstrations
- Industry-oriented and interdisciplinary research activities

Faculty members, technical staff, and research supervisors provide continuous guidance to students during laboratory experimentation and research activities. The College continuously upgrades laboratory infrastructure and equipment to meet emerging technological advancements and research requirements in agricultural engineering. Overall, the postgraduate laboratories and research facilities at Dr. N.T.R. College of Agricultural Engineering, Bapatla, provide a strong platform for advanced learning, innovation, technology development, and quality research in Agricultural Engineering disciplines.

6.5.6.2. Research Contingency

The following is the PG/Ph D Research contingency expenditure for the years 2021-22 to 2025-26.

Table 25. PG/Ph.D. Research contingency during 2021 to 2026

Year	2021-22	2022-23	2023-24	2024-25	2025-26
Expenditure (Rs.)	9,72,564/-	11,70,778/-	9,91,276/-	25,96,739/-	18,82,765/-

6.5.7. Outcome/Output

6.5.7.1. Student Performance in National Examination

Students of Dr. N.T.R. College of Agricultural Engineering, Bapatla, have consistently demonstrated academic excellence and competitive spirit by securing commendable ranks and achievements in various national-level competitive examinations and professional activities. The College provides continuous academic guidance, technical training, and motivation to students for pursuing higher education, research, management studies, and professional careers through competitive examinations. Students of the College have secured impressive ranks in the Graduate Aptitude Test in Engineering (GATE) examination over the years, including single-digit and top national ranks. Several students

secured ranks within the top 10, top 100, and top 250 categories, reflecting the strong academic foundation and technical competency imparted by the institution. These achievements have enabled students to pursue postgraduate studies and research programmes in premier institutions such as IITs, NITs, ICAR institutes, and other reputed universities across the country. In addition to GATE achievements, students have also performed well in national-level management aptitude examinations such as:

- NMAT (Narsee Monjee Aptitude Test)
- XAT (Xavier Aptitude Test)
- CMAT (Common Management Admission Test)

Students secured notable percentile scores and ranks in these examinations, thereby gaining opportunities for higher education and professional career advancement in management and allied sectors. The students at the College also achieved recognition at the national level through:

- Selection as Agricultural Research Scientists in examinations conducted by ICAR
- Participation in National Level Ideation Competitions organized by ICAR–NAARM, Hyderabad and innovation development organizations
- Recognition as Best Outgoing Student for outstanding academic and extracurricular performance.

These accomplishments demonstrate the academic excellence, analytical ability, technical knowledge, innovation skills, and competitive preparedness of the students of Dr. N.T.R. College of Agricultural Engineering, Bapatla. The achievements also reflect the effective teaching-learning environment, mentoring support, practical exposure, and career guidance provided by the College for the holistic development of students.

Table 26. List of students selected in competitive examinations during 2021 – 2026

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Year 2020-21

GATE Rank Holders

S.NO	Name of the Students	I.D.NO	Rank
1	Sudhir Kumar K	BE/2017-069	4
2	Venkata RamPrakash M	BE/2017-056	8
3	Yogesh Singh D	BE/2017-070	23

4	Narendra Kumar Chaudhary	BE/2017-009	27
5	Soubhagya Kumar S	BE/2017-068	51
6	Gulshan Kumar	BE/2017-066	67
7	Bhargavi K	BE/2017-007	96
8	Mikhilya N	BE/2017-024	96
9	Rama Kumar J	BE/2017-038	126

ICAR/JRF STUDENTS

S.NO	Name of the Students	I.D.NO	ICAR JRF Rank
1	R. Naveen	BE/2017-034	17
2	S. Archana	BE/2017-005	53
3	A. Vyshnavi	BE/2017-063	63
4	V.V. Brahmananda Reddy	BE/2017-055	78

Year 2021-22

GATE Rank Holders

S.NO	Name of the Students	I.D.NO	Rank Card
1	Shashwat Maurya	BE-18-067	2
2	P.G. V. S Naga Durga	BE-18-014	105
3	Pranav Deevi	BE-18-030	163
4	S Poojith	BE-18-028	184
5	N Sai Sumanth	BE-18-037	229

ICAR/JRF STUDENTS

S.NO	Name of the Students	I.D.NO	ICAR JRF Rank
1	Dosuri Anil Kumar	BE-18-007	64
2	L Navya Tejaswini	BE-18-026	108
3	S Poojith	BE-18-028	152
4	D. Vasu	BE-18-54	64
5	K. V. L. Supraja	BE-18-057	141
			ICAR JRF (ABM)
1	D Swathi Kiran	BE-18-048	125
2	Sk. M. Junaid	BE 18-22	136
3	Neelam. Revathi	BE-18-033	182

Year 2023-24

Gate ranks and Admission in PG Details

S No.	Name of the student	ID NO	Rank
1	Raunak Kumar	BE/2020-027	15
2	Suman Kumar Saha	BE/2020-048	31
3	Mounika Miriyala	BED/2021-007	225
4	Srinivasa Rao Krosuri	BE/2020-046	IIPM Bengluru
5	Sai Deepika Vallabhudasu	BE/2020-030	IIPM Bengluru
6	Narasimha Reddy Munnam	BE/2020-063	IIPM Bengluru

ICAR JRF

S.No.	Name of the Student	I.D.No.	Rank
1	Lakshmi Narayana Vadakoppula	BE/2020-018	97
2	Maruthi Naik Ramavath	BE/2020-022	135
3	Siri Chandana Jeldi	BE/2020-038	380
4	Sravani Unkili	BE/2020-042	
5	Sushma Raj Vempa	BE/2020-050	348
6	Uday Kiran Gorige	BE/2020-055	60 (ABM)
7	Kavya Devi Madaka	BE/2020-064	258
8	Durgaprasad Badagala	BED/2021-001	95
9	Haripriya Kalapa	BED/2021-004	75
10	Parvin Shaik	BED/2021-008	194

Year 2024-25

GATE Rank Holders

S.NO	Name of the Students	I.D.NO	Rank
1	V. Bhavya Sai	BE/2021-010	63
2	Prakash Kumar	BE/2021-038	66
3	Anshuman Gosh	BE/2021-004	159
4	Rimjhim Kumari	BE/2021-041	234

Year 2025-26

Gate ranks and Admission in PG Details

S No.	Name of the student	ID NO	GATE ranks
1	Mr. Prakash Kumar	BE/2021-038	2
2	K. Deepthi	BED/2023-008	16

3	Gulshan Kumar	BE/2022-029	29
4	B. Keerthika	BE/2022-018	56
5	Ujma Anzum	BED/2023-042	83
6	Vishal Kumar Mandal	BE/2022-40	83
7	P. Anjani	BED/2023-002	117
8	Stella Prashanthi	BED/2023-037	151
9	G. Yamuna	BED/2023-051	169
10	Durganand Yadav	BE/2022-028	179
11	Vijay Shankar	BE/2023-028	195
12	B. Ramyasri	BED/2023-031	222
13	Hussain M.	BED/2023-016	532
14	Piyush Keshav Jha	BE/2022-036	Secured admission Manipal B School through CAT
15	Ankit Rahad	BE/2022-033	Secured admission ABM in Pant Nagar through CMAT

Academic year	Name of the student	Student performance
2022 – 23	Raja Sekhar M BEM/2015-010	Selected as an Agricultural Research Scientist at National level examination conducted by ICAR in the Dept. of Processing and Food Engineering
	Shilpa S Selvan BEM/2017-005	
	Sivamma P BEP/2019-004	
	Tejas Shrivas Bearing ID No BE/19-58	Secured 98 Percentile in NMAT & 93.72 Percentile in XAT examinations
	Mr. Akash Kumar Singh and Mr. Tatwadarshi Hota	Participated in National Level Ideation Competition organized by IDEA (Association for Innovation Development for Entrepreneurship in Agriculture ICAR - NAARM, Hyderabad)

Students Performance in National Examinations

Year	GATE	ICAR (JRF)	Management

2020-21	10	5	6
2021-22	5	6	-
2022-23	5	5	3
2023-24	3	4	5
2024-25	5	4	-
2025-26	13		4

6.5.7.2. Students Placement profile

Dr. N.T.R. College of Agricultural Engineering, Bapatla, has consistently maintained a commendable placement record for both undergraduate and postgraduate students. The College actively supports students through career guidance programmes, campus recruitment drives, industrial interactions, skill development training, Student READY programmes, and placement assistance activities. Students at the College are being placed in public sector organizations, private industries, banking sectors, research organizations, and higher education institutions across the country. The placement profile reflects the growing demand for Agricultural Engineering graduates in sectors such as farm machinery industries, irrigation companies, food processing industries, agri-tech companies, research institutions, banking services, and higher education institutions. Many students also pursue postgraduate studies and research programmes in premier institutions including IITs, ICAR institutes, NITs, and State Agricultural Universities. The year-wise placement particulars of undergraduate and postgraduate students are presented below:

Table 27. Percentage of students selected in Placements during 2021 - 2026

Year	No. of Outgoing Students	Public Sector	Private Sector	Banking Sector	Others (Higher Studies)	Total Placements	% to Outgoing students
U. G.							
2021-22	77	03	10	04	10	27	35%
2022-23	74	--	17	01	12	30	41%
2023-24	56	01	31	--	10	42	75%
2024-25	73	--	32	--	13	45	61%
2025-26	84	--	--	--	--	--	--
P. G.							
2021-22	11	--	11	--		11	100%
2022-23	15	--	12	--	03	11	100%
2023-24	08	--	06	--	02	15	100%
2024-25	01	01	01	--	-	08	100%
2025-26	14	--	07	--	07	01	100%
Total (UG+PG)	**413**	**05**	**127**	**05**	**57**	**190**	**46.97%**

Table 28. Firm-wise placement profile of the students

Year	Name of the firm	No. of students placed	Salary package (LPA)	Name of the students
2021 - 22	Farm Gate Technologies	3	3.5	P. Shyam Kumar B.Veera Babu G.Lokesh
	Infosys	5	3.5	Yamini K Vijaya Naga Sirisha Venkata RamaKrishna E Lakshmi Sailaja A Priyanka Kumari
2022 - 23	Tractors and Farm Equipment Limited (TAFE), Chennai	2	6.0	Rahul P Mohan kumar.G
	Mahindra & Mahindra, Mumbai	2	6.5	D. Surendra S.Harshavardhan Reddy
	Dhan Foundation, Madurai	9	4.0	T Surendra Vani sree G Usha Sri K Bhargavi K Chandini Priya Venkata Sai K Naga Deekshitha V Sai Charan Teja T Maheswari Devi S
	Tafe, Hyderabad	2	3.5	M.Nag Charan K.Murali
	Skipper &Metzer India Llp, Hyderabad	1	3.5	N. Naveen
2023 - 24	Mahindra Tractors	3	6.50	M. Aravaind Ch.Swetha P. Anitha
	Dhan Foundation	10	3.00	G.Rohith P.Vyshnavi M.Shanmukha Vyshnavi A. Shantipriya K Chandrisha P. Kusala S. Manindra' P.Suprajasuma B.Satya Sowmya K. Swetha
	Sids farm, Hyderabad	10	4.50	B.Durga Prasad

				U.Sravani V.Lakshminarayana D.Srilakshmi J.Sirichandana S. Manindra M.Narasimha Reddy M. Maruti Naik D. Sriram Rahul M.Venkatesh Varma
	Lation crafts, Vijayawada	1	4.00	Navya. G
	Lactalis India	3	4.50	J. Ravve Teja K. Haripriya V. Sushma Raj
	ITC-SEARCH NGO	3	4.00	V. Venkatesh M. Venkataesh varma
2024 - 25	Skipper & Metzer	2	3.50	B.Yashwanth B. Aravind
	Dhan Foundation	9	3.00 during probation 5.50 post probation	P.Jaya N.Maneesha B.Srikanya Y.Sireesha Ch.Ajay John Srujana T.Santhoshini S. Aruna B.Dellieswari
	Nabhitha Engineering, Hyderabad	5	3.00	M. Sai Lakshmi M. Neeraja Blessy A. Siiva Teja Shaik Reshma K. Karuaya
	John Deere	1	4.00	Shaik Sameer
	Kothari group – Drip Irrigation	1	3.50	D. Sai Srinivasa Rao
	ITC-SEARCH NGO	3	3.00	Ch. Jhansi K. Swaroopa M. Lavanya
	Rasi Seeds	3		A. Maruthi Ch. Ajay K. Venkatanarayana
	Sam Agri Pvt Ltd	1	3.00	Ch. Tirumala Lokesh
	Akshayakalpa Farms Pvt. Ltd.,	2	3.00	V. Padmini K. Reshma
	Indira Krishna Biotech	1	3.00	A. Kusuma
	Croplinks Pvt ltd	2	3.50	M. Sai Lakshmi
	Heartfulness foundation as development officer	2	4.50	Uma Maheswara Rao Satyanarayana K
	Saveer biotech ltd	3	3.50	Tatwadarshi Hota Rohit Reddy Prasanth Gaurav

2025-26	Mahindra Swaraj Tractors	1	5.75	Metta Leela Rani
	Akshaya kalpa farm and dairy, bangalore	5	3.50	Ch. Gayathri K.Naga Jyothsa Y.Mahati B.Daisy Dumbeswari
	Akshaya kalpa farms and dairy hyderabad	3	3.50	Sindhu Vaishnavi Divya Ande K.Chytanya Durga
	ML Agro industries pvt ltd	3	4.00	Nithin Gunupuru P Gopi Chand K Malikarjuna Rao
	Tafe	3	3.50	Nithin Gunupuru Tarun Kumar Bandi Murali Kanth Ch
	ITC aquabusiness	3	3.00	Vasavi T Harshaswi Sk. Imrana
	Dhan Foundation	8	4.50	P.Asritha T.Dharani K Gayatri Sai T. Harini M.Lavanya K. Manisha chandrika T.Maheswari K. Supraja

Overall, a total of 413 undergraduate and postgraduate students graduated during the assessment period, out of which 190 students secured placements in various sectors or pursued higher studies, resulting in an overall placement percentage of 46.97 percent. The consistent placement performance of the College demonstrates the effectiveness of the academic curriculum, practical exposure, industrial training, and skill-oriented teaching-learning environment provided at Dr. N.T.R. College of Agricultural Engineering, Bapatla. The College continues to strengthen industry linkages and employability enhancement initiatives to improve placement opportunities for students in the agricultural engineering sector.

6.5.7.3. Awards/Recognitions/Certificates

Awards, Recognitions, and Achievements of Faculty Members

The faculty members of Dr. N.T.R. College of Agricultural Engineering, Bapatla, have received several prestigious awards, recognitions, patents, fellowships, and professional

honours at University, State, National, and international levels for their contributions in teaching, research, extension, innovation, and academic excellence. These achievements reflect the academic strength, research capability, and professional commitment of the institution. Faculty members of the College have been honoured with several Meritorious Teacher Awards and Best Teacher Awards at both University and State levels in recognition of excellence in teaching and academic service. The College faculty also received the prestigious State Best Teacher Award from the Government of Andhra Pradesh for outstanding contributions to higher education and academic administration. The institution has also achieved recognition in the field of research and innovation through awards such as:

- Outstanding Women Researcher in Hydroponics
- Young Professional Award
- Best Paper Award
- Best Oral Presentation Award
- Best Agri Engineer Award
- Prof. N.G. Ranga and Bharathi Devi Ranga Memorial Gold Medal Award for securing highest OGPA in Ph.D. programme.
- Best Agri Engineer Award
- Netaji Subhas – ICAR – International Fellowship – 2025, ICAR – New Delhi

Faculty members have also received professional recognition for innovation and intellectual property development through the grant of an Indian Design Patent for the development of a *Mechanical Dried Chillies Compaction cum Bagging Unit*. In addition, faculty members have been awarded prestigious fellowships and professional recognitions, including the Netaji Subhas – ICAR International Fellowship awarded by the Indian Council of Agricultural Research (ICAR), New Delhi. These awards and recognitions demonstrate the continued commitment of the faculty towards academic excellence, quality teaching, innovative research, technology development, and professional service in the field of Agricultural Engineering and allied sciences. The achievements of the faculty members significantly contribute to the academic reputation, institutional development, and research excellence of Dr. N.T.R. College of Agricultural Engineering, Bapatla.

Table 29. The list of awards secured by the staff members

Academic year	Name of the staff member	Award received

2020 – 21	Dr. M. Sardar Baig Associate Dean	Meritorious Teacher Award
2021 – 22	Dr. B. Sreenivasula Reddy Associate Professor	Meritorious Teacher Award
2022 – 23	Dr. K. Krupavathi Assistant Professor	Meritorious Teacher Award (Agricultural Engineering & Technology)
	Dr. Rega Ganesh Babu Associate Professor	Meritorious Teacher Award (Agricultural Engineering & Technology)
	Dr. Rega Ganesh Babu Associate Professor	Prof. N.G. Ranga and Bharathi Devi Ranga memorial gold medal award for securing the highest OGPA in Ph.D.
2023 – 24	Dr. G. Ravi Babu Professor & Head	"State Best Teacher" for the year 2023 by Government of Andhra Pradesh
	Dr. K. Lavanya Assistant Professor	Meritorious Teacher Award (Agricultural Engineering & Technology)
	Dr. S. Vishnu Vardhan, Associate Professor Dr B. John Wesley, Professor	Indian Design Patent on "Mechanical dried chillies compaction cum bagging unit"
2024 – 25	Dr. S. Vishnu Vardhan Associate Professor	"Best Teacher" for the year 2024 for his meritorious service as Teacher at State Level by Govt. of Andhra Pradesh.
	Dr. K. Krupavathi Assistant Professor	"Outstanding Women Researcher in Hydroponics" for the contribution and achievement in the discipline of engineering on 01.03.2025 by Venus International Foundation, Chennai, Tamil Nadu.
	Dr. N. Ashok Kumar Assistant Professor	"Young Professional Award" in 6th International Conference "Holistic Innovations and Technologies Advances for Sustainable Agriculture" on 4 to 6thMarch 2024 Jointly Organized by SR University, Hyderabad and Just Agriculture Education Group & ISAHRD, Chandigarh.
2025 – 26	Dr. B. Sreenivasula Reddy Professor and Head	Best Paper Award 2025 by the Indian Society of Agricultural Engineers, New Delhi
	Dr. B. Lavanya, Associate Professor	Best Teacher Award – 2025 by Glacier Journal Research Foundation, Global Management Council, Ahmedabad
	Dr. K. Krupavathi Associate Professor & Head	Best oral presentation award in 2nd National Seminar on Emerging Trends in Agriculture and Allied Sciences: A Pathway to Food Security. ENCOURAGE society, Bengalure & AgriTech Publication, Bengaluru
	Dr. A. Srinivasa Rao Assistant Professor	Meritorious Teacher Award for the year 2023 from Acharya N.G. Ranga

		Agricultural University during the 57th Annual Convocation
	Er. N. Kishore Assistant Professor	Best Agri Engineer Award from ICFRE–HFRI, Shimla, in association with the National Medicinal Plant Board, RRS, Himachal Pradesh (Govt. of India) and ISAHARD Society, New Delhi, during 13th International Conference
	Er. N. Kishore Assistant Professor	Netaji Subhas- ICAR International Fellowship-2025 by Agricultural Education Division, Indian Council of Agricultural Research, New Delhi

Student Achievements in Cultural and Literary Activities

Students of Dr. N.T.R. College of Agricultural Engineering, Bapatla, actively participate in various cultural, literary, and extracurricular activities conducted at University and All India levels. The College encourages students to showcase their talents and creativity through participation in competitions organized during intercollegiate sports, games, cultural, and literary meets conducted by Acharya N.G. Ranga Agricultural University and other institutions. Over the years, students of the College have secured several prizes and recognitions in a wide range of events such as:

- Cartooning
- Poster Making
- Solo Singing
- Solo Instrumental Music
- Solo Dance
- Folk Dance
- Patriotic Poem Competitions
- Extempore (English)
- One-Act Play Competitions
- Literary and Cultural Events

Students have won First Prize, Second Prize, Third Prize, and Best Performer Awards in various competitions conducted during Inter-Collegiate Sports, Games, Cultural and Literary Meets organized at institutions such as Agricultural College, Bapatla and S.V. Agricultural College, Tirupati. The consistent achievements of students in cultural and literary events reflect the College's commitment towards the holistic development of students

by encouraging participation in co-curricular and extracurricular activities along with academic excellence. These activities help students improve communication skills, creativity, confidence, leadership qualities, teamwork, and overall personality development.

The College provides continuous encouragement and support to students for participation in cultural and literary programmes at University, State, and National levels, thereby promoting a vibrant and student-centered campus environment.

Table 30. Students who participated and won prizes in various cultural and literary activities at University and All India level

Name of the Student	Name of the Activity	Merit/Place	Name and Place of Event
2025-26			
B Sri Spandan	Cartooning	I Prize	II phase Inter collegiate sports, games, cultural and literary meet, S.V. Agricultural College, Tirupathi From 05-12-2025 to 08-12-2025
M Leela Rani	Solo Singing	II Prize	
B Sri Spandan	Poster Making	III Prize	
B. Narasimha Rao	One act play	Best performer	
2024-25			
Leela Rani M	Solo Song	I Prize	II Phase Inter collegiate sports, games, cultural and literary meet, Agricultural College, Bapatla from 17-11-2024 to 20-11-2024
Pravallika S	Patriotic Poem	III Prize	
Piyush Keshav	Extempore-English	III Prize	
Varaprasad	Solo Instrument	III Prize	
2023-24			
P Susan Raju	Solo Instrumental	I Prize	II phase Inter collegiate sports, games, cultural and literary meet, S.V. Agricultural College, Tirupathi from 12-12-2023 to 15-12-2023
T Harsha Sri	Solo Flok Dance	II Prize	
Piyush Keshav Jha	Extempore - English	II Prize	
2022-23			
M. Aravind	Patriotic Poem	I Prize	II phase Inter collegiate sports, games, cultural and literary meet, Agricultural College, Bapatla from 08-12-2022 to 11-12-2022
2021-22			
M Aravind	Patriotic Poem	I Prize	II phase Inter collegiate sports, games, cultural and literary meet, Agricultural
Sashwath	Solo Instrumental	I Prize	

CH V N Sirisha	Solo Dance	II Prize	College, Bapatla from 19-11-2021 to 23-11-2021

Few glimpses of the cultural activities of students at Inter-Collegiate Level



Fig. 137. Cultural activities of students at Inter-Collegiate Level

6.5.7.4. Employability

Dr. N.T.R. College of Agricultural Engineering, Bapatla, places significant emphasis on enhancing the employability of students through a comprehensive and industry-oriented teaching–learning process. The College continuously strives to equip students with technical knowledge, practical exposure, problem-solving ability, communication skills, leadership qualities, and professional ethics required for successful careers in Agricultural Engineering and allied sectors. The curriculum is designed in accordance with ICAR recommendations and industry requirements, integrating theoretical instruction with practical training, laboratory work, field exposure, industrial visits, experiential learning, and Student READY programmes. Students are provided with hands-on training in agricultural machinery, irrigation systems, food processing technologies, renewable energy applications, soil and water management, precision agriculture, and modern engineering tools. Such practical exposure enhances technical competency and improves students' confidence in handling real-time field and industrial challenges.

The College also encourages participation in seminars, workshops, technical competitions, project work, internships, skill development programmes, and research activities to strengthen analytical thinking, innovation, creativity, and teamwork among students. Training programmes conducted in industries, ICAR institutes, government organizations, and private companies expose students to modern technologies, workplace culture, and emerging trends in agricultural engineering. Special attention is given to developing soft skills and professional attributes such as communication skills, interpersonal relations, leadership, decision-making ability, time management, ethics, and adaptability. Students are encouraged to participate in cultural, literary, sports, NSS, NCC, and extension activities for their holistic personality development and social responsibility.

The College also motivates students to prepare for competitive examinations such as GATE, ICAR-JRF/SRF, banking, management entrance examinations, and public sector recruitment examinations. Career guidance programmes, placement drives, entrepreneurship awareness programmes, and interactions with industry experts further enhance employment opportunities and self-employment potential among students. As a result of these efforts, students of the College have been securing placements in public sector organizations, private industries, research institutions, banking sectors, agri-tech companies, and higher education institutions. Many students pursue postgraduate studies and research programmes in premier national institutions such as IITs, NITs, ICAR institutes, and State Agricultural Universities.

Thus, the College focuses on developing technically competent, socially responsible, innovative, and professionally skilled graduates capable of contributing effectively to the workforce, farming community, agricultural development, and the overall economy of the nation.

6.5.9. Certificate

I, the Associate Dean Dr. D.D. Smith hereby certify that the information contained in sections 6.4 and 6.5.1 to 6.5.7.4 are furnished as per the records available in the college and degree awarding university.

Signature of the Associate Dean of the college with date & seal